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(54) **BINDING MOIETY FOR CONDITIONAL
ACTIVATION OF IMMUNOGLOBULIN
MOLECULES**

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(57) **ABSTRACT**

Disclosed herein are binding moieties that comprise non-
CDR loops for masking the binding of a binding molecule
to its target and CDRs for binding bulk serum proteins.
Conditionally active target binding proteins that contain the
binding moieties are also provided. Pharmaceutical compo-
sitions comprising the binding proteins disclosed herein and
methods of using such formulations are further provided.

20 Claims, 35 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1

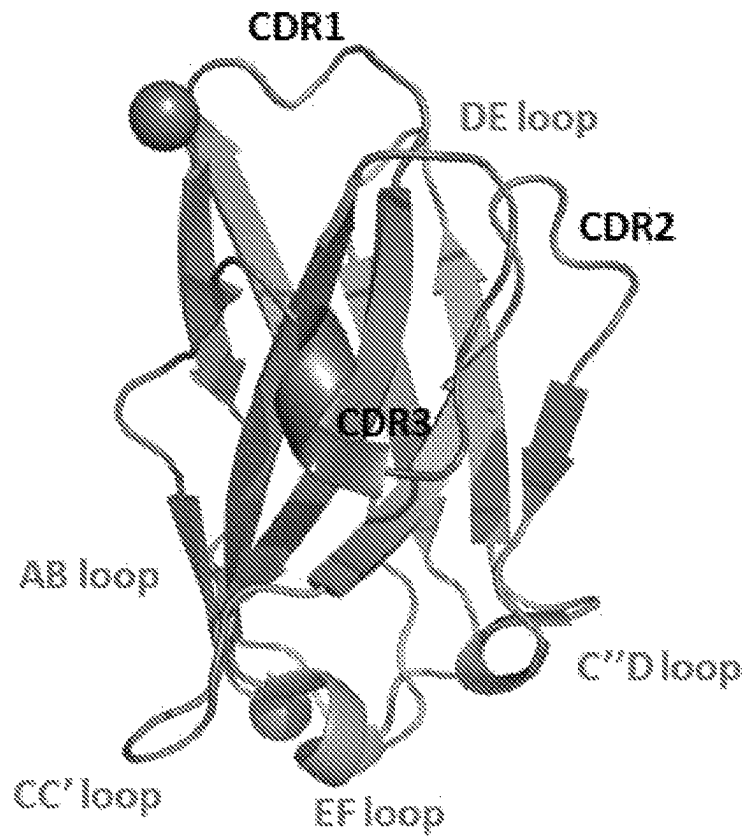


FIG. 2

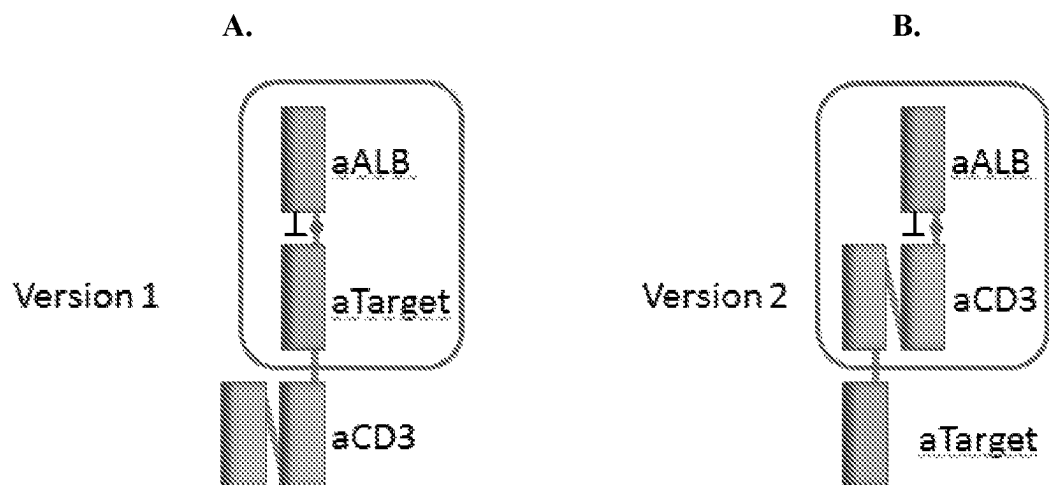


FIG. 3

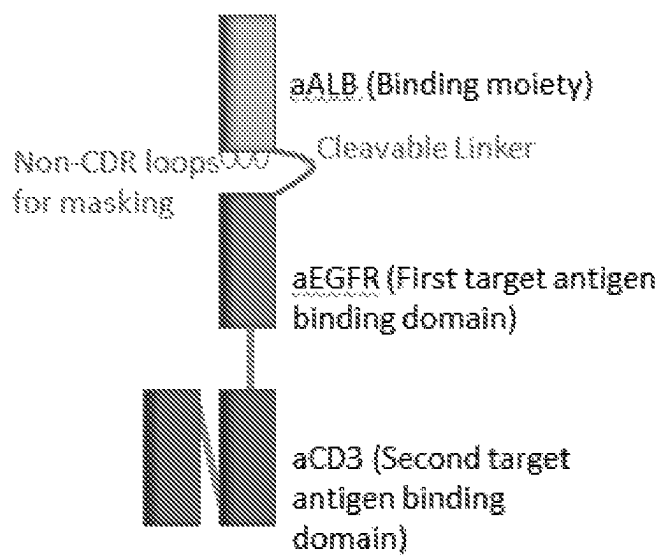
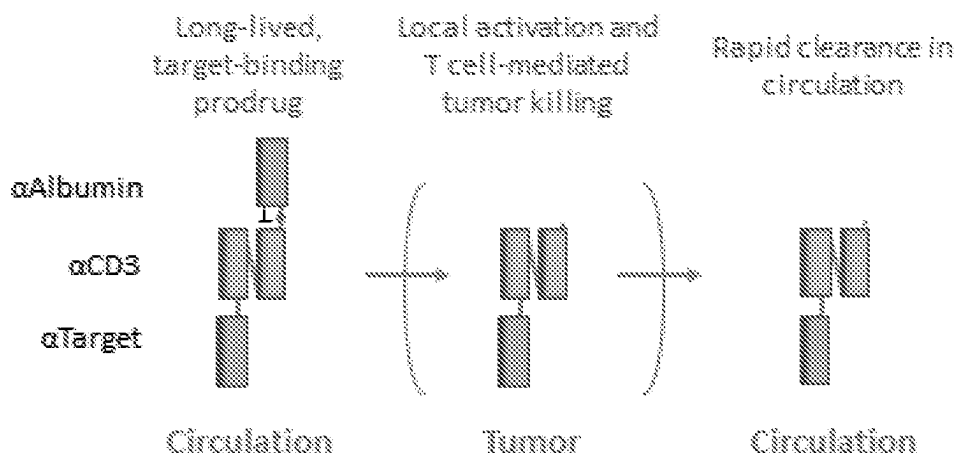
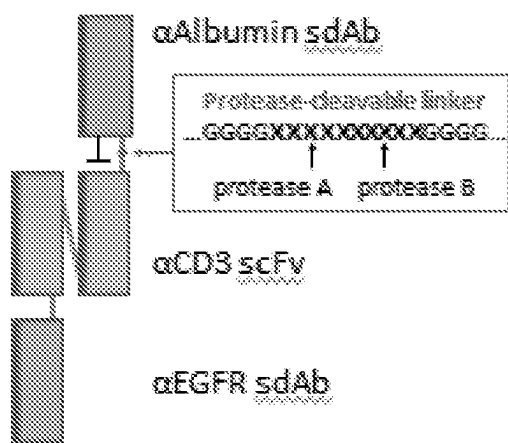


FIG. 4

A.



B.



C.

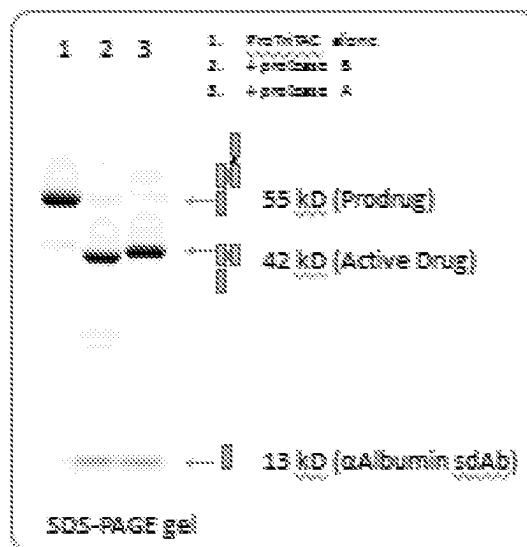
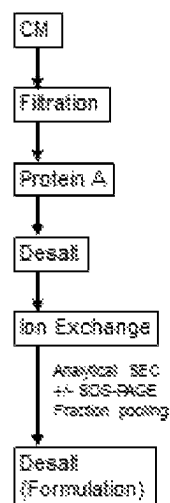
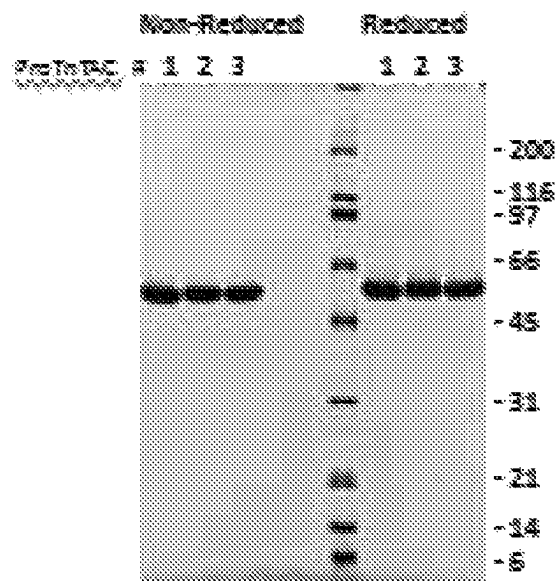


FIG. 5

A.



B.



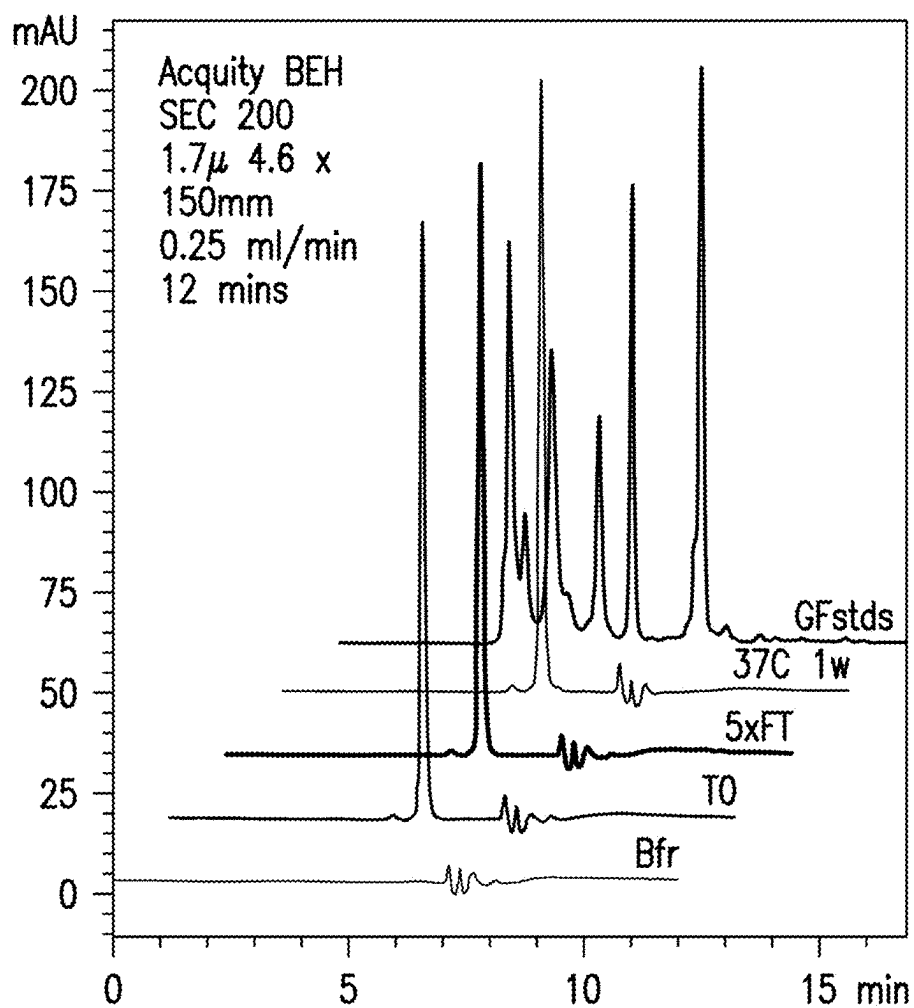


FIG.6A

Condition	% HMW	% Main	% LMW
T0	1.7	96.0	2.3
5xFT	1.6	97.0	1.4
37C 1w	2.6	95.4	2.0

FIG.6B

FIG. 7

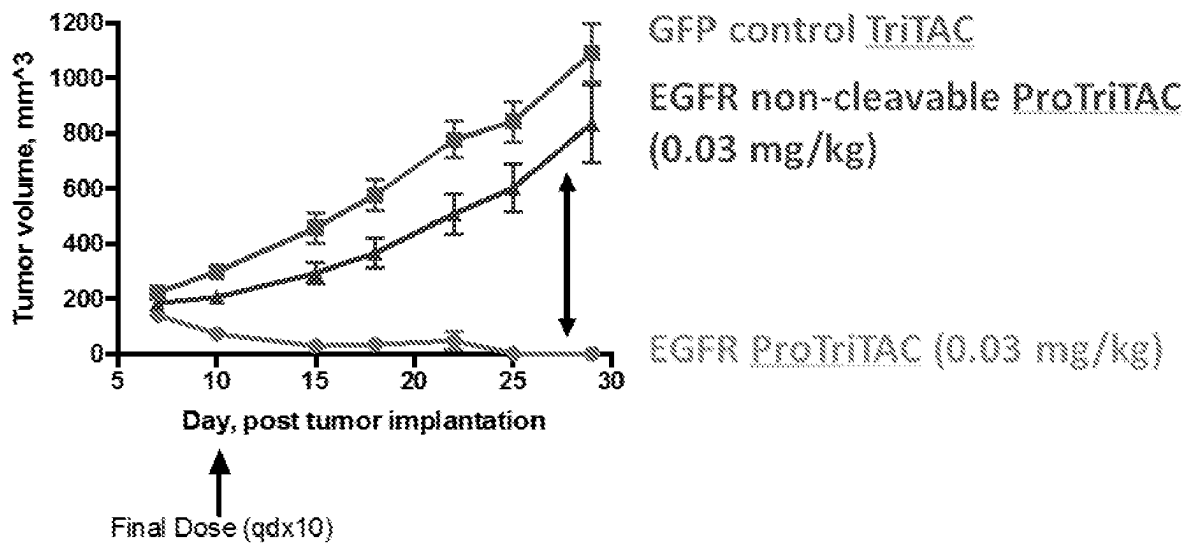


FIG. 8

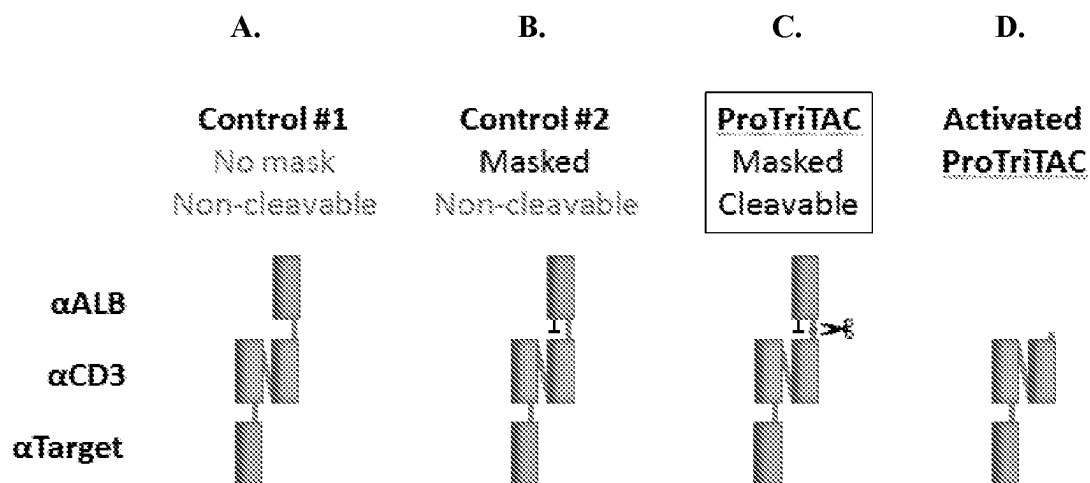


FIG. 9

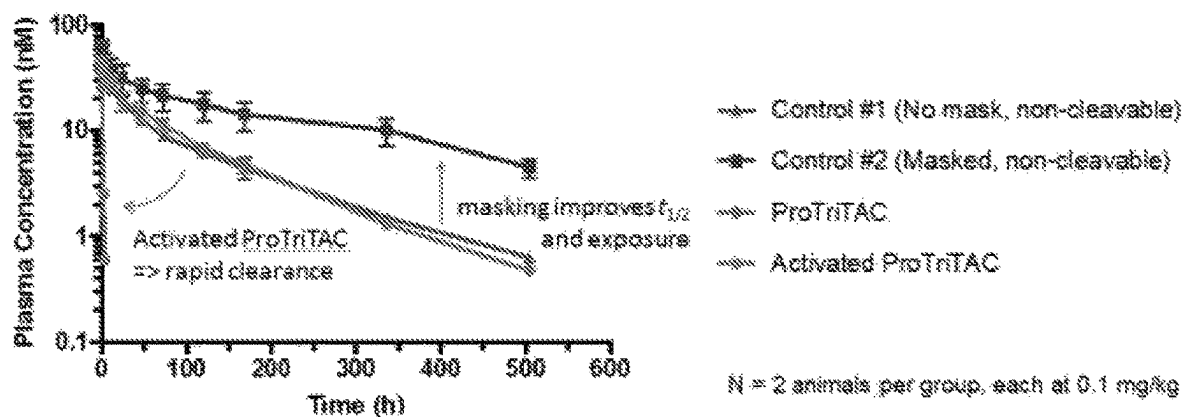


FIG. 10

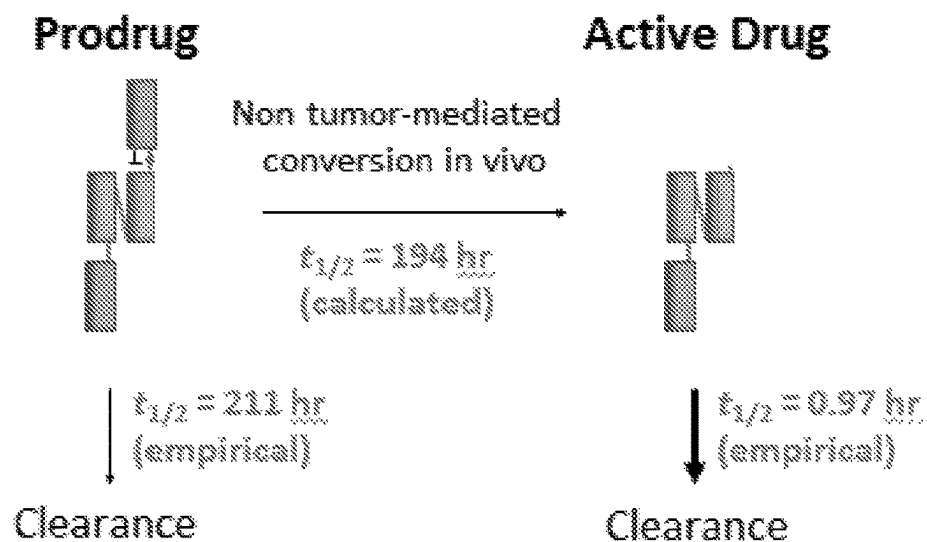


FIG. 11

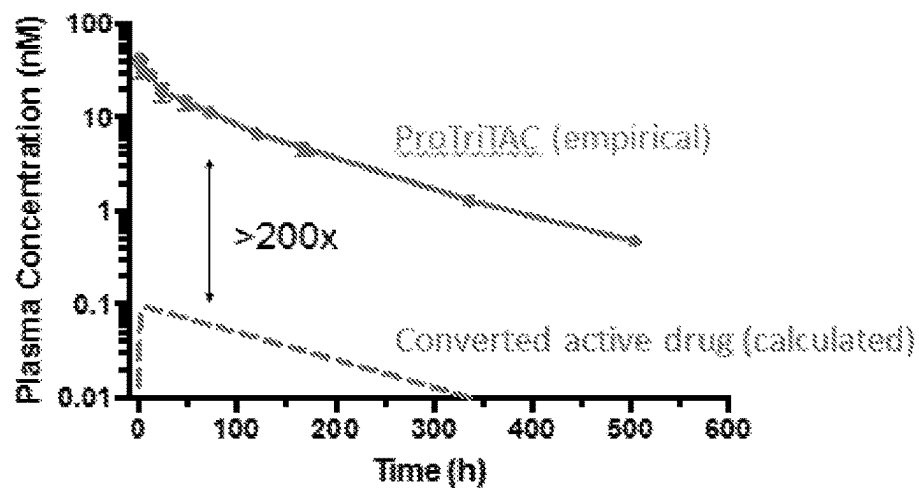


FIG. 12

250x CD3 Binding Differential in ELISA

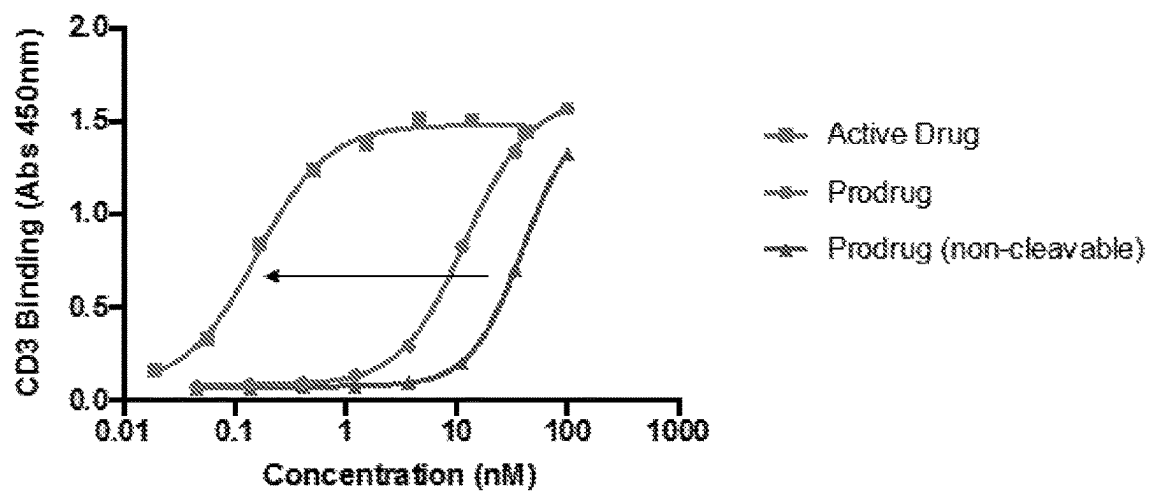


FIG. 13

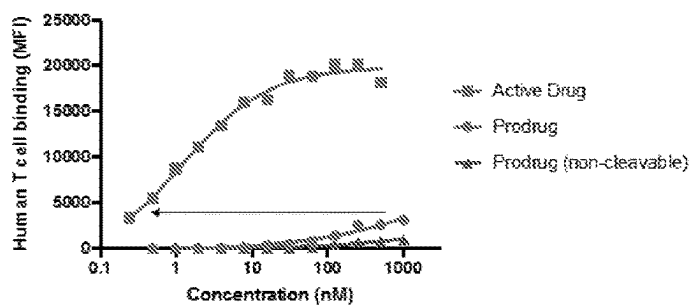
>1000x Human Primary T Cell Binding Differential in Flow Cytometry

FIG. 14

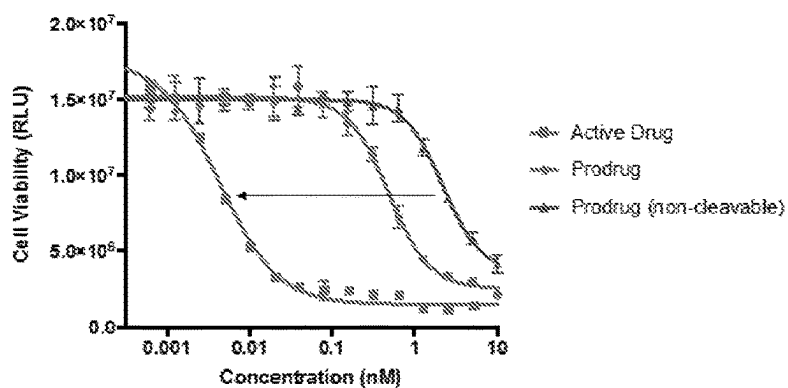
550x Functional Differential in T Cell Killing Assay

FIG. 15

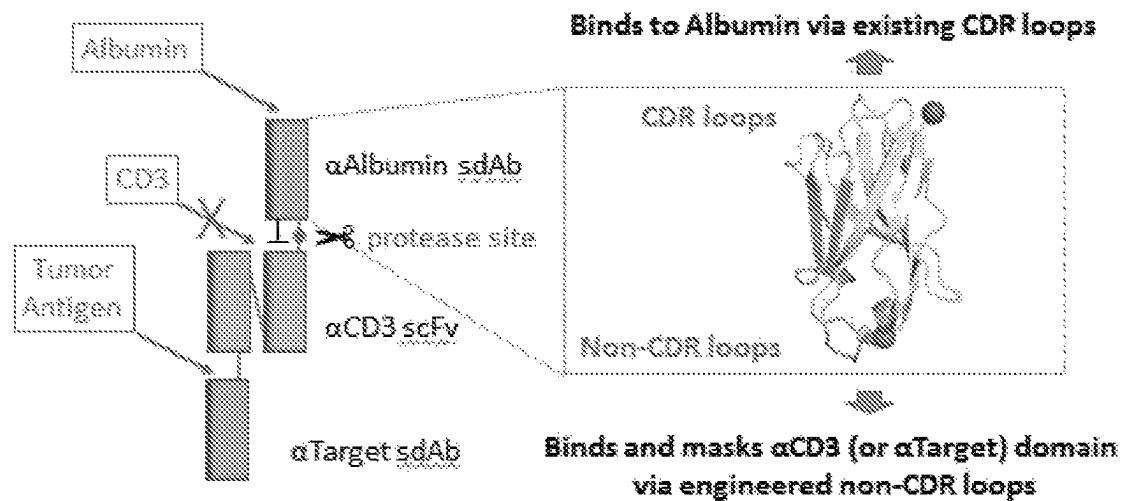


FIG. 16

Linking Half-Life Extension with Functional Masking (Generic)

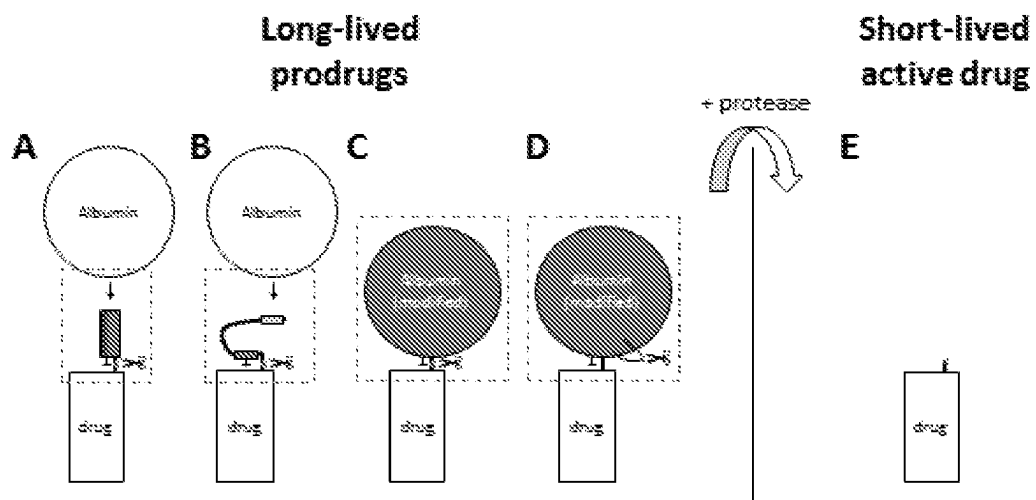


FIG. 17

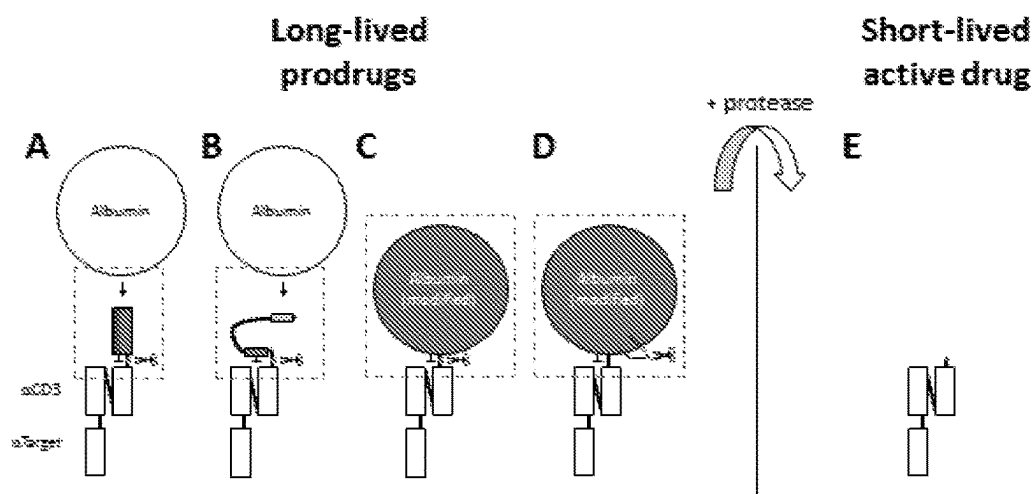
Linking Half-Life Extension with Functional Masking
(T cell engager)

FIG. 18

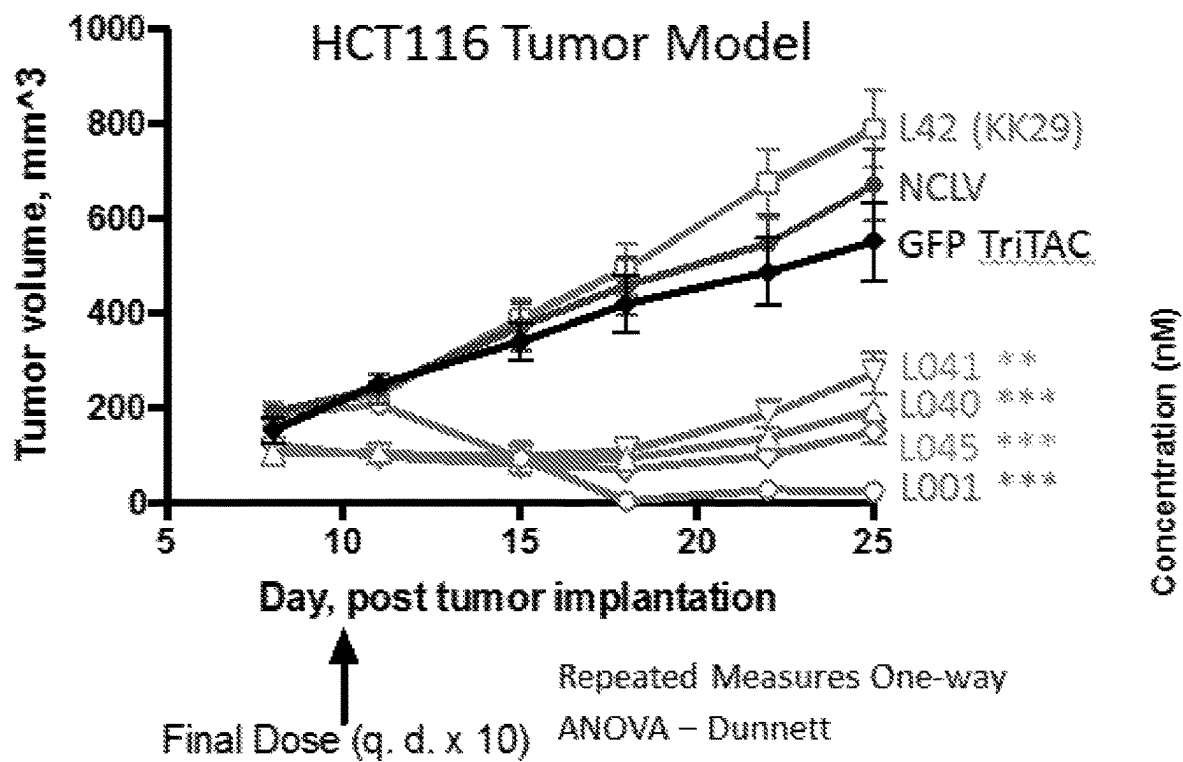
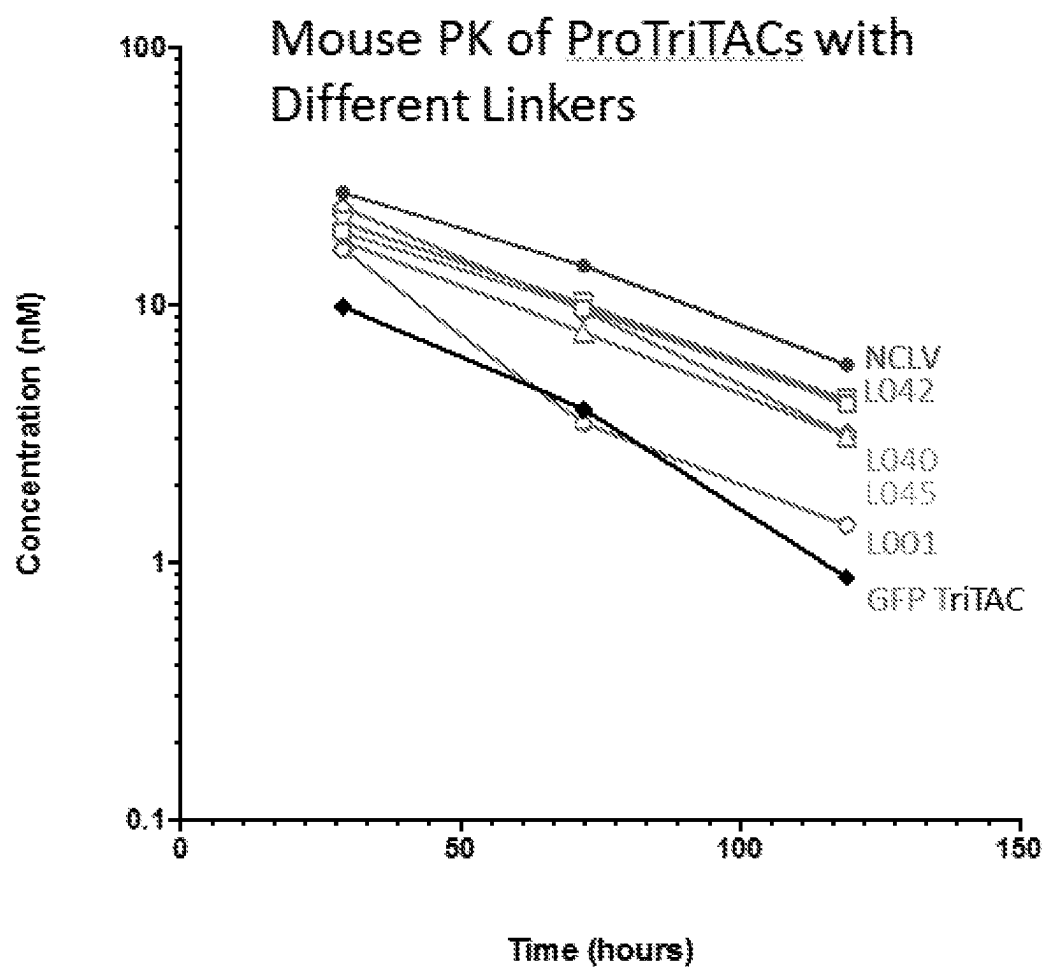


FIG. 19



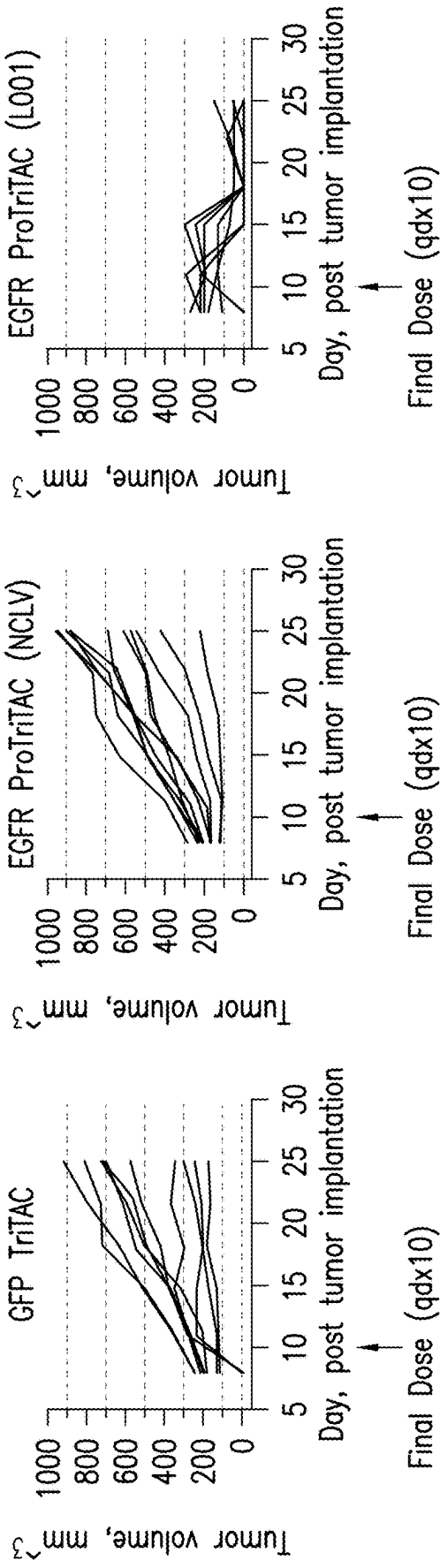


FIG. 20A

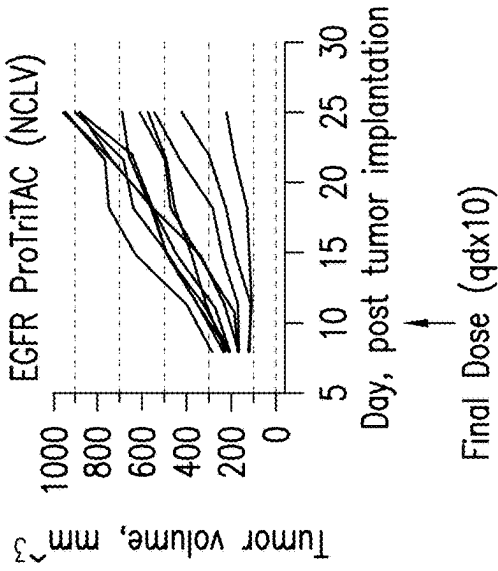


FIG. 20B

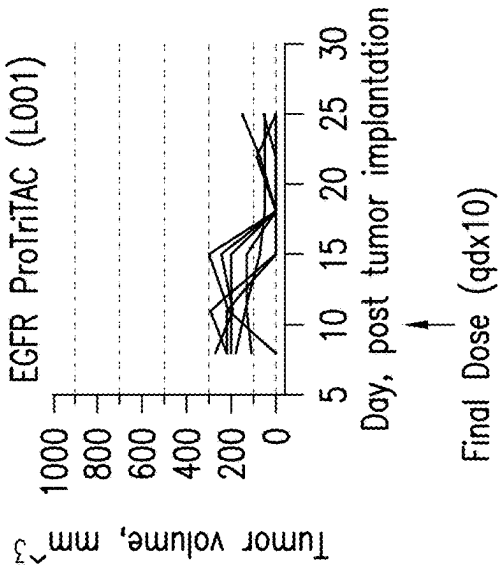


FIG. 20C

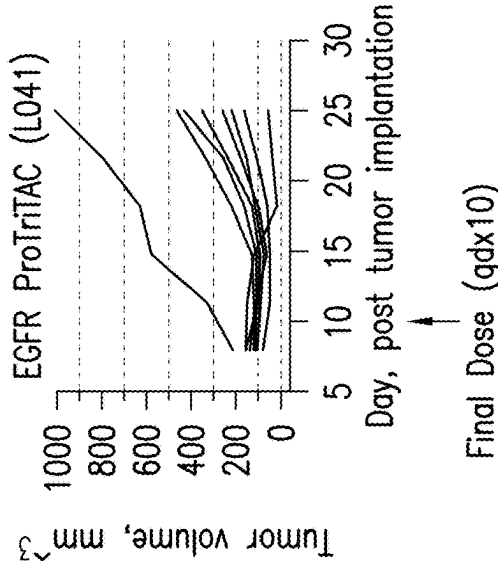


FIG. 20D

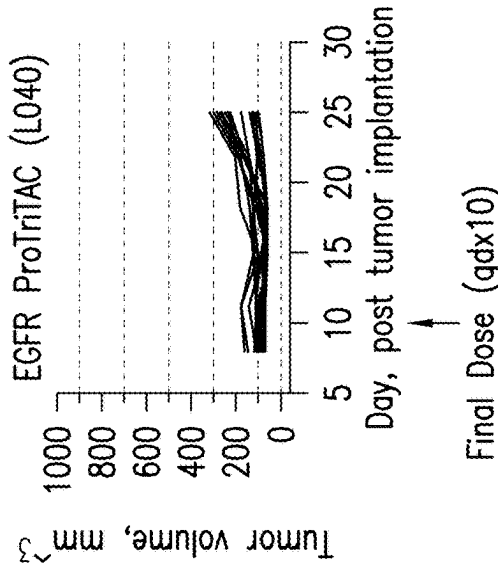


FIG. 20E

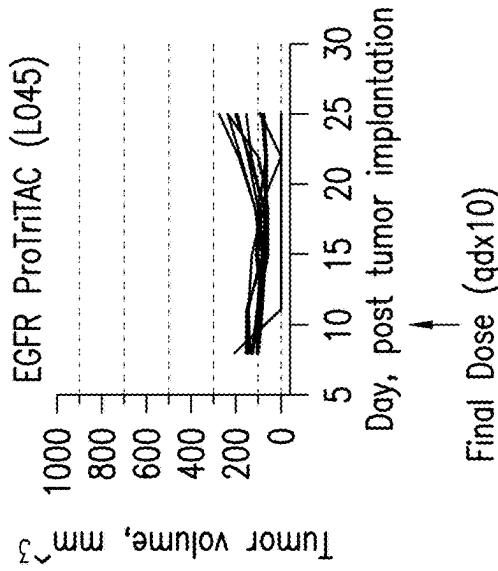


FIG. 20F

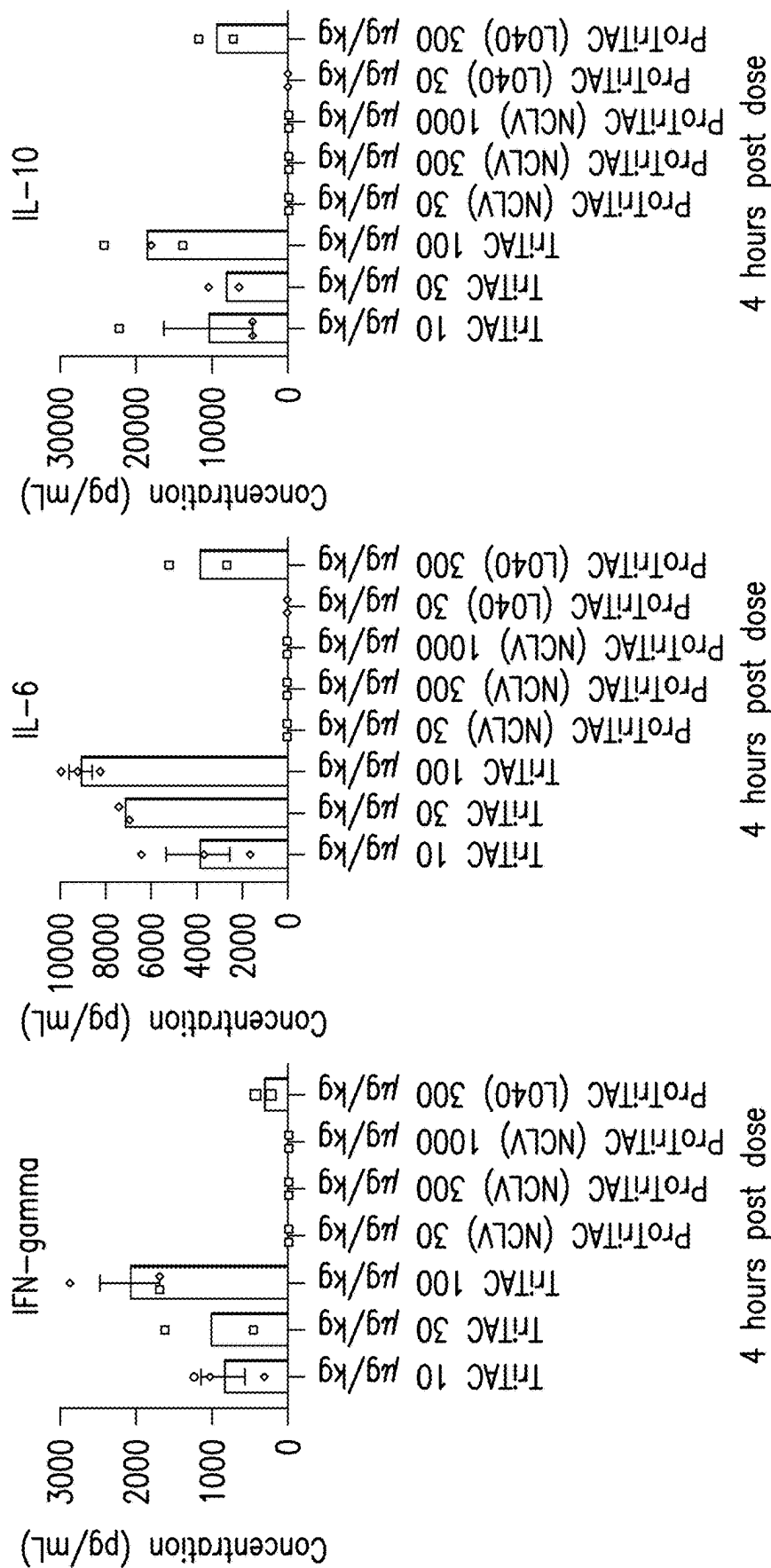


FIG. 21C

FIG. 21B

FIG. 21A

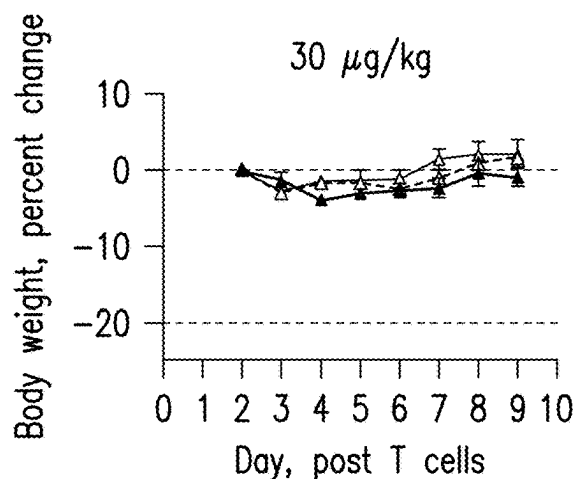


FIG. 22A

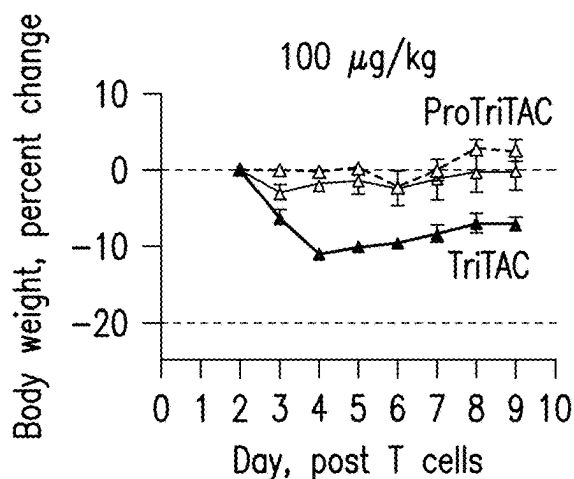


FIG. 22B

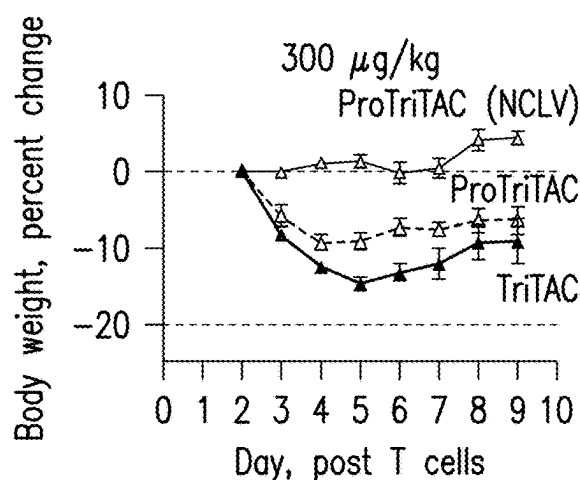


FIG. 22C

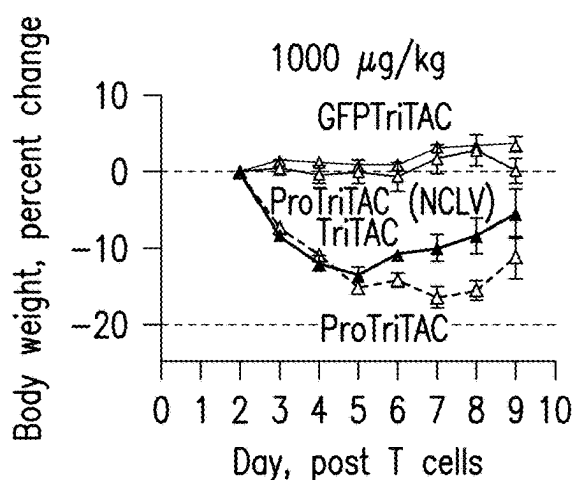


FIG. 22D

Dose Level ($\mu\text{g/kg}$)	TriTAC	ProTriTAC (L001)	ProTriTAC (NCLV)
30	✓	✓	✓
100		✓	✓
300			✓
1000			✓
Fold Protection	—	3X	$\geq 30\text{X}$

FIG. 22E

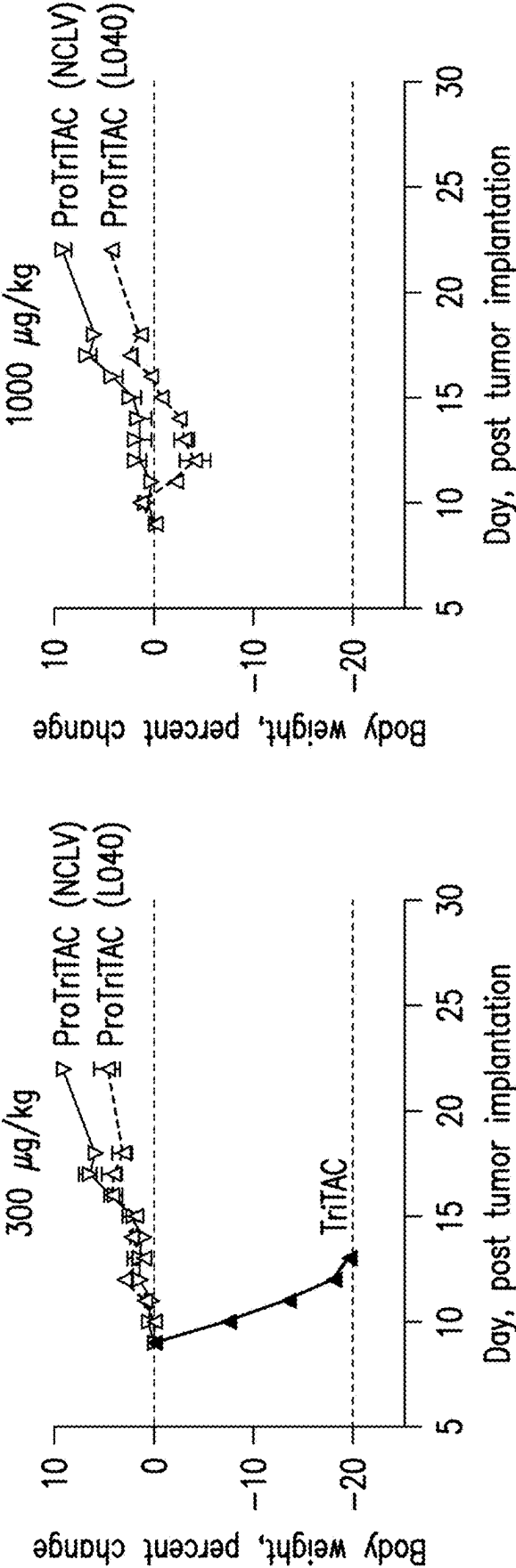


FIG. 23B

FIG. 23A

Dose Level (ug/kg)	TriTAC	ProTriTAC (1001)	ProTriTAC (L040)	ProTriTAC (NCLV)
30	✓	✓	✓	✓
100		✓	✓	✓
300			✓	✓
1000			✓	✓
Fold Protection	-	3X	≥30X	≥30X

FIG. 23C

FIG. 24A

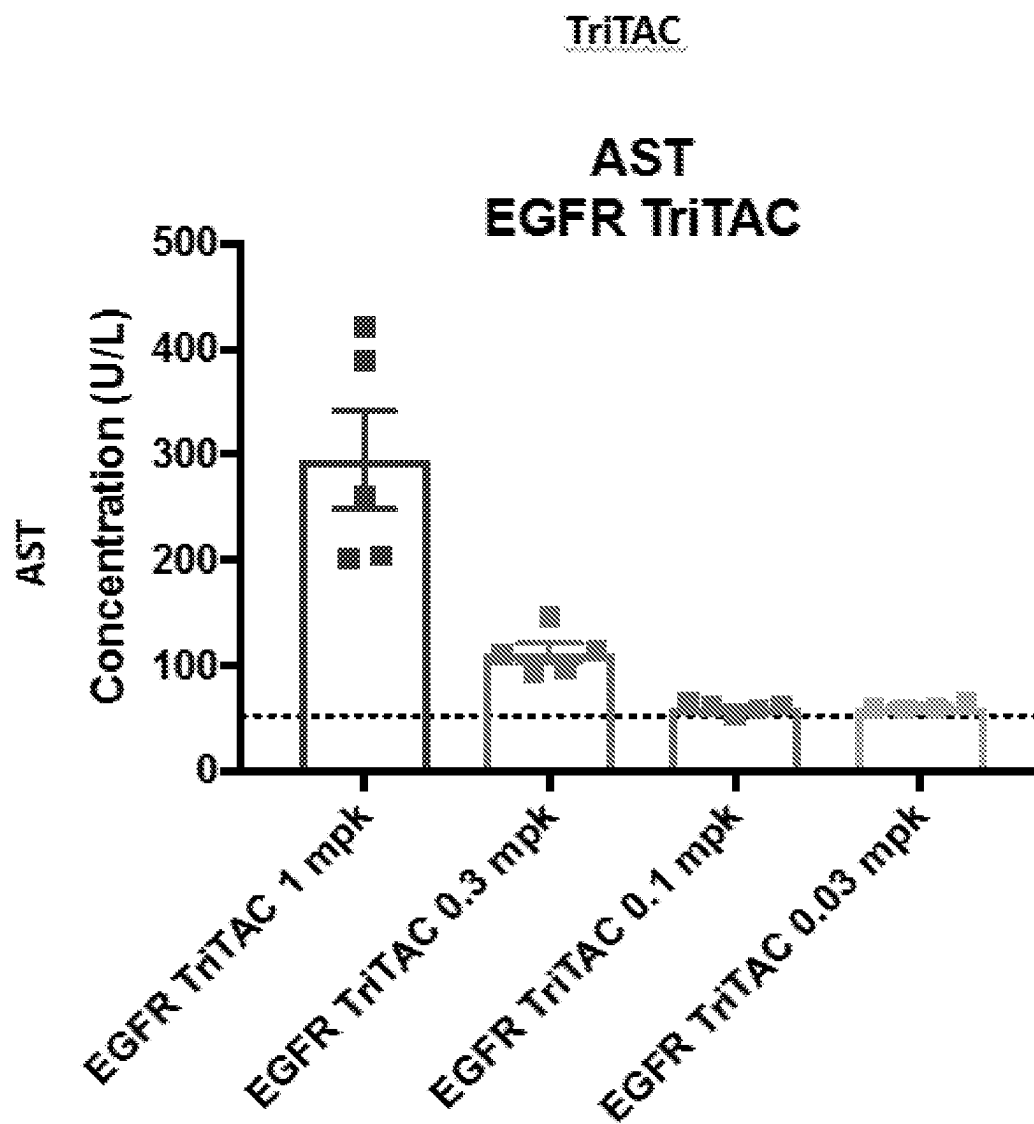


FIG. 24B

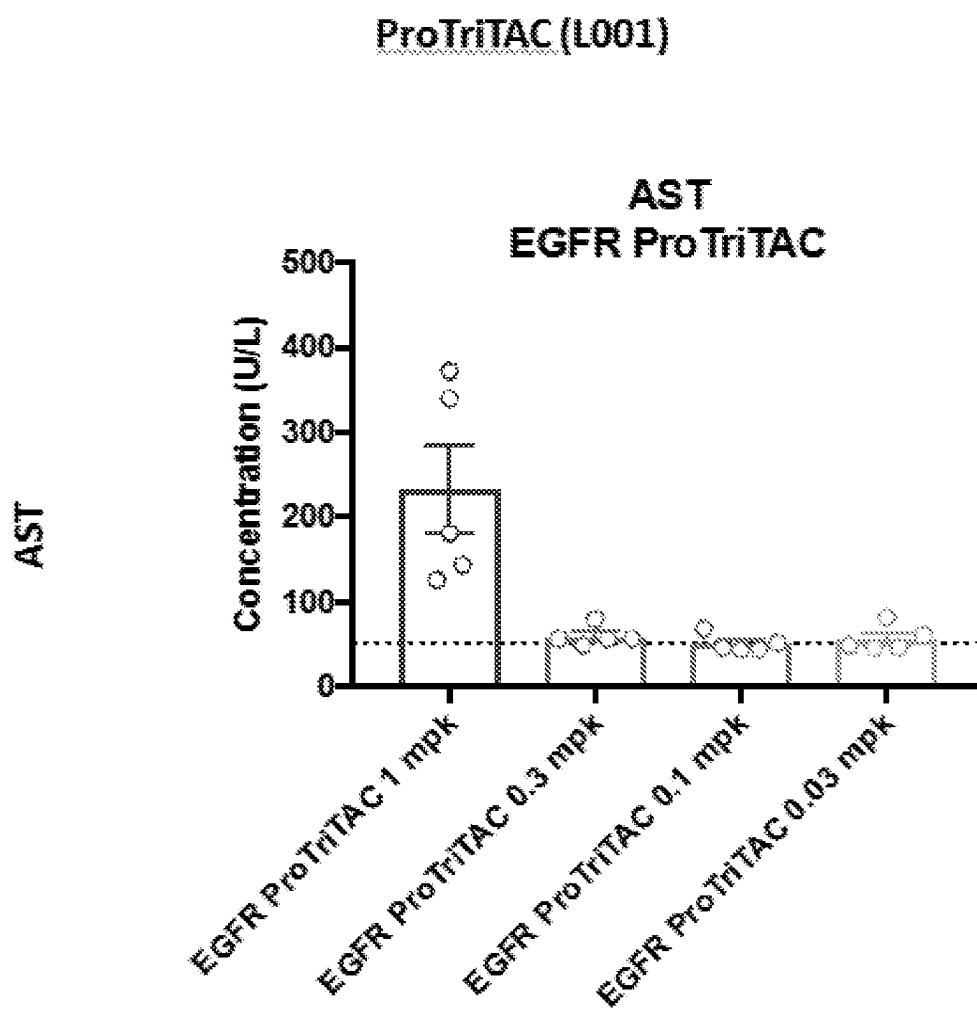


FIG. 24C

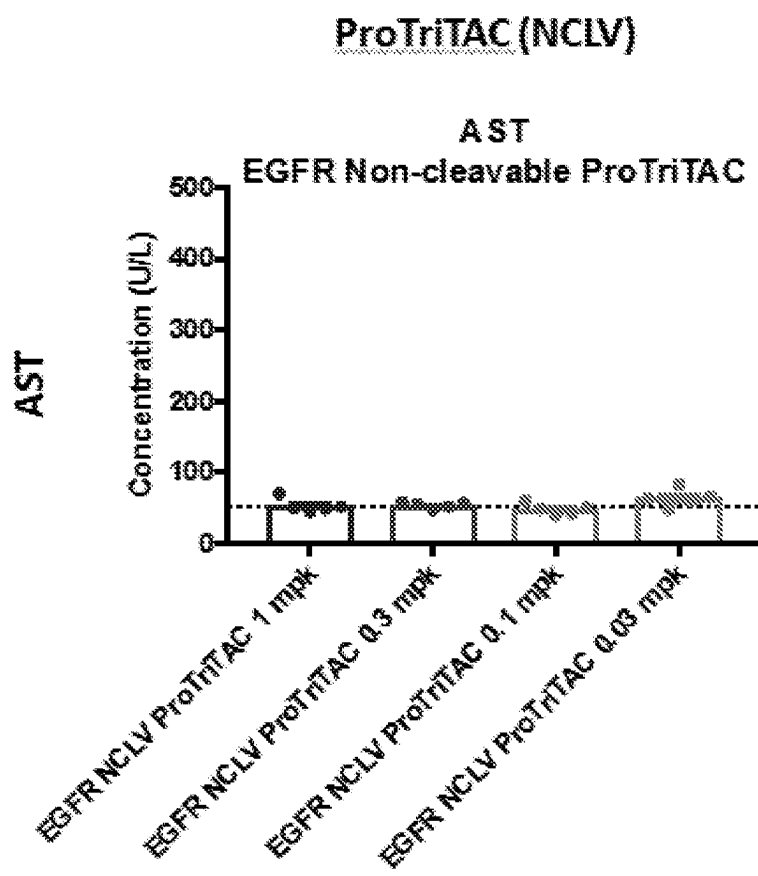


FIG. 25A

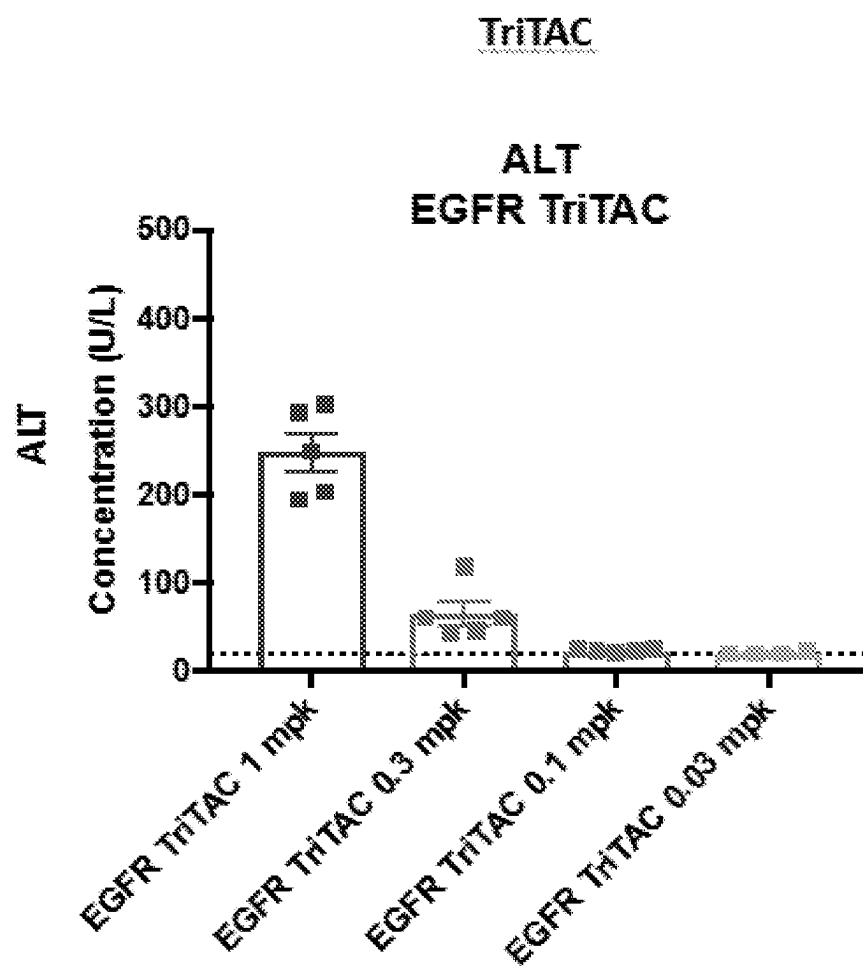


FIG. 25B

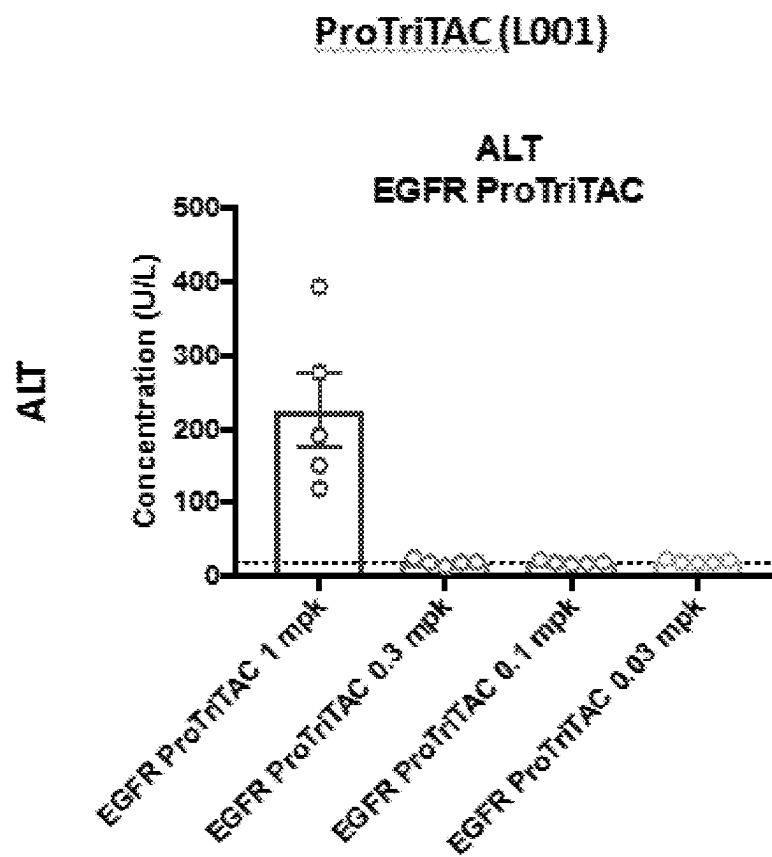


FIG. 25C

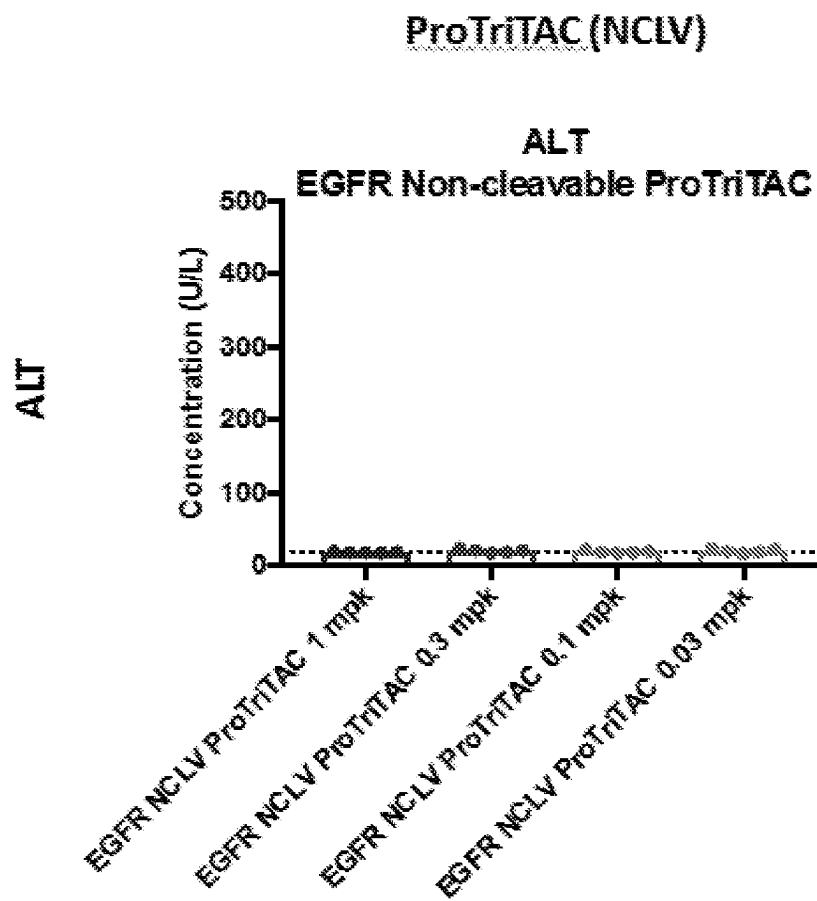


FIG. 26

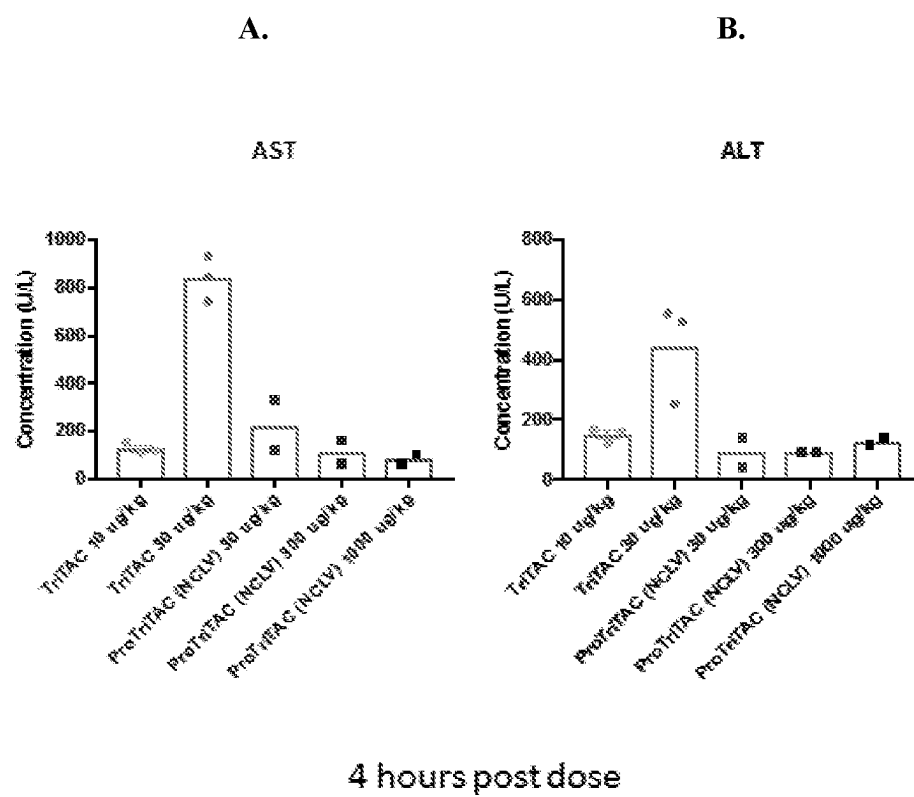


FIG. 27A

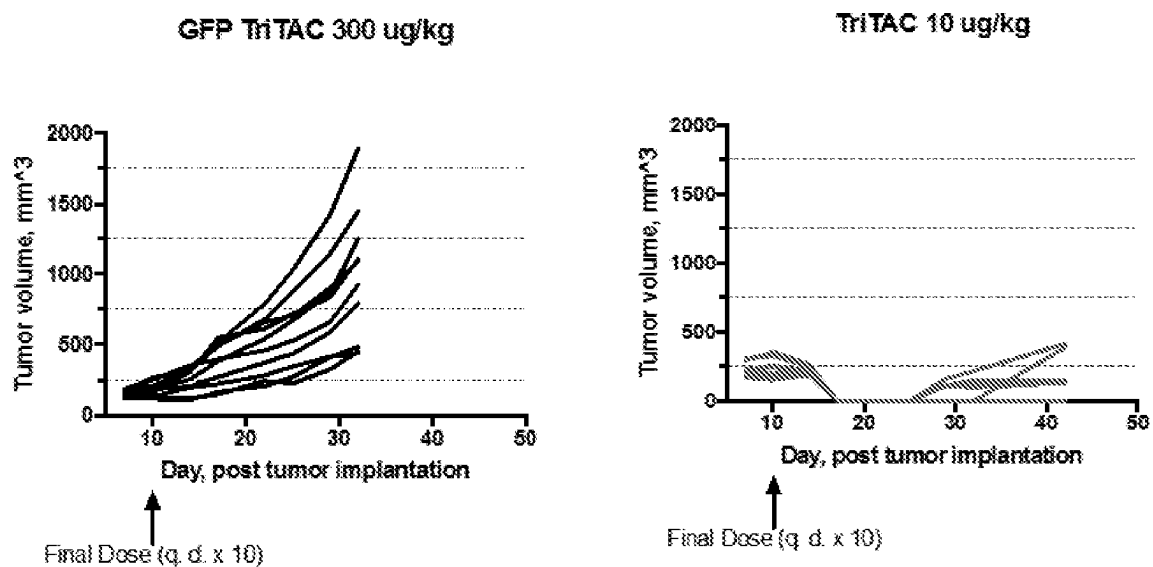


FIG. 27B

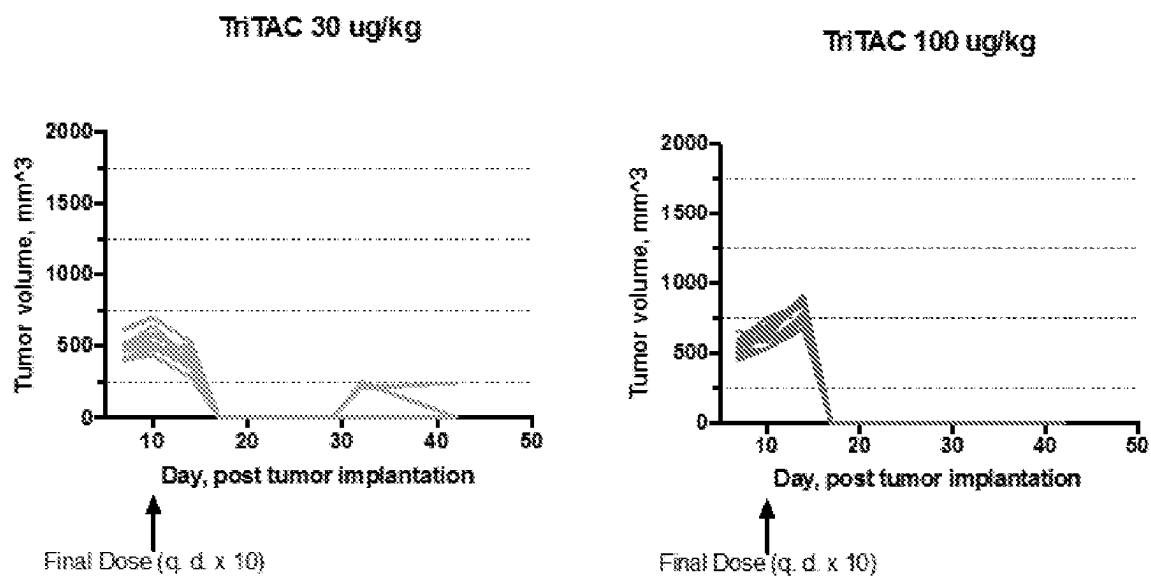


FIG. 27C

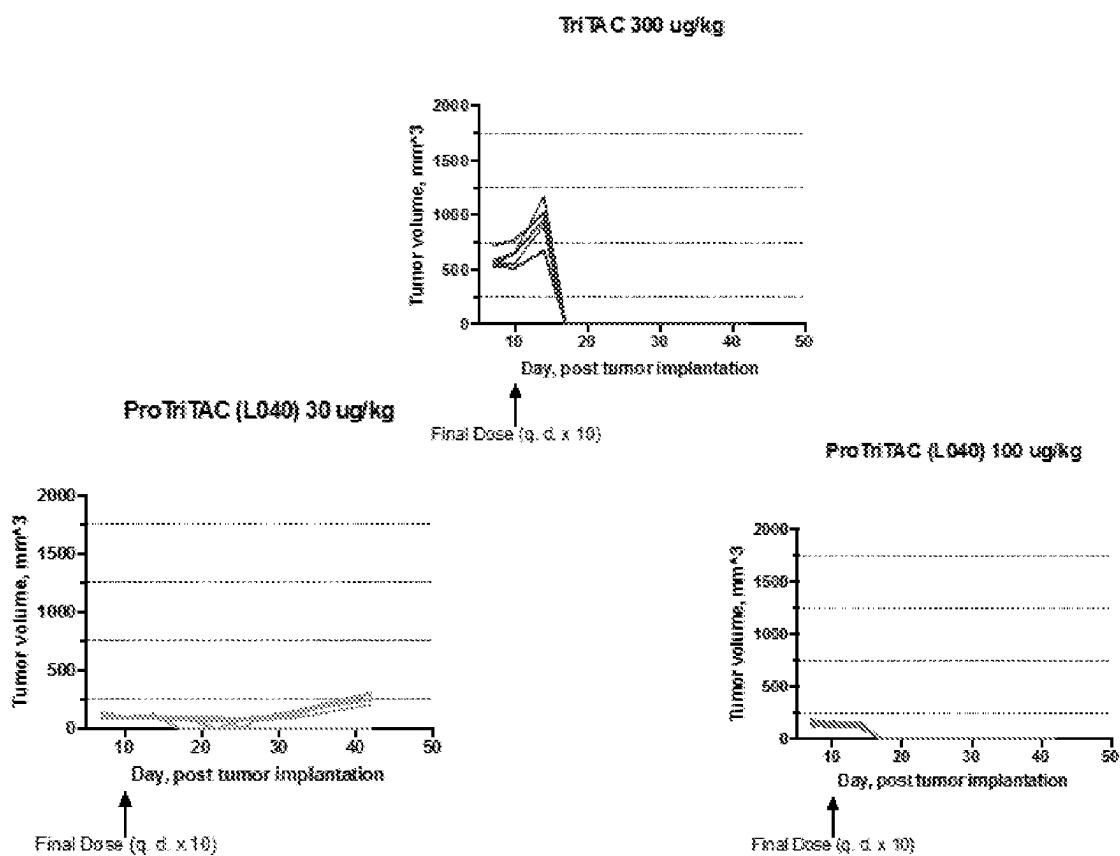


FIG. 27D

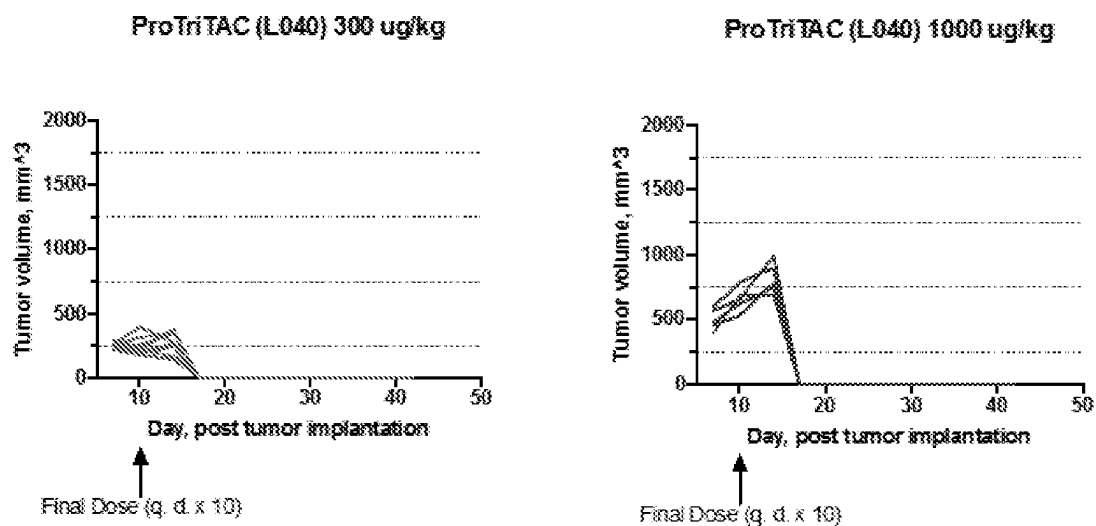
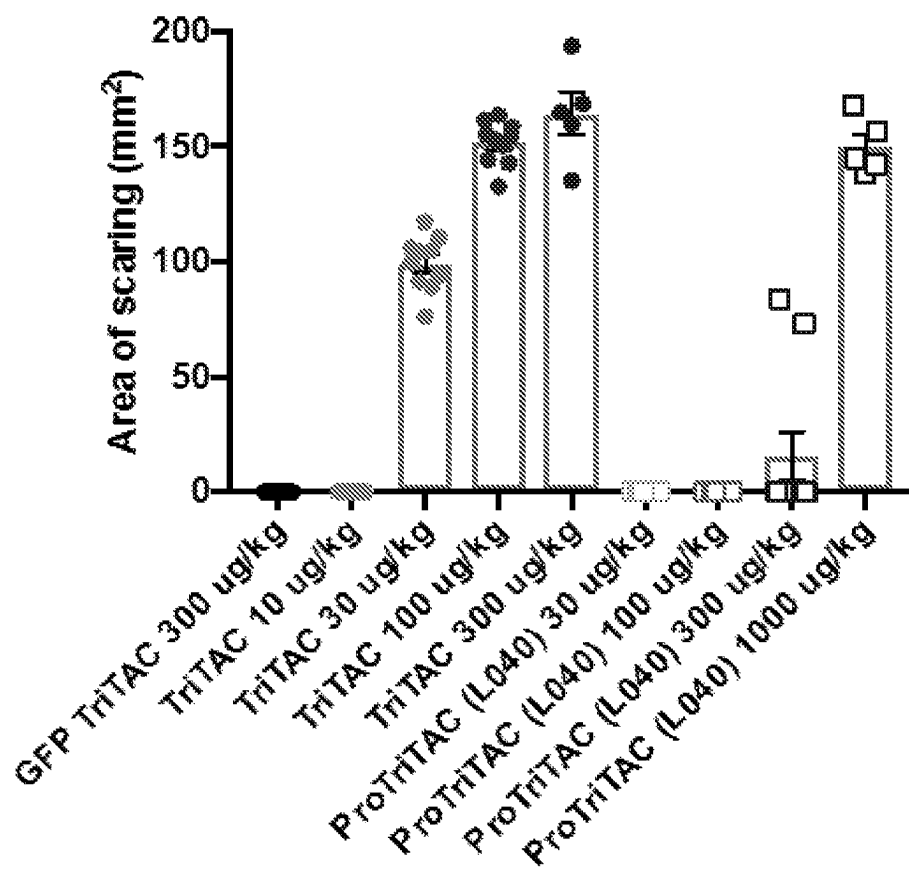
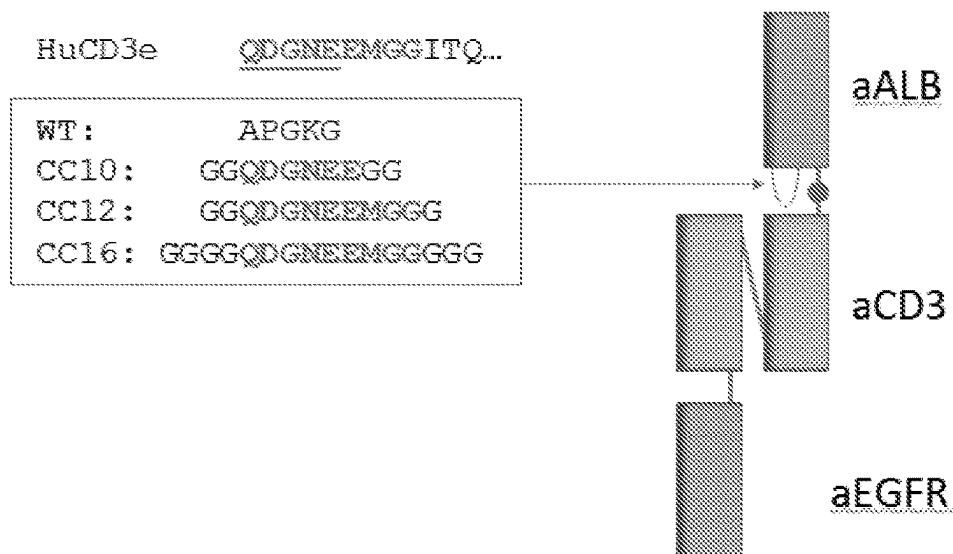


FIG. 28



Alternate Parallel Approach: Grafting aCD3 Epitope into CC' Loop to Compete for Binding to Adjacent aCD3 scFv



All ProTriTAC Candidates Were Tested with or without Pre-activation Using the Tumor-Associated Protease Matriptase

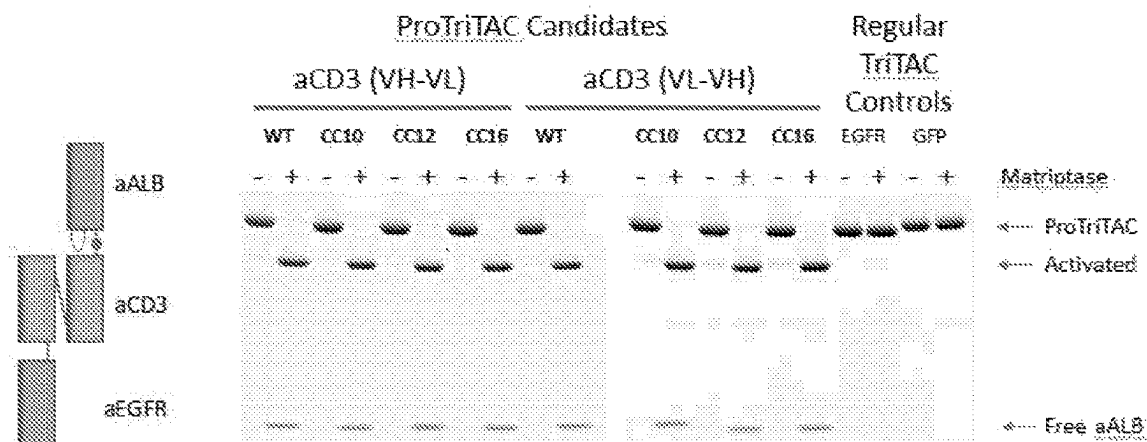


FIG. 31

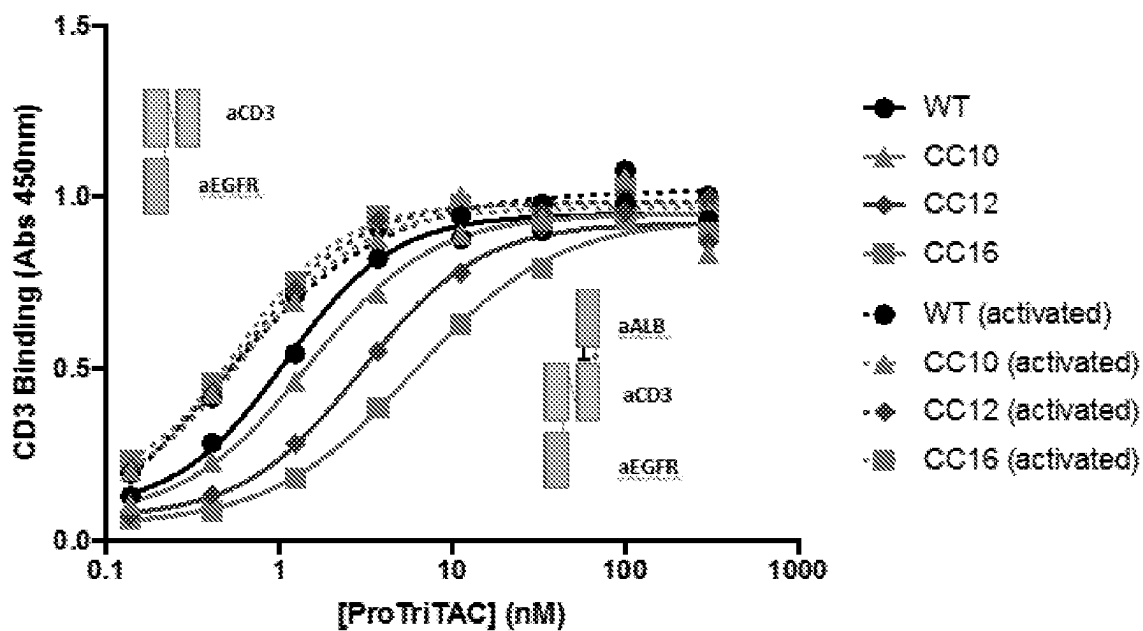
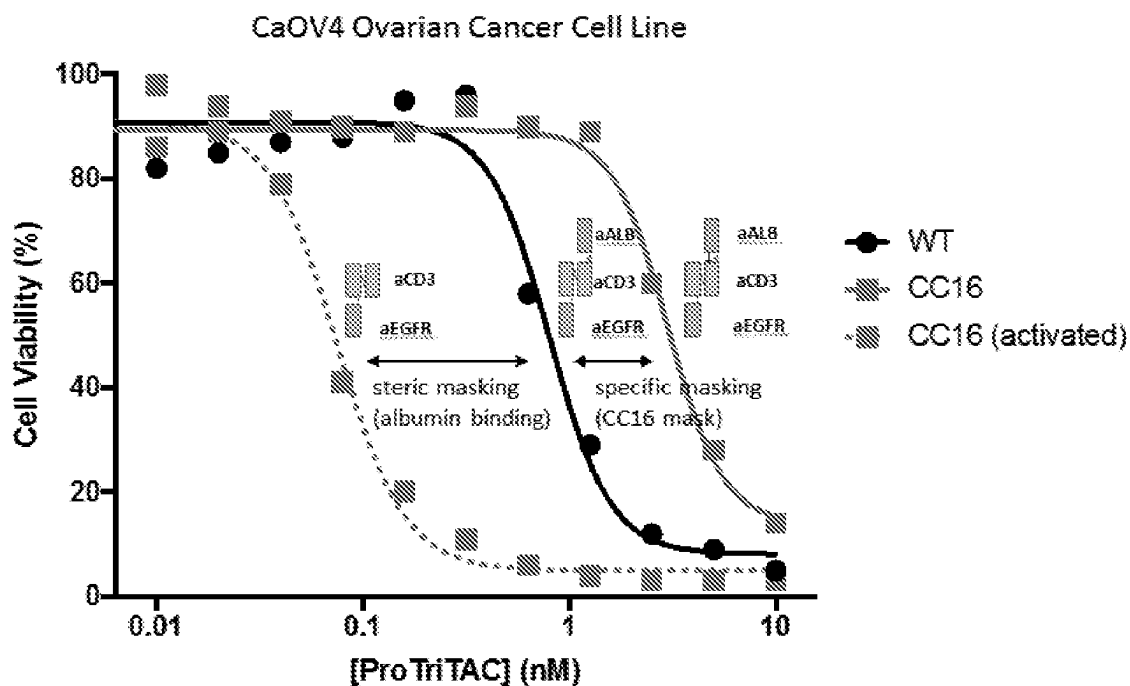


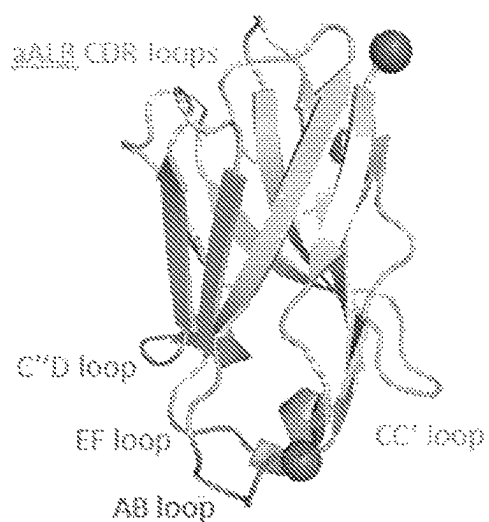
FIG. 32



total masking = steric masking + specific masking

FIG. 33

Systematically Exploring Non-CDR Loops by Library Mutagenesis to Engineer Bifunctional Inhibitory α ALB sdAb



Make Libraries by Randomizing Non-CDR Loops

	AB	CC'	C'D	EF
Library A	✓	✓	✓	✓
Library B	✓			✓
Library C		✓		

Pan against HSA

Pan against α CD3 or α Target

Screen for bifunctional binders

Soft Library Mutagenesis Identified CC' Loop as Most
Amenable to Modification

Naive Library				After panning with HSA			
AB	CC'	C''D	EF	AB	CC'	C''D	EF
LVQPGN----	APGKG----	DSVKGR----	SLRPED----	LVQPGN----	YKEWD----	DSVKGR----	SLRPED----
CFCXLCFR---	STQDM----	DSVKGR----	SLRPED----	LVQPGN----	APGKG----	DSDKGR----	SLPPQD----
LVQPGN----	KQHSN----	DSVKGR----	SLRPED----	LVQPGN----	ARR*G----	DSVKGR----	SLRPED----
IVLYEW----	*APLV----	SSMIQM----	SLRPED----	GLINSIA---	RCSEGG----	DSVKGR----	SLRPED----
CFMCYW----	DMWGANCS--	DSVKGR----	SLRPED----	LVQPGN----	APGKG----	DSVKGR----	SLRPED----
FYLAWAA---	INVGSPAY---	DSVKGR----	SLHPED----	LVQPGN----	XLHGGS----	DSVKGR----	SLRPED----
GLSNRSGC---	CSVGISRQ---	LLIFKF----	LMRI*MX----	LVQPGN----	LTNTS----	RPG*GLG----	SLRPED----
FFNSEQF*	APGKG----	FLILALTKP--	QLMI*HFI----	LVQPGN----	APGKG----	GAYFTGR----	SLRPED----
LVQPGN----	LRGGLHRV---	SKWACC----	SVS-----	LVQPGN----	KQQSHIM---	DSVKGR----	SLRPED----
VYMFNLVD---	QLETPC----	NYRWISL----	SLRPED----	LVQPGN----	LPXMKR----	DSVKGR----	SLRPED----
LVQPGN----	APGKG----	SVLFGVAYL---	SLRPED----	LVQPGN----	VGHPK----	DSVKGR----	SLRPED----
FGPYNLVS---	KLSTIWM---	MRRPCVIS---	VPSYRVA----	LVQPGN----	APGKG----	DSVKGR----	SLRPED----
*F*IC*	DSGLTLR---	LLWHWD----	VTEMNAMS----	LVQPGN----	TVTTP----	DSVKGR----	SLRPED----
GXTDGGT---	FLIXF*X---	DSVKGR----	LKRTVY----	LVQPGN----	SNSLNS----	DSVKGR----	SLRPED----
SICIVSFY---	TNTAS----	RGLIDNDYQ---	SPD*RQYSV----	LVQPGN----	TSSRT----	DSVKGR----	FPNKAIN*
LSRPXX----	CPPAIMP---	DSVKGL----	SLRPED----	LVQPGN----	SNSLNS----	DSVKGR----	SLRPED----
FALQCPXL---	IVGN*T----	LINLITE----	SLCPED----	LVQPGN----	VANLY----	DSVKGR----	SLRPED----
S*MSMLF---	NVTL*WV---	SDNILY----	SLRPED----	LVQPGN----	SEAKT----	DSVKGR----	SLRPED----
GF*IDVFIG--	ISLPVMS---	TPRTPTRP---	SLRPED----	LVQPGN----	APGEG----	DSVKGR----	SLRPED----
LFQPGN----	LYEPRAE---	FLCFDLQ----	*FLM*Y----	LVQPGN----	NMIRYPK---	DSVKGR----	SLRPED----
HY*RISAM---	YGNNCVLX---	VARAYD----	YRNHNRN----	LVQPGN----	APGKG----	DSVKGR----	SLRPED----
LVQ*GN----	A*GKG----	DSV*GR----	SLR*ED----	VAVLRGE---	RALNL----	DSVKGR----	SLRPED----
LVQPGN----	APGKG----	*VACGR----	NTTAR*C----	AAADNGV---	APGKG----	DSVKGR----	SLRPED----
MLTGFPGF---	*EIIW----	DSVKGR----	SLRPED----	LVQPGN----	TSRNG----	DSVKGR----	SLRPED----
				LVQPGN----	TNLSY----	DSVKGR----	SLRPED----
				LVQPGN----	VKKSQNS----	DSVKGR----	SLRPED----
				LVQPGN----	ADDRTH----	YSVKGR----	SLRPED----
				LVQPGN----	APGKG----	DSVKGR----	SLRPED----
				LVQPGN----	APNXV----	*SADXC----	SXNPQD----
				PGGEGX----	XSISP----	DSVKGR----	SLPPED----

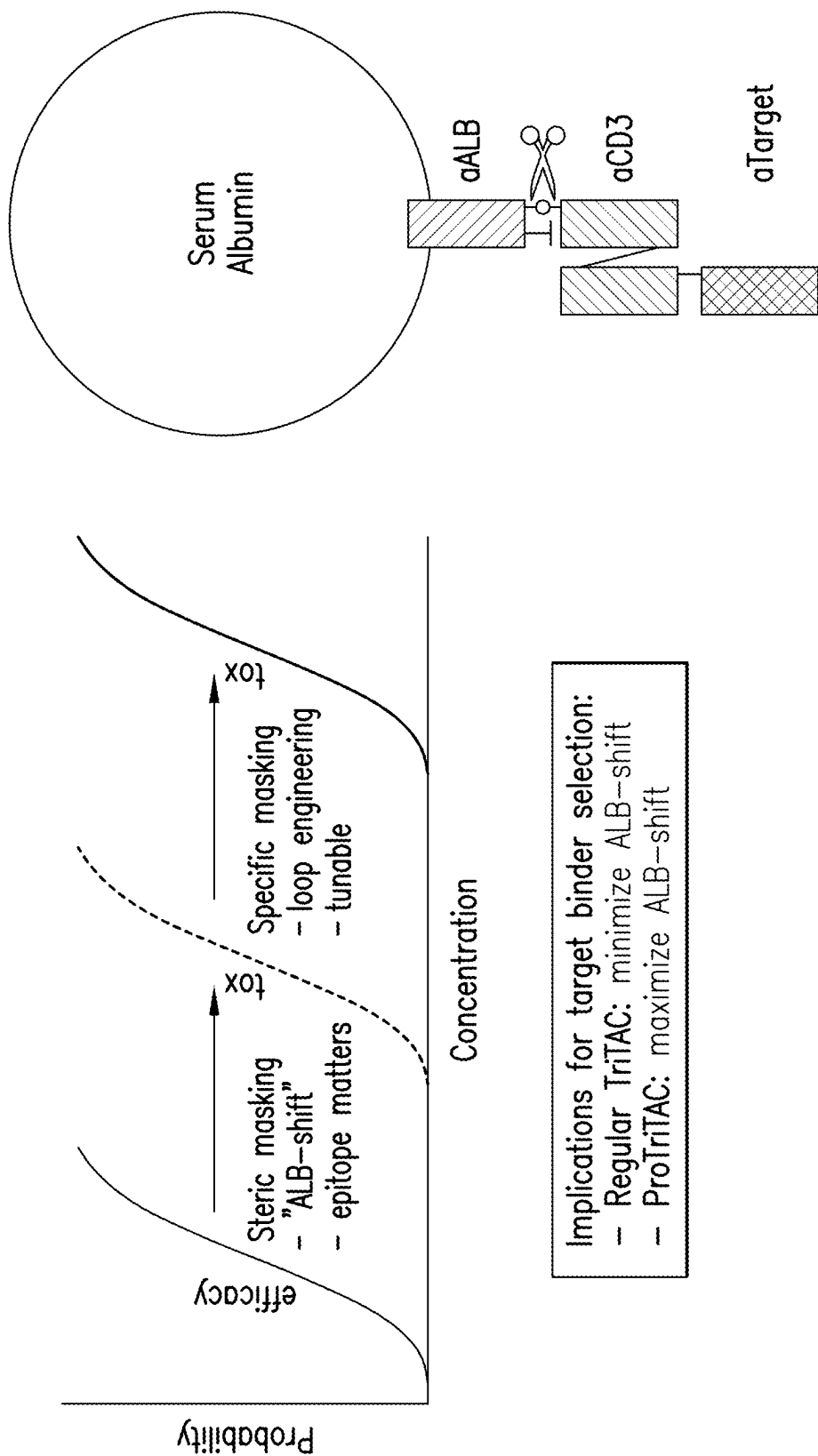
CC' loop tolerates diverse sequence
compositions and lengths

(LVQPGN) (APGKG) (DSVKGR) (SLRPED)

WT seas

FIG.34

ProTriTAC Conceptually Expands Therapeutic Window by
Both Steric Masking and Specific Masking



Implications for target binder selection:

- Regular TriTAC: minimize ALB-shift
- ProTriTAC: maximize ALB-shift

FIG.35

FIG. 36

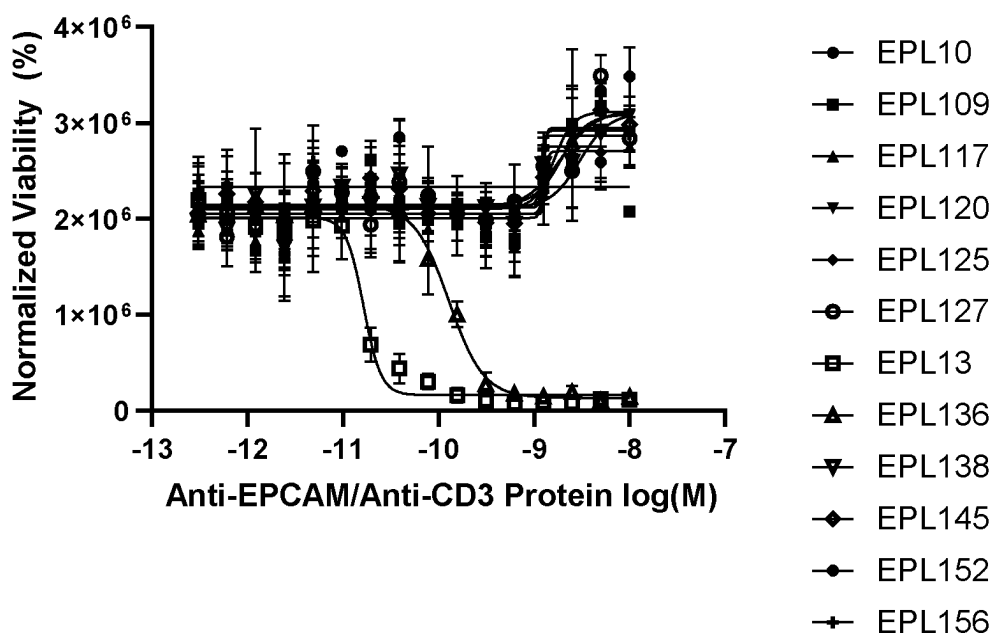


FIG. 37

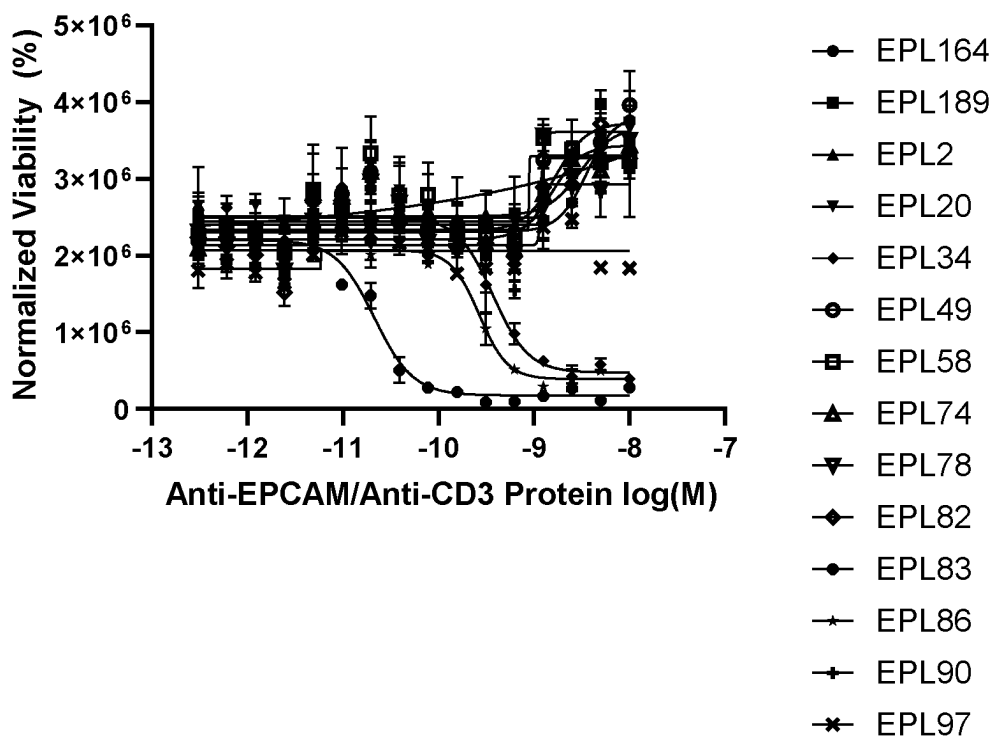


FIG. 38

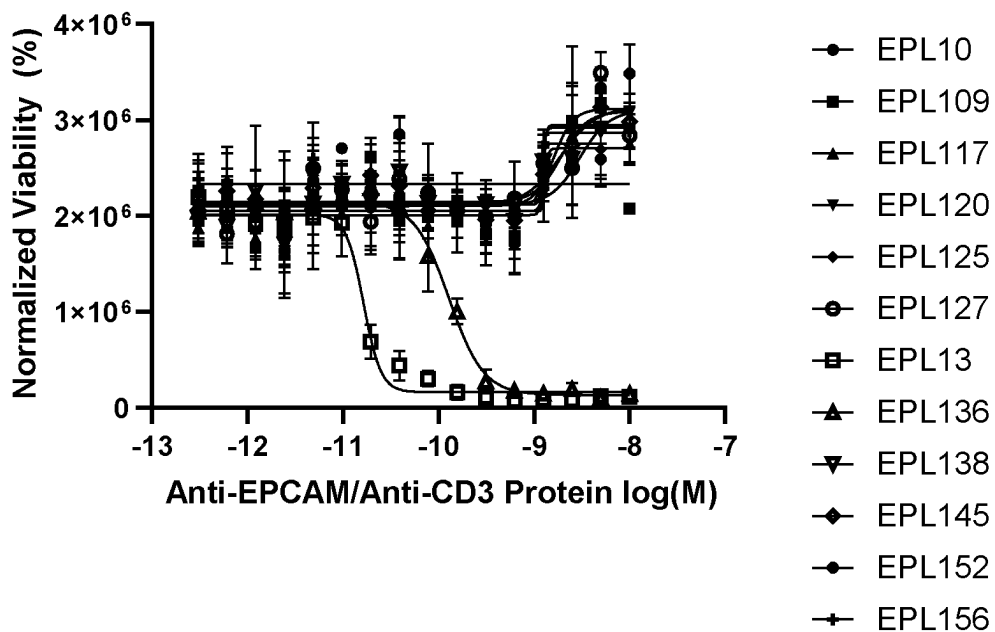


FIG. 39

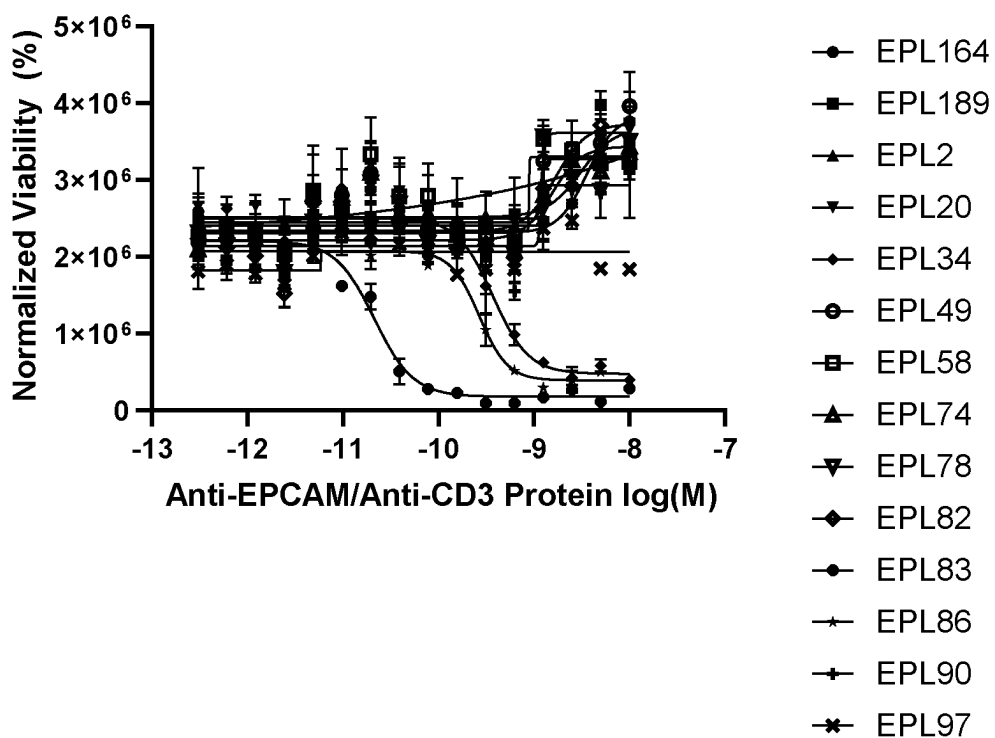


FIG. 40

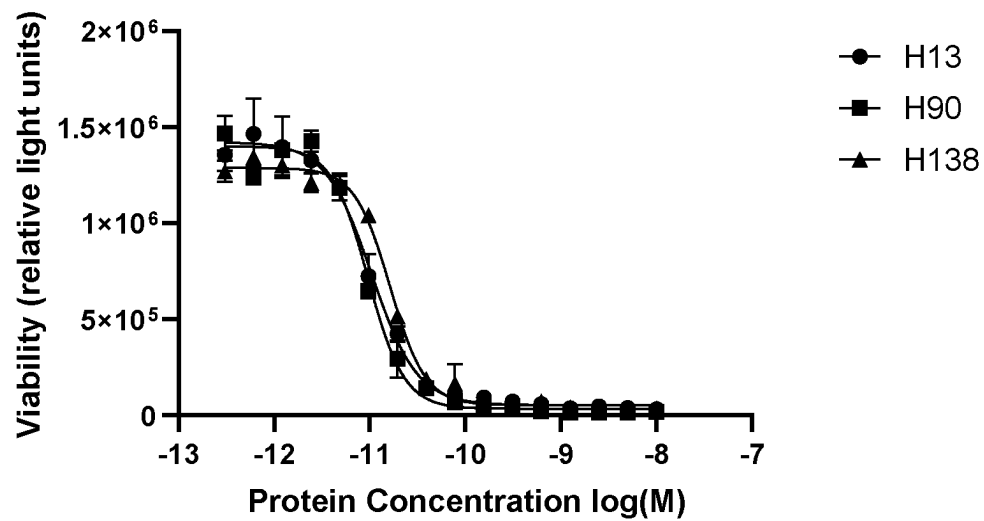
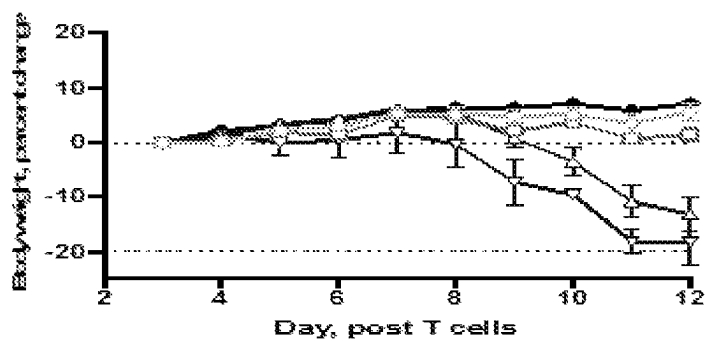


FIG. 41A

EpCAM TriTAC



Start dose (q.d. x 10)

- EpCAM TriTAC 0.03 mg/kg
- EpCAM TriTAC 0.1 mg/kg
- EpCAM TriTAC 0.3 mg/kg
- EpCAM TriTAC 1 mg/kg
- GFP TriTAC 1 mg/kg

FIG. 41B
EpCAM ProTriTAC (NCLV)

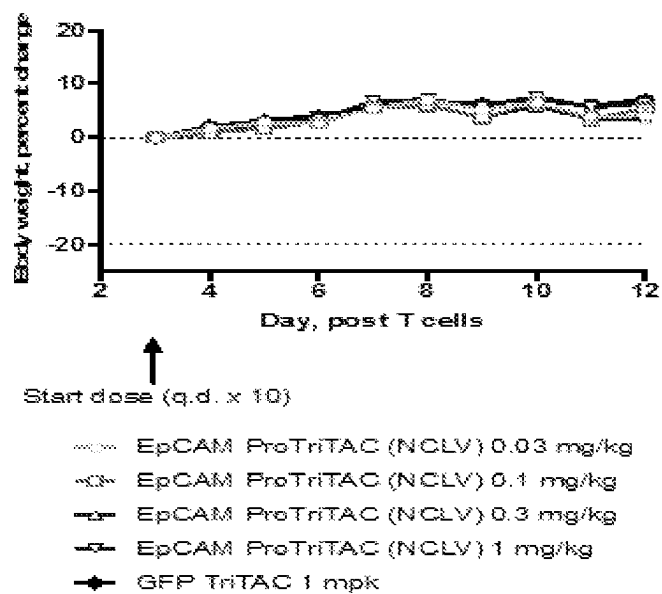
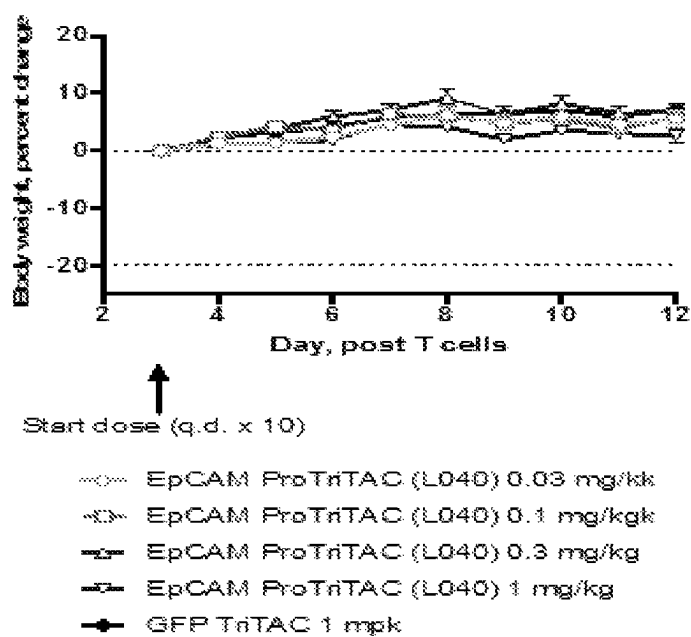


FIG. 41C
EpCAM ProTriTAC (L040)



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BINDING MOIETY FOR CONDITIONAL ACTIVATION OF IMMUNOGLOBULIN MOLECULES

CROSS-REFERENCE

This application is the U.S. National Stage Application of International Application No. PCT/US2019/032307, filed May 14, 2019, and claims the benefit of U.S. Provisional Application Nos. 62/671,344, filed May 14, 2018; 62/671,349, filed May 14, 2018; 62/756,429, filed Nov. 6, 2018; and 62/756,453 filed Nov. 6, 2018, all of which are incorporated by reference herein in their entirety.

INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference, and as if set forth in their entireties.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Dec. 6, 2023, is named 47517-729_831_SL.txt and is 1,222,504 bytes in size.

BACKGROUND OF THE INVENTION

T cell engagers transiently tether T cells to tumor cells and mediate T cell-directed tumor killing. T cell engagers, such as blinatumomab (BLINCYTO®), have demonstrated clinical activity in several hematological malignancies. Adoption of T cell engagers in solid tumors is limited by the scarcity of tumor antigens with sufficient differential expression between tumor and normal tissue. T cell engagers that are preferentially active in the tumor microenvironment may enable the safe targeting of more solid tumor antigens.

There is a need to extend the half-life of a therapeutic, diagnostic, or imaging molecule in circulation and also improve its ability to reach its target within an intended location (e.g., a tumor cell) without non-specific binding.

SUMMARY OF THE INVENTION

One embodiment provides a binding moiety comprising a non-CDR loop and a cleavable linker, wherein the moiety is capable of masking the binding of binding molecule to its target, wherein the binding molecule comprises an immunoglobulin molecule or a non-immunoglobulin molecule. In some embodiments, the moiety is a natural peptide, a synthetic peptide, an engineered scaffold, or an engineered bulk serum protein. In some embodiments, the engineered scaffold comprises a sdAb, a scFv, a Fab, a VHH, a fibronectin type III domain, immunoglobulin-like scaffold, DARPin, cystine knot peptide, lipocalin, three-helix bundle scaffold, protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold. In some embodiments, the moiety is capable of binding to a bulk serum protein. In some embodiments, the non-CDR loop is from a variable domain, a constant domain, a C1-set domain, a C2-set domain, an I-domain, or any combinations thereof. In some embodiments, the moiety further comprises complementarity deter-

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mining regions (CDRs). In some embodiments, the moiety is capable of binding to the bulk serum protein. In some embodiments, the bulk serum protein is a half-life extending protein. In some embodiments, the bulk serum protein is albumin, transferrin, IgG1, IgG2, IgG4, IgG3, IgA monomer, Factor XIII, Fibrinogen, IgE, or pentameric IgM. In some embodiments, the bulk serum protein is albumin, transferrin, Factor XIII, or Fibrinogen. In some embodiments, the CDRs within the binding moiety provide binding site specific for the bulk serum protein. In some embodiments, the binding moiety is capable of masking the binding of the target antigen binding domain (such as an immunoglobulin molecule) or a non-immunoglobulin binding molecule to its target via specific intermolecular interactions between the binding moiety and the target antigen binding domain or the non-immunoglobulin binding moiety. In some embodiments, the non-CDR loop within the binding moiety provides a binding site specific for binding of the binding moiety to the target antigen binding domain (such as an immunoglobulin molecule) or the non-immunoglobulin binding molecule.

In some embodiments, the binding moiety comprises a binding site specific for an immunoglobulin light chain. In some embodiments, the immunoglobulin light chain is an Igx free light chain. In some embodiments, the CDRs provide the binding site specific for the bulk serum protein or the immunoglobulin light chain. In some embodiments, the immunoglobulin molecule is a target antigen binding domain. In some embodiments, the moiety is bound to the target antigen binding domain. In some embodiments, the moiety is covalently linked to the target antigen binding domain. In some embodiments, the moiety is capable of masking the binding of the target antigen binding domain to its target via specific intermolecular interactions between the binding moiety and the target antigen binding domain. In some embodiments, the non-CDR loop provides a binding site specific for binding of the moiety to the target antigen binding domain. In some embodiments, upon cleavage of the cleavable linker, the binding moiety is separated from the target antigen binding domain and the target antigen binding domain binds to its target. In some embodiments, the target antigen domain binds to a tumor antigen. In some embodiments, the tumor antigen comprises EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, and CEA. 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, ROR1, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1. In some embodiments, the target antigen domain binds to an immune checkpoint protein. In some embodiments, the immune checkpoint protein is CD27, CD137, 2B4, TIGIT, CD155, ICOS, HVEM, CD40L, LIGHT, OX40, DNAM-1, PD-L1, PD1, PD-L2, CTLA-4, CD8, CD40, CEACAM1, CD48, CD70, A2AR, CD39, CD73, B7-H3, B7-H4, BTLA, IDO1, IDO2, TDO, KIR, LAG-3, TIM-3, or VISTA. In some embodiments, the target antigen binding domain binds to a T-cell. In some embodiments, the target antigen binding domain binds to CD3. In some embodiments, the cleavable linker comprises a cleavage site. In some embodiments, the cleavage site is recognized by a protease. In some embodiments, the protease cleavage site is recognized by a serine protease, a cysteine protease, an

aspartate protease, a threonine protease, a glutamic acid protease, a metalloproteinase, a gelatinase, or a asparagine peptide lyase. In some embodiments, the protease cleavage site is recognized by a Cathepsin B, a Cathepsin C, a Cathepsin D, a Cathepsin E, a Cathepsin K, a Cathepsin L, a kallikrein, a hK1, a hK10, a hK15, a plasmin, a collagenase, a Type IV collagenase, a stromelysin, a Factor Xa, a chymotrypsin-like protease, a trypsin-like protease, a elastase-like protease, a subtilisin-like protease, an actinidin, a bromelain, a calpain, a caspase, a caspase-3, a MircP, a papain, a HIV-1 protease, a HSV protease, a CMV protease, a chymosin, a renin, a pepsin, a matriptase, a legumain, a plasmepsin, a nepenthesin, a metalloexopeptidase, a metalloendopeptidase, a matrix metalloprotease (MMP), a MMP1, a MMP2, a MMP3, a MMP7, a MMP8, a MMP9, a MMP10, a MMP11, a MMP12, a MMP13, a MMP14, an ADAM9, an ADAM10, an ADAM12, an urokinase plasminogen activator (uPA), an enterokinase, a prostate-specific target (PSA, hK3), an interleukin-1 β converting enzyme, a thrombin, a FAP (FAP- α), a dipeptidyl peptidase, a type II transmembrane serine protease (TTSP), a neutrophil elastase, a cathepsin G, a proteinase 3, a neutrophil serine protease 4, a mast cell chymase, and a mast cell tryptase.

One embodiment provides a conditionally active binding protein comprising a binding moiety (M) which comprises a non-CDR loop, a cleavable linker (L), a first target antigen binding domain (T1), and a second target antigen binding domain (T2), wherein the first target antigen binding domain (T1) comprises an immunoglobulin molecule, wherein the non-CDR loop is capable of binding to the first target antigen binding domain, and wherein the binding moiety is capable of masking the binding of the first target antigen binding domain to its target. In some embodiments, the binding moiety is capable of binding to a half-life extending protein. In some embodiments, the binding moiety is a natural peptide, a synthetic peptide, an engineered scaffold, or an engineered serum bulk protein. In some embodiments, the engineered scaffold comprises a sdAb, a scFv, a Fab, a VHH, a fibronectin type III domain, immunoglobulin-like scaffold, DARPin, cystine knot peptide, lipocalin, three-helix bundle scaffold, protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold. In some embodiments, the non-CDR loop is from a variable domain, a constant domain, a C1-set domain, a C2-set domain, an I-domain, or any combinations thereof. In some embodiments, the binding moiety further comprises complementarity determining regions (CDRs). In some embodiments, the binding moiety comprises a binding site specific for a bulk serum protein. In some embodiments, the bulk serum protein is albumin, transferrin, IgG1, IgG2, IgG4, IgG3, IgA monomer, Factor XIII, Fibrinogen, IgE, or pentameric IgM. In some embodiments, the binding moiety further comprises a binding site specific for an immunoglobulin light chain. In some embodiments, the immunoglobulin light chain is an Igk free light chain. In some embodiments, the CDRs provide the binding site specific for the bulk serum protein or the immunoglobulin light chain, or any combinations thereof. In some embodiments, the binding moiety is capable of masking the binding of the first target antigen binding domain to its target via specific intermolecular interactions between the binding moiety and the first target antigen binding domain. In some embodiments, the non-CDR loop provides a binding site specific for binding of the binding moiety to the first target antigen binding domain. In some embodiments, the first or the second target antigen binding domain binds to a tumor antigen). In some embodi-

ments, the tumor antigen comprises at least one of. EpCAM (exemplary protein sequence comprises UniProtKB ID No. P16422), EGFR (exemplary protein sequence comprises UniProtKB ID No. P00533), HER-2(exemplary protein sequence comprises UniProtKB ID No. P04626), HER-3 (exemplary protein sequence comprises UniProtKB ID No. P21860), c-Met (exemplary protein sequence comprises UniProtKB ID No. P08581), FoIR (exemplary protein sequence comprises UniProtKB ID No. P15238), PSMA (exemplary protein sequence comprises UniProtKB ID No. Q04609), CD38 (exemplary protein sequence comprises UniProtKB ID No. P28907), BCMA (exemplary protein sequence comprises UniProtKB ID No. Q02223), and CEA (exemplary protein sequence comprises UniProtKB ID No. P06731, 5T4 (exemplary protein sequence comprises UniProtKB ID No. Q13641), AFP (exemplary protein sequence comprises UniProtKB ID No. P02771), B7-H3 (exemplary protein sequence comprises UniProtKB ID No. Q5ZPR3), CDH-6 (exemplary protein sequence comprises UniProtKB ID No. P97326), CAIX (exemplary protein sequence comprises UniProtKB ID No. Q16790), CD117 (exemplary protein sequence comprises UniProtKB ID No. P10721), CD123 (exemplary protein sequence comprises UniProtKB ID No. P26951), CD138 (exemplary protein sequence comprises UniProtKB ID No. P18827), CD166 (exemplary protein sequence comprises UniProtKB ID No. Q13740), CD19 (exemplary protein sequence comprises UniProtKB ID No. P15931), CD20 (exemplary protein sequence comprises UniProtKB ID No. P11836), CD205 (exemplary protein sequence comprises UniProtKB ID No. 060449), CD22 (exemplary protein sequence comprises UniProtKB ID No. P20273), CD30 (exemplary protein sequence comprises UniProtKB ID No. P28908), CD33 (exemplary protein sequence comprises UniProtKB ID No. P20138), CD352 (exemplary protein sequence comprises UniProtKB ID No. Q96DU3), CD37 (exemplary protein sequence comprises UniProtKB ID No. P11049), CD44 (exemplary protein sequence comprises UniProtKB ID No. P16070), CD52 (exemplary protein sequence comprises UniProtKB ID No. P13588), CD56 (exemplary protein sequence comprises UniProtKB ID No. P13591), CD70 (exemplary protein sequence comprises UniProtKB ID No. P32970), CD71 (exemplary protein sequence comprises UniProtKB ID No. P02786), CD74 (exemplary protein sequence comprises UniProtKB ID No. P04233), CD79b (exemplary protein sequence comprises UniProtKB ID No. P40259), DLL3 (exemplary protein sequence comprises UniProtKB ID No. Q9NYJ7), EphA2 (exemplary protein sequence comprises UniProtKB ID No. P29317), FAP (exemplary protein sequence comprises UniProtKB ID No. Q12884), FGFR2 (exemplary protein sequence comprises UniProtKB ID No. P21802), FGFR3 (exemplary protein sequence comprises UniProtKB ID No. P22607), GPC3 (exemplary protein sequence comprises UniProtKB ID No. P51654), gpA33 (exemplary protein sequence comprises UniProtKB ID No. Q99795), FLT-3 (exemplary protein sequence comprises UniProtKB ID No. P36888), gpNMB (exemplary protein sequence comprises UniProtKB ID No. Q14956), HPV-16 E6 (exemplary protein sequence comprises UniProtKB ID No. P03126), HPV-16 E7 (exemplary protein sequence comprises UniProtKB ID No. P03129), ITGA2 (exemplary protein sequence comprises UniProtKB ID No. P17301), ITGA3 (exemplary protein sequence comprises UniProtKB ID No. P26006), SLC39A6 (exemplary protein sequence comprises UniProtKB ID No. Q13433), MAGE (exemplary protein sequence comprises UniProtKB ID No. Q9HC15), mesothelin (exemplary protein sequence comprises UniPro-

tkB ID No. Q13421), Muc1 (exemplary protein sequence comprises UniProtkB ID No. P15941), Muc16 (exemplary protein sequence comprises UniProtkB ID No. Q8WX17), NaPi2b (exemplary protein sequence comprises UniProtkB ID No. 095436), Nectin-4 (exemplary protein sequence comprises UniProtkB ID No. Q96918), CDH-3 (exemplary protein sequence comprises UniProtkB ID No. Q8WX17), CDH-17 (exemplary protein sequence comprises UniProtkB ID No. E5RJT3), EPHB2 (exemplary protein sequence comprises UniProtkB ID No. P29323), ITGAV (exemplary protein sequence comprises UniProtkB ID No. P06756), ITGB6 (exemplary protein sequence comprises UniProtkB ID No. P18564), NY-ESO-1 (exemplary protein sequence comprises UniProtkB ID No. P78358), PRLR (exemplary protein sequence comprises UniProtkB ID No. P16471), PSCA (exemplary protein sequence comprises UniProtkB ID No. 043653), PTK7 (exemplary protein sequence comprises UniProtkB ID No. Q13308), ROR1 (exemplary protein sequence comprises UniProtkB ID No. Q01973), SLC44A4 (exemplary protein sequence comprises UniProtkB ID No. Q53GD3), SLITRK5 (exemplary protein sequence comprises UniProtkB ID No. Q8IW52), SLITRK6 (exemplary protein sequence comprises UniProtkB ID No. Q9HY7), STEAP1 (exemplary protein sequence comprises UniProtkB ID No. Q9UHE8), TIM1 (exemplary protein sequence comprises UniProtkB ID No. Q96D42), Trop2 (exemplary protein sequence comprises UniProtkB ID No. P09758), or WT1 (exemplary protein sequence comprises UniProtkB ID No. P19544), or any combinations thereof. In some embodiments, the first or the second target antigen binding domain binds to an immune checkpoint protein. In some embodiments, the immune checkpoint protein is at least one of: CD27 (exemplary protein sequence comprises UniProtkB ID No. P26842), CD137 (exemplary protein sequence comprises UniProtkB ID No. Q07011), 2B4 (exemplary protein sequence comprises UniProtkB ID No. Q9bZW8), TIGIT (exemplary protein sequence comprises UniProtkB ID No. Q495A1), CD155 (exemplary protein sequence comprises UniProtkB ID No. P15151), ICOS (exemplary protein sequence comprises UniProtkB ID No. Q9Y6W8), HVEM (exemplary protein sequence comprises UniProtkB ID No. 043557), CD40L (exemplary protein sequence comprises UniProtkB ID No. P29965), LIGHT (exemplary protein sequence comprises UniProtkB ID No. 043557), OX40 (exemplary protein sequence comprises UniProtkB ID No.), DNAM-1 (exemplary protein sequence comprises UniProtkB ID No. Q15762), PD-L1 (exemplary protein sequence comprises UniProtkB ID No. Q9ZQ7), PD1 (exemplary protein sequence comprises UniProtkB ID No. Q15116), PD-L2 (exemplary protein sequence comprises UniProtkB ID No. Q9BQ51), CTLA-4 (exemplary protein sequence comprises UniProtkB ID No. P16410), CD8 (exemplary protein sequence comprises UniProtkB ID No. P10966, P01732), CD40 (exemplary protein sequence comprises UniProtkB ID No. P25942), CEACAM1 (exemplary protein sequence comprises UniProtkB ID No. P13688), CD48 (exemplary protein sequence comprises UniProtkB ID No. P09326), CD70 (exemplary protein sequence comprises UniProtkB ID No. P32970), AA2AR (exemplary protein sequence comprises UniProtkB ID No. P29274), CD39 (exemplary protein sequence comprises UniProtkB ID No. P49961), CD73 (exemplary protein sequence comprises UniProtkB ID No. P21589), B7-H3 (exemplary protein sequence comprises UniProtkB ID No. Q5ZPR3), B7-H4 (exemplary protein sequence comprises UniProtkB ID No. Q7Z7D3), BTLA (exemplary protein sequence comprises UniProtkB ID No. Q76A9), IDO1 (ex-

emplary protein sequence comprises UniProtkB ID No. P14902), IDO2 (exemplary protein sequence comprises UniProtkB ID No. Q6ZQW0), TDO (exemplary protein sequence comprises UniProtkB ID No. P48755), KIR (exemplary protein sequence comprises UniProtkB ID No. Q99706), LAG-3 (exemplary protein sequence comprises UniProtkB ID No. P18627), TIM-3 (also known as HAVCR2, exemplary protein sequence comprises UniProtkB ID No. Q8TDQ0), or VISTA (exemplary protein sequence comprises UniProtkB ID No. Q9D659). In some embodiments, the first or the second target antigen binding domain binds to an immune cell.

In some embodiments, the first or the second target antigen binding domain binds to a T-cell. In some embodiments, the first or the second target antigen binding domain binds to CD3. In some embodiments, the binding moiety (M), the cleavable linker (L), the first target antigen binding domain (T1), and the second target antigen binding domain (T2) are in one of the following configurations: M:L:T1:T2, and T2:T1:L:M. In some embodiments, the binding moiety comprises an albumin binding domain (anti-Alb), the first target antigen binding domain (T1) comprises a CD3 binding domain (e.g., an anti-CD3 scFV), and a ProTriTAC molecule has the following orientation: anti-Alb: anti-CD3: T2. In some embodiments, the binding moiety comprises an albumin binding domain (anti-Alb), the second target antigen binding domain (T2) comprises a CD3 binding domain (e.g., an anti-CD3 scFV), and a ProTriTAC molecule has the following orientation: anti-Alb: T1: anti-CD3. The T1 domain, in certain examples, is a tumor antigen binding domain, such as, but not limited to, an anti-EGFR domain, an anti-MSLN domain, an anti-BCMA domain, an anti-EpCAM domain, an anti-PSMA domain, or an anti-DLL3 domain.

In some embodiments, the cleavable linker comprises a cleavage site. In some embodiments, the cleavage site is recognized by a protease. In some embodiments, the protease cleavage site is recognized by a serine protease, a cysteine protease, an aspartate protease, a threonine protease, a glutamic acid protease, a metalloproteinase, a gelatinase, or an asparagine peptide lyase. In some embodiments, the protease cleavage site is recognized by a Cathepsin B, a Cathepsin C, a Cathepsin D, a Cathepsin E, a Cathepsin K, a Cathepsin L, a kallikrein, a hK1, a hK10, a hK15, a plasmin, a collagenase, a Type IV collagenase, a stromelysin, a Factor Xa, a chymotrypsin-like protease, a trypsin-like protease, an elastase-like protease, a subtilisin-like protease, an actinidain, a bromelain, a calpain, a caspase, a caspase-3, a Mir1-CP, a papain, a HIV-1 protease, a HSV protease, a CMV protease, a chymosin, a renin, a pepsin, a matriptase, a legumain, a plasmepsin, a nepenthesin, a metalloexopeptidase, a metalloendopeptidase, a matrix metalloprotease (MMP), a MMP1, a MMP2, a MMP3, a MMP7, a MMP8, a MMP9, a MMP10, a MMP11, a MMP12, a MMP13, a MMP14, an ADAM9, an ADAM10, an ADAM12, an urokinase plasminogen activator (uPA), an enterokinase, a prostate-specific target (PSA, hK3), an interleukin-1 β converting enzyme, a thrombin, a FAP (FAP- α), a dipeptidyl peptidase, a type II transmembrane serine protease (TTSP), a neutrophil elastase, a cathepsin G, a proteinase 3, a neutrophil serine protease 4, a mast cell chymase, and a mast cell tryptase. In some embodiments, the conditionally active protein further comprises a half-life extension domain bound to the binding moiety, wherein the half-life extension domain provides the binding protein with a safety switch, and wherein upon cleavage of the linker the binding protein is activated by separation of the binding moiety and the

half-life extension domain from the first target antigen binding domain, and the binding protein is thereby separated from the safety switch. In some embodiments, the cleavage of the linker is in a tumor microenvironment.

One embodiment provides a conditionally active binding protein, comprising a binding moiety linked to a target antigen binding domain via a non-CDR loop within the binding moiety, wherein the binding moiety is further linked to a half-life extension domain and comprises a cleavable linker, wherein the target antigen binding domain comprises an immunoglobulin molecule, wherein the binding protein has an extended half-life prior to its activation by cleavage of the linker, and wherein upon activation the binding moiety and the half-life extension domain are separated from the target antigen binding domain, and wherein the binding protein, in its activated state, does not have an extended half-life. In some embodiments, the cleavage of the linker is in a tumor microenvironment.

In some embodiments, the non-CDR loop comprises a CC' loop of at least one of: a camelid VHH domain, a human VH domain, a humanized VH domain, or a single domain antibody. In some embodiments, the binding moiety comprises a binding site specific for a CD3e domain, and wherein the binding site for the CD3e domain comprises at least one of the following motifs: QDGNE (SEQ ID NO: 921), QDGNEE (SEQ ID NO: 801), DGNE (SEQ ID NO: 922), and DGNEE (SEQ ID NO: 923).

BRIEF DESCRIPTION OF THE DRAWINGS

The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which.

FIG. 1 illustrates a variable domain of an exemplary immunoglobulin domain, comprising complementarity determining regions (CDR1, CDR2, and CDR3), and non-CDR loops connecting the beta strand (AB, CC', C" D, EF, and DE).

FIGS. 2A-2B provide exemplary arrangements of various domains of a conditionally active binding protein of this disclosure. FIG. 2A: Version 1. FIG. 2B: Version 2.

FIG. 3 shows an exemplary conditionally active target binding protein of this disclosure.

FIGS. 4A-4C show activation and possible mode of action of trispecific molecules (ProTriTAC). FIG. 4A shows ProTriTAC molecules in circulation, in tumor environment, and in circulation. FIG. 4B shows an exemplary sequence (SEQ ID NO: 929) for the protease cleavable site in a linker tethered to an anti-albumin binding moiety and FIG. 4C shows an SDS-PAGE gel showing the ProTriTAC in its activatable (prodrug) and activated (active drug) states.

FIGS. 5A-B illustrate a process for making and purifying molecules described herein.

FIG. 5A shows a schematic flowchart for manufacturing a ProTriTAC molecule and FIG. 5C shows an SDS-PAGE gel showing three purified ProTriTAC molecules.

FIGS. 6A-B show analytical size exclusion chromatograms on a ProTriTAC molecule exposed to different stress conditions in graph form in FIG. 6A with the corresponding data in FIG. 6B. FIG. 6B provides the data for FIG. 6A.

FIG. 7 shows protease-dependent, anti-tumor activity of exemplary ProTriTAC molecules in HCT116 Colorectal Tumor Xenograft Model in NSG Mice.

FIGS. 8A-8D show various designs of exemplary ProTriTAC and control molecules.

FIG. 8A: Control #1; FIG. 8B: Control #2; FIG. 8C: ProTriTAC; and FIG. 8D: Activated ProTriTAC.

FIG. 9 shows pharmacokinetic profiles for exemplary ProTriTAC and control molecules.

FIG. 10 shows conversion and half-life of exemplary ProTriTAC molecule.

FIG. 11 shows plasma clearance of an exemplary ProTriTAC molecule and its converted active drug format.

FIG. 12 shows CD3 binding potential of an exemplary ProTriTAC molecule, its converted active drug format, and a control non-cleavable ProTriTAC molecule.

FIG. 13 shows human primary T cell binding potential of an exemplary ProTriTAC molecule, its converted active drug format, and a control non-cleavable ProTriTAC molecule.

FIG. 14 shows T cell killing potential of an exemplary ProTriTAC molecule, its converted active drug format, and a control non-cleavable ProTriTAC molecule.

FIG. 15 shows the schematic structure of an exemplary trispecific molecule containing a binding moiety as described herein (also referred to herein as ProTriTAC or activatable ProTriTAC).

FIGS. 16A-16E shows exemplary schematic structures for Prodrug molecules combining functional masking and half-life extension. FIG. 16A shows a ProDrug molecule comprising an anti-albumin moiety which includes a masking moiety, and a cleavable linker connecting the anti-albumin moiety and the drug. FIG. 16B shows a ProDrug molecule comprising an anti-albumin moiety comprising two peptide motifs linked by linker, one of which includes a masking moiety, and a cleavable linker connecting the albumin binding moiety to a drug. FIG. 16C shows a ProDrug molecule comprising a modified albumin (containing a masking moiety) linked to a drug by a cleavable linker. FIG. 16D shows a ProDrug molecule comprising a modified albumin (containing a masking moiety and a protease cleavage site) linked to a drug. FIG. 16E shows an activated ProDrug. In each schematic structure (FIGS. 16A-16D) the drug molecule is functionally masked by the anti-albumin moiety or the modified albumin from binding its target or from being activated at an undesired site or from binding at non-target sites and thereby creating a drug sink.

FIGS. 17A-17E show exemplary schematic structures for ProTriTAC molecules combining functional masking and half-life extension. FIG. 17A shows a ProTriTAC molecule comprising an anti-albumin moiety which includes a masking moiety, and a cleavable linker connecting the anti-albumin moiety and a T cell engager molecule. FIG. 17B shows a ProTriTAC molecule comprising an anti-albumin moiety comprising two peptide motifs linked by linker, one of which includes a masking moiety, and a cleavable linker connecting the albumin binding moiety to a T cell engager molecule. FIG. 17C shows a ProTriTAC molecule comprising a modified albumin (containing a masking moiety) linked to a T cell engager by a cleavable linker. FIG. 17D shows a ProTriTAC molecule comprising a modified albumin (containing a masking moiety and a protease cleavage site) linked to a T cell engager. FIG. 17E shows an activated ProTriTAC. In each schematic structure (FIGS. 17A-17D) a target binding interface within the ProTriTAC molecule is functionally masked by the anti-albumin moiety or the modified albumin from binding its target or from being activated at an undesired site or from binding at non-target sites and thereby creating a sink.

FIG. 18 shows anti-tumor activity of exemplary ProTriTAC molecules and TriTAC molecules of this disclosure.

FIG. 19 shows pharmacokinetic profile of exemplary ProTriTAC molecules and TriTAC molecules of this disclosure.

FIGS. 20A-20F show admix xenograft individual tumor volumes following administering exemplary ProTriTAC molecules or TriTAC molecules of this disclosure. FIG. 20A illustrates results from a GFP TriTAC. FIG. 20B illustrates results from EGFR ProTriTAC (NCLV). FIG. 20C illustrates results from EGFR ProTriTAC (L001). FIG. 20D illustrates results from EGFR ProTriTAC (L041). FIG. 20E illustrates results from EGFR ProTriTAC (L040). FIG. 20F illustrates results from EGFR ProTriTAC (L045).

FIGS. 21A-21C shows cytokine levels (IFN-gamma (FIG. 21A), IL-6 (FIG. 21B), and IL-10; FIG. 21C) following administering exemplary ProTriTAC molecules or TriTAC molecules of this disclosure.

FIGS. 22A-22E show body weight percent change in mice, following administering exemplary ProTriTAC molecules and TriTAC molecules of this disclosure. FIG. 22A illustrates results of 30 µg/kg; FIG. 22B illustrates results of 100 µg/kg; FIG. 22C illustrates results of 300 µg/kg; FIG. 22D illustrates results of 1000 µg/kg; and FIG. 22E provides fold protection at the various concentrations.

FIGS. 23A-23C show body weight percent change in mice, following administering varying concentrations of exemplary ProTriTAC molecules of this disclosure, containing non-cleavable or cleavable linkers. FIG. 23A illustrates results of 300 µg/kg; FIG. 23B illustrates results of 1000 µg/kg; FIG. 23C provides fold protection at the various concentrations.

FIGS. 24A-24C show serum concentrations of aspartate aminotransferase (AST), in mice, following administering varying concentrations of a ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) (FIG. 24C), a TriTAC molecule (FIG. 24A), or a ProTriTAC molecule (FIG. 24B) containing a cleavable linker.

FIGS. 25A-25C show serum concentrations of alanine aminotransferase (ALT), in mice, following administering varying concentrations of a ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) (FIG. 25C), a TriTAC molecule (FIG. 25A), or a ProTriTAC molecule (FIG. 25B) containing a cleavable linker.

FIGS. 26A-26B show serum concentrations of ALT (right panel; FIG. 26B) or AST (left panel; FIG. 26A), in cynomolgus monkeys, following administering varying concentrations of an EGFR ProTriTAC molecule, or an EGFR ProTriTAC (NCLV) molecule.

FIGS. 27A-27D show tumor volume in mice following administration of a GFP TriTAC molecule, an EGFR TriTAC molecule, or an EGFR ProTriTAC molecule, in varying concentrations. FIG. 27A shows GFP TriTAC (at 300 µg/kg) and an EGFR TriTAC (at 10 µg/kg). FIG. 27B shows the EGFR TriTAC (at 30 µg/kg and at 100 µg/kg). FIG. 27C shows the EGFR TriTAC (at 300 µg/kg) and an EGFR ProTriTAC (at 30 µg/kg and 100 µg/kg). FIG. 27D shows the EGFR ProTriTAC (at 300 µg/kg and 1000 µg/kg).

FIG. 28 shows serum concentrations of ALT and AST, in mice, following administration of varying concentrations of a GFP TriTAC, an EGFR TriTAC, and an EGFR ProTriTAC molecule.

FIG. 29 shows grafting of a CD36 epitope into the CC' loop of a binding moiety of this disclosure. HuCD3e: SEQ ID NO: 902; CC10: SEQ ID NO: 260; CC12: SEQ ID NO: 259; and CC16: SEQ ID NO: 261. WT disclosed as SEQ ID NO: 795.

FIG. 30 shows separation of a binding moiety of this disclosure, from a ProTriTAC molecule that contained the binding moiety, upon tumor associated protease activation by matriptase.

FIG. 31 shows CD3 binding of ProTriTAC molecules, with or without activation, containing an exemplary binding moiety of this disclosure.

FIG. 32 shows cell killing potential of a ProTriTAC molecule, with or without activation, containing an exemplary binding moiety of this disclosure.

FIG. 33 shows the soft library mutagenesis approach carried out to explore the non-CDR loops within an exemplary binding moiety of this disclosure.

FIG. 34 illustrates the results of the soft library mutagenesis approach carried out to explore the non-CDR loops within an exemplary binding moiety of this disclosure, after panning against HSA (human serum albumin or also referred to herein as albumin). Figure discloses "Naïve Library" sequences as SEQ ID NOS 930-1018, respectively, in order of appearance and discloses "After panning with HSA" sequences as SEQ ID NOS 1019-1140, respectively, in order of appearance.

FIG. 35 illustrates that a binding moiety of this disclosure is able to expand the therapeutic window of a molecule containing it (e.g., a ProTriTAC molecule) by both steric masking and specific masking.

FIG. 36 provides results from a representative T cell dependency cellular cytotoxicity assay with NCI-H508 cells using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.

FIG. 37 provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.

FIG. 38 provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.

FIG. 39 provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing an anti-EpCAM domain as described herein and an anti-CD3 domain.

FIG. 40 provides results from a representative T cell dependency cellular cytotoxicity assay using exemplary fusion proteins of this disclosure containing a humanized anti-EpCAM domain as described herein and an anti-CD3 domain.

FIGS. 41A-41C show body weight percent change in mice, following administering exemplary EpCAM ProTriTAC molecules and EpCAM TriTAC molecules of this disclosure.

DETAILED DESCRIPTION OF THE INVENTION

While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that the following claims define

the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

Provided herein in certain embodiments are ProTriTAC molecules (also referred to herein as protrispecific molecules) that are T cell engager prodrugs designed to be conditionally active in a tumor microenvironment. In some cases, this enables targeting of a wider selection of tumor antigens (e.g., solid tumor antigens). The ProTriTAC molecules, in some examples, combine the desirable attributes of several prodrug approaches, including, but not limited to: combination of steric and specific masking, wherein the steric masking is, in some cases, is through albumin that is recognized by an anti-albumin domain in a ProTriTAC molecule, and the specific masking, in some cases, is through specific intermolecular interactions between an anti-albumin domain (in some examples) and a target antigen binding domain of the ProTriTAC molecule (such as, an anti-CD3 scFv domain, in some examples); additional safety imparted by half-life differential of prodrug versus an active drug, derived by activation of the conditionally activated ProTriTAC molecule; ability to plug-and-play with different tumor target binders.

Certain Definitions

The terminology used herein is for the purpose of describing particular cases only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the given value. Where particular values are described in the application and claims, unless otherwise stated the term “about” should be assumed to mean an acceptable error range for the particular value.

The terms “individual,” “patient,” or “subject” are used interchangeably. None of the terms require or are limited to situation characterized by the supervision (e.g. constant or intermittent) of a health care worker (e.g. a doctor, a registered nurse, a nurse practitioner, a physician’s assistant, an orderly, or a hospice worker).

A “single chain Fv” or “scFv”, as used herein, refers to a binding protein in which the variable domains of the heavy chain and of the light chain of a traditional two chain antibody are joined to form one chain. Typically, a linker peptide is inserted between the two chains to allow for proper folding and creation of an active binding site.

A “cleavage site for a protease,” or “protease cleavage site”, as meant herein, is an amino acid sequence that can be cleaved by a protease, such as, for example, a matrix metalloproteinase or a furin. Examples of such sites include Gly-Pro-Leu-Gly-Ile-Ala-Gly-Gln (SEQ ID NO: 924) or Ala-Val-Arg-Trp-Leu-Leu-Thr-Ala (SEQ ID NO: 925), which can be cleaved by metalloproteinases, and Arg-Arg-Arg-Arg-Arg (SEQ ID NO: 926), which is cleaved by a furin. In therapeutic applications, the protease cleavage

site can be cleaved by a protease that is produced by target cells, for example cancer cells or infected cells, or pathogens.

As used herein, “elimination half-time” is used in its ordinary sense, as is described in *Goodman and Gillman’s The Pharmaceutical Basis of Therapeutics* 21-25 (Alfred Goodman Gilman, Louis S. Goodman, and Alfred Gilman, eds., 6th ed. 1980). Briefly, the term is meant to encompass a quantitative measure of the time course of drug elimination. The elimination of most drugs is exponential (i.e., follows first-order kinetics), since drug concentrations usually do not approach those required for saturation of the elimination process. The rate of an exponential process may be expressed by its rate constant, k , which expresses the fractional change per unit of time, or by its half-time, $t_{1/2}$, the time required for 50% completion of the process. The units of these two constants are time^{-1} and time, respectively. A first-order rate constant and the half-time of the reaction are simply related ($k \times t_{1/2} = 0.693$) and may be interchanged accordingly. Since first-order elimination kinetics dictates that a constant fraction of drug is lost per unit time, a plot of the log of drug concentration versus time is linear at all times following the initial distribution phase (i.e. after drug absorption and distribution are complete). The half-time for drug elimination can be accurately determined from such a graph.

A “therapeutic agent,” as used herein, includes a “binding molecule.”

The term “binding molecule,” as used herein is any molecule, or portion or fragment thereof, that can bind to a target molecule, cell, complex and/or tissue, and which includes proteins, nucleic acids, carbohydrates, lipids, low molecular weight compounds, and fragments thereof, each having the ability to bind to one or more of a soluble protein, a cell surface protein, a cell surface receptor protein, an intracellular protein, a carbohydrate, a nucleic acid, a hormone, or a low molecular weight compound (small molecule drug), or a fragment thereof. The binding molecule, in some instances, is a protein belonging to the immunoglobulin superfamily, or a non-immunoglobulin binding molecule. The “binding molecule” does not include a cytokine.

The term “proteins belonging to immunoglobulin superfamily,” or “immunoglobulin molecules,” as used herein, include proteins that comprise an immunoglobulin fold, such as antibodies and target antigen binding fragments thereof, antigen receptors, antigen presenting molecules, receptors on natural killer cells, antigen receptor accessory molecules, receptors on leukocytes, IgSF cellular adhesion molecules, growth factor receptors, and receptor tyrosine kinases/phosphatases.

The term “antibodies” include antibodies or immunoglobulins of any isotype, fragments of antibodies that retain specific binding to antigen, including, but not limited to, Fab, Fv, scFv, and Fd fragments, chimeric antibodies, humanized antibodies, single-chain antibodies (scAb), single domain antibodies (dAb), single domain heavy chain antibodies, a single domain light chain antibodies, bi-specific antibodies, multi-specific antibodies, and fusion proteins comprising an antigen-binding (also referred to herein as antigen binding) portion of an antibody and a non-antibody protein. The antibodies, in some examples, are detectably labeled, e.g., with a radioisotope, an enzyme that generates a detectable product, a fluorescent protein, and the like. The antibodies, in some cases, are further conjugated to other moieties, such as members of specific binding pairs, e.g., biotin (member of biotin-avidin specific binding pair), and the like. The antibodies, in some cases, are bound to a

solid support, including, but not limited to, polystyrene plates or beads, and the like. Also encompassed by the term are Fab', Fv, F(ab')₂, and or other antigen binding fragments that retain specific binding to antigen, and monoclonal antibodies. As used herein, a monoclonal antibody is an antibody produced by a group of identical cells, all of which were produced from a single cell by repetitive cellular replication. That is, the clone of cells only produces a single antibody species. While a monoclonal antibody can be produced using hybridoma production technology, other production methods known to those skilled in the art can also be used (e.g., antibodies derived from antibody phage display libraries). An antibody, in some instances, is monovalent or bivalent. An antibody, in some instances, is an Ig monomer, which is a "Y-shaped" molecule that consists of four polypeptide chains: two heavy chains and two light chains connected by disulfide bonds.

The term "non-immunoglobulin binding molecules," as used herein, include, but is not limited to examples such as a growth factor, a hormone, a signaling protein, an inflammatory mediator, ligand, a receptor, or a fragment thereof, a native hormone or a variant thereof being able to bind to its natural receptor; a nucleic acid or polynucleotide sequence being able to bind to complementary sequence or a soluble cell surface or intracellular nucleic acid/polynucleotide binding proteins, a carbohydrate binding moiety being able to bind to other carbohydrate binding moieties, cell surface or intracellular proteins, a low molecular weight compound (drug) that binds to a soluble or cell surface or intracellular target protein. The non-immunoglobulin binding molecules, in some cases, include coagulation factors, plasma proteins, fusion proteins, and imaging agents. The non-immunoglobulin binding molecules do not include a cytokine.

A "cytokine," as meant herein, refers to intercellular signaling molecules, and active fragments and portions thereof, which are involved in the regulation of mammalian somatic cells. A number of families of cytokines, for example, interleukins, interferons, and transforming growth factors are included.

As used herein, "non-CDR loops" within immunoglobulin (Ig) molecules are regions of a polypeptide other than the complementarity determining regions (CDRs) of an antibody. These regions may be derived from an antibody or an antibody fragment. These regions may also be synthetically or artificially derived, such as through mutagenesis or polypeptide synthesis.

In an Ig, Ig-like, or beta-sandwich scaffold that has 9 beta-strands (e.g., a VH, a VL, a camelid VHH, a sdAb), the non-CDR loops can refer to the AB, CC', C'D, EF loops or loops connecting beta-strands proximal to the C-terminus. In an Ig, Ig-like, or beta-sandwich scaffold that has 7 beta-strands (e.g., a CH, a CL, an adnectin, a Fn-III), the non-CDR loops can refer to the AB, CD, and EF loops or loops connecting beta-strands proximal the C-terminus. In other Ig-like or beta-sandwich scaffolds, the non-CDR loops are the loops connecting beta-strands proximal to the C-terminus or topologically equivalent residues using the framework established in the Halaby 1999 publication (Prot Eng Des Sel 12:563-571).

In a non-beta-sandwich scaffold (e.g., a DARPin, an affimer, an affibody), the "non-CDR loops" refer to an area that is (1) amenable for sequence randomization to allow engineered specificities to a second antigen, and (2) distal to the primary specificity determining region(s) typically used on the scaffold to allow simultaneous engagement of the scaffold to both antigens without steric interference. For this purpose, the primary specificity determining region(s) can

be defined using the framework established in the Skrllec 2015 publication (Trends in Biotechnol, 33:408-418). An excerpt of the framework is listed below.

Scaffold	Primary specificity determining region(s)
Affibody	13 residues in two helices
Affimer	12-36 residues
Anticalin	Four loops (up to 24 aa)
Avimer	11 residues
Scaffold	Primary specificity determining region(s)
Centyrin	13 residues
DARPin	7 residues in each n-repeat, or 8 residues in each n-repeat
Fynomer	6 residues in the RT- and n-Src-loop
Kunitz domain	1-2 loops

"Target antigen binding domain", as used herein, refers to a region which targets a specific antigen. A target antigen binding domain comprises, for example an sdAb, an scFv, a variable heavy chain antibody (VHH), a variable heavy (VH) or a variable light domain (VL), a full length antibody, or any other peptide that has a binding affinity towards a specific antigen. The target antigen binding domain does not include a cytokine.

"TriTAC," as used herein refers to a trispecific binding protein that is not conditionally activated. Binding Moiety, Cleavable Linker and Conditionally Active Binding Proteins

This disclosure provides, in some embodiments, binding moieties that are capable of masking the interaction of binding molecules with their targets. In some embodiments, a binding moiety of this disclosure comprises a masking moiety and a cleavable linker, such as a protease cleavable linker. In some embodiments, a binding moiety of this disclosure comprises a masking moiety (e.g., a modified non-CDR loop sequence) and a non-cleavable linker. As illustrated in FIG. 35, the binding moiety is capable of synergistically expanding a therapeutic window of a molecule that comprises the moiety, by both steric masking and specific masking. In some examples, the binding molecule is a protein belonging to an immunoglobulin superfamily, such as a target antigen binding domain comprising an immunoglobulin fold. In some embodiments, the binding molecule is a non-immunoglobulin protein. In some embodiments, the binding moiety combines both steric masking (for example, via binding to a bulky serum albumin) and specific masking (for example, via non-CDR loops binding to the CDRs of an anti-CD3 scFv domain). In some cases, modifying the non-CDR loops within the binding moiety does not affect albumin binding. The protease cleavable linker, in some cases, enables activation of a prodrug molecule comprising the binding moiety (such as a ProTriTAC molecule comprising a binding moiety as described herein, a CD3 binding domain, and an albumin binding domain), in a single proteolytic event, thereby allowing more efficient conversion of the prodrug molecule in tumor microenvironment. Further, tumor-associated proteolytic activation, in some cases, reveals active T cell engager (such as a ProTriTAC molecule comprising a binding moiety as described herein, a CD3 binding domain, and an albumin binding domain) with minimal off-tumor activity after activation. The present disclosure, in some embodiments, provides a half-life extended T cell engager format (ProTriTAC) comprising a binding moiety as described herein, which in some cases represents a new and improved approach to engineer conditionally active T cell engagers.

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FIG. 4B provides a schematic for an exemplary ProTriTAC molecule comprising an exemplary binding moiety as described herein (the α albumin sdAb) and a gel showing the ProTriTAC before and after activation by cleaving of the protease cleavable linker, and FIG. 4A shows a possible mode of action of the same. FIG. 15 shows the schematic structure of an exemplary trispecific molecule containing a binding moiety as described herein (also referred to herein as ProTriTAC or activatable ProTriTAC), with engineered non-CDR loops. The exemplary trispecific molecule contains an anti-albumin domain comprising a cleavable linker (such a linker comprising a protease cleavable site, also referred to herein as a substrate linker) and a masking domain; an anti-CD3 binding domain; and an anti-target domain (specific for a tumor antigen) which in some cases is a non-immunoglobulin molecule. In some cases, non-CDR loops in the anti-albumin domain is capable of binding and masking the anti-target domain. In some cases, non-CDR loops in the anti-albumin domain is capable of binding and masking the anti-CD3 domain. The binding moiety, in some embodiments, comprises a CDR loop specific for binding albumin.

Provided herein, in a first embodiment, is a binding moiety that masks the binding of a target antigen binding domain, and is capable of binding to a bulk-serum protein, such as a half-life extending protein. The binding moiety of the first embodiment, in certain instances, further comprises a cleavable linker attached to it. The cleavable linker, for example, comprises a protease cleavage site or a pH dependent cleavage site. The cleavable linker, in certain instances, is cleaved only in a tumor micro-environment. Thus, in some examples, the binding moiety of the first embodiment, bound to the half-life extending protein, connected to the cleavable linker, and further bound to the target antigen binding domain, maintains the target antigen binding domain in an inert state in circulation until the cleavable linker is cleaved off in a tumor microenvironment. The half-life of the target antigen binding domain, such as an antibody or an antigen binding fragment thereof, is thus extended in systemic circulation by using the binding moiety of the first embodiment which acts as a safety switch that keeps the target antigen binding moiety in an inert state until it reaches the tumor microenvironment where it is conditionally activated by cleavage of the linker and is able to bind its target antigen.

In a second embodiment is provided a binding moiety that masks the interaction between a non-immunoglobulin binding molecule and its target. The binding moiety of the second embodiment, in certain instances, is capable of binding to a bulk serum protein. In some instances, the binding moiety of the second embodiment further comprises a cleavable linker attached to it. The cleavable linker, for example, comprises a protease cleavage site or a pH dependent cleavage site. The cleavable linker, in certain instances, is cleaved only in a tumor micro-environment. The non-immunoglobulin binding molecule is, in some cases, maintained in an inert state by the binding moiety of the second embodiment and activated by cleavage of the linker, for example in a target environment. In some instances, the cleavable linker is cleaved off in a tumor microenvironment and in such cases the tumor microenvironment is the target environment. The half-life of the non-immunoglobulin binding molecule is thus extended in systemic circulation by using the binding moiety of the second embodiment which acts as a safety switch that keeps the non-immunoglobulin binding molecule in an inert state until it reaches the target environment where it is conditionally activated by cleavage

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of the linker. In some examples of the second embodiment where the non-immunoglobulin binding molecule is an imaging agent, said agent is activated in a target environment upon cleavage of the cleavable linker. The target environment, in such cases, is a tissue or a cell or any biological environment that is to be imaged using the imaging agent.

The safety switch described above provides several advantages; some examples including (i) expanding the therapeutic window of the immunoglobulin molecule, such as a target antigen binding domain, a non-immunoglobulin binding molecule; (ii) reducing target-mediated drug disposition by maintaining the immunoglobulin molecule, such as a target antigen binding domain, the non-immunoglobulin binding molecule, in an inert state when a conditionally active protein comprising a binding moiety according to the first or second embodiments is in systemic circulation; (iii) reducing the concentration of undesirable activated proteins in systemic circulation, thereby minimizing the spread of chemistry, manufacturing, and controls related impurities, e.g., pre-activated drug product, endogenous viruses, host-cell proteins, DNA, leachables, anti-foam, antibiotics, toxins, solvents, heavy metals; (iv) reducing the concentration of undesirable activated proteins in systemic circulation, thereby minimizing the spread of product related impurities, aggregates, breakdown products, product variants due to: oxidation, deamidation, denaturation, loss of C-term Lys in MABs; (v) preventing aberrant activation of the immunoglobulin molecule, such as a target antigen binding domain, or the non-immunoglobulin binding molecule in circulation; (vi) reducing the toxicities associated with the leakage of activated species from diseased tissue or other pathophysiological conditions, e.g., tumors, autoimmune diseases, inflammations, viral infections, tissue remodeling events (such as myocardial infarction, skin wound healing), or external injury (such as X-ray, CT scan, UV exposure); and (vii) reducing non-specific binding of the immunoglobulin molecule, such as a target antigen binding domain, or the non-immunoglobulin binding molecule. Furthermore, post-activation, or in other words post breaking of the safety switch, the immunoglobulin molecule, such as a target antigen binding domain, the non-immunoglobulin binding molecule is separated from the safety switch which provided extended half-life, and thus is cleared from circulation.

In addition, the binding moieties of the first, second, and the third embodiments, in some cases, are used to generate a "biobetter" version of a biologic. Generally, preparing a biobetter form of a molecule, e.g., an antibody or an antigen binding fragment thereof, involves taking the originator molecule and making specific alterations in it to improve its parameters and thereby make it a more efficacious, less frequently dosed, better targeted, and/or a better tolerated drug. Thus, a target antigen binding domain masked by the binding moiety of the first embodiment which is bound to a half-life extending protein, and conditionally activated in a tumor microenvironment by cleavage of the cleavable linker, gives the target antigen binding domain a significantly longer serum half-life and reduces the likelihood of its undesirable activation in circulation, thereby producing a "biobetter" version of the target antigen binding domain. Similarly, the binding moieties of the second embodiment are, in some cases, utilized to generate biobetter versions of the non-immunoglobulin binding molecules. Accordingly, in various embodiments, biobetter versions of immunoglobulin molecules, non-immunoglobulin binding molecules are pro-

vided, wherein the biobetter function is attributed to a binding moiety, respectively, according to the first or second embodiments.

The binding moieties described herein comprise at least one non-CDR loop. In some embodiments, a non-CDR loop provides a binding site for binding of the binding moiety of the first embodiment to a target antigen binding domain. In some examples of the first embodiment, a non-CDR loop provides a binding site for binding of the binding moiety of the first embodiment to an immunoglobulin molecule, such as a target antigen binding domain. In some examples of the second embodiment, a non-CDR loop provides a binding site for binding of the binding moiety of the second embodiment to a non-immunoglobulin binding molecule. In some cases, the binding moiety of the first embodiment masks binding of the target binding domain to the target antigen, e.g. via steric occlusion, via specific intermolecular interactions, or a combination of both. The binding moieties of the second embodiment also, in some cases, mask binding of a non-immunoglobulin binding molecule to their targets, via steric occlusion, via specific intermolecular interactions, or a combination of both.

In some embodiments, the binding moieties described herein further comprise complementarity determining regions (CDRs). In some instances, the binding moieties are domains derived from an immunoglobulin molecule (Ig molecule). The Ig may be of any class or subclass (IgG1, IgG2, IgG3, IgG4, IgA, IgE, IgM etc). A polypeptide chain of an Ig molecule folds into a series of parallel beta strands linked by loops. In the variable region, three of the loops constitute the "complementarity determining regions" (CDRs) which determine the antigen binding specificity of the molecule. An IgG molecule comprises at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, or an antigen binding fragment thereof. Each heavy chain is comprised of a heavy chain variable region (abbreviated herein as VH) and a heavy chain constant region. The heavy chain constant region is comprised of three domains, CH1, CH2 and CH3. Each light chain is comprised of a light chain variable region (abbreviated herein as VL) and a light chain constant region. The light chain constant region is comprised of one domain, CL. The VH and VL regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDRs) with are hypervariable in sequence and/or involved in antigen recognition and/or usually form structurally defined loops, interspersed with regions that are more conserved, termed framework regions (FR). Each VH and VL is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. In some embodiments of this disclosure, at least some or all of the amino acid sequences of FR1, FR2, FR3, and FR4 are part of the "non-CDR loop" of the binding moieties described herein. As shown in FIG. 1, a variable domain of an immunoglobulin molecule has several beta strands that are arranged in two sheets. The variable domains of both light and heavy immunoglobulin chains contain three hypervariable loops, or complementarity-determining regions (CDRs). The three CDRs of a V domain (CDR1, CDR2, CDR3) cluster at one end of the beta barrel. The CDRs are the loops that connect beta strands B-C, C'-C'', and F-G of the immunoglobulin fold, whereas the bottom loops that connect beta strands AB, CC', C''-D and E-F of the immunoglobulin fold, and the top loop that connects the D-E strands of the immunoglobulin fold are the non-CDR loops. In some embodiments of this disclosure, at least some amino

acid residues of a constant domain, CH1, CH2, or CH3, are part of the "non-CDR loop" of the binding moieties described herein. Non-CDR loops comprise, in some embodiments, one or more of AB, CD, EF, and DE loops of a C1-set domain of an Ig or an Ig-like molecule; AB, CC', EF, FG, BC, and EC' loops of a C2-set domain of an Ig or an Ig-like molecule; DE, BD, GF, A(A1A2)B, and EF loops of I(Intermediate)-set domain of an Ig or Ig-like molecule.

Within the variable domain, the CDRs are believed to be responsible for antigen recognition and binding, while the FR residues are considered a scaffold for the CDRs. However, in certain cases, some of the FR residues play an important role in antigen recognition and binding. Framework region residues that affect Ag binding are divided into two categories. The first are FR residues that contact the antigen, thus are part of the binding-site, and some of these residues are close in sequence to the CDRs. Other residues are those that are far from the CDRs in sequence, but are in close proximity to it in the 3-D structure of the molecule, e.g., a loop in heavy chain.

In some embodiments, the non-CDR loop is modified to generate an antigen binding site specific for a bulk serum protein, such as albumin. It is contemplated that various techniques can be used for modifying the non-CDR loop, e.g., site-directed mutagenesis, random mutagenesis, insertion of at least one amino acid that is foreign to the non-CDR loop amino acid sequence, amino acid substitution. An antigen peptide is inserted into a non-CDR loop, in some examples. In some examples, an antigenic peptide is substituted for the non-CDR loop. The modification, to generate an antigen binding site, is in some cases in only one non-CDR loop. In other instances, more than one non-CDR loop are modified. For instance, the modification is in any one of the non-CDR loops shown in FIG. 1, i.e., AB, CC', C'' D, EF, and D-E. In some cases, the modification is in the DE loop. In other cases the modifications are in all four of AB, CC', C''-D, E-F loops. In certain examples, the binding moieties described herein are bound to the immunoglobulin molecules, e.g., a target antigen binding domain, the non-immunoglobulin binding molecules via their AB, CC', C'' D, or EF loop and are bound to a bulk-serum protein, such as albumin, via their B-C, C'-C'', or F-G loop. In certain examples, the binding moiety of the first embodiment is bound to the target antigen binding domain via its AB, CC', C'' D, and EF loop and is bound to a bulk-serum protein, such as albumin, via its BC, C'C'', and FG loop.

In certain examples, the binding moiety of the first embodiment is bound to a target antigen binding domain via one or more of AB, CC', C'' D, and E-F loop and is bound to a bulk-serum protein, such as albumin, via one or more of BC, C'C'', and FG loop. In certain examples, the binding moiety of the first embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC', C'' D, or EF loop and is bound to a target antigen binding domain via its BC, C'C'', or FG loop. In certain examples, the binding moiety of the first embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC', C'' D, and EF loop and is bound to a target antigen binding domain via its BC, C'C'', and FG loop. In certain examples, the binding moiety of the first embodiment is bound to a bulk serum protein, such as albumin, via one or more of AB, CC', C'' D, and E-F loop and is bound to the target antigen binding protein, via one or more of BC, C'C'', and FG loop.

In certain examples, the binding moiety of the second embodiment is bound to a non-immunoglobulin molecule via one or more of AB, CC', C'' D, and E-F loop and is bound to a bulk-serum protein, such as albumin, via one or more of

BC, C'C", and FG loop. In certain examples, the binding moiety of the second embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC', C" D, or EF loop and is bound to a non-immunoglobulin molecule via its BC, C'C", or FG loop. In certain examples, the binding moiety of the second embodiment is bound to a bulk serum protein, such as albumin, via its AB, CC', C" D, and EF loop and is bound to a non-immunoglobulin molecule via its BC, C'C", and FG loop. In certain examples, the binding moiety of the second embodiment is bound to a bulk serum protein, such as albumin, via one or more of AB, CC', C" D, and E-F loop and is bound to a non-immunoglobulin molecule, via one or more of BC, C'C", and FG loop.

The binding moieties are any kinds of polypeptides. For example, in certain instances the binding moieties are natural peptides, synthetic peptides, or fibronectin scaffolds, or engineered bulk serum proteins. The bulk serum protein comprises, for example, albumin, fibrinogen, or a globulin. In some embodiments, the binding moieties are engineered scaffolds. Engineered scaffolds comprise, for example, sdAb, a scFv, a Fab, a VHH, a fibronectin type III domain, immunoglobulin-like scaffold (as suggested in Halaby et al., 1999. Prot Eng 12(7):563-571), DARPin, cystine knot peptide, lipocalin, three-helix bundle scaffold, protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold.

In some cases, the binding moiety of the first embodiment binds to at least one target antigen binding domain. In further embodiments, the non-CDR loops within the binding moiety of the first embodiment provide a binding site for the at least one target antigen binding domain. The target antigen binding domain, in some cases, binds to target antigens expressed on the surface of a diseased cell or tissue, for example a tumor or a cancer cell. Target antigens include but are not limited to EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, and CEA. 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, ROR1, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1.

In some cases, the binding moiety of the first embodiment is bound to a first target antigen binding domain via its non-CDR loops and the first target antigen binding domain is further connected to a second target antigen binding domain. Examples of first and second target antigen binding domains include, but are not limited to, a T cell engager, a bispecific T cell engager, a dual-affinity re-targeting antibody, a variable heavy domain (VH), a variable light domain (VL), a scFv comprising a VH and a VL domain, a soluble TCR fragment comprising a Valpha and Vbeta domain, a single domain antibody (sdAb), or a variable domain of camelid derived nanobody (VHH), a non-Ig binding domain, i.e., antibody mimetic, such as anticalins, affilins, affibody molecules, affimers, affitins, alphabodies, avimers, DARPins, fynomers, kunitz domain peptides, and monobodies, a ligand or peptide. In some examples, the first or the second target antigen binding domain is a VHH domain. In some examples, the first or the second target antigen binding domain is a sdAb. In some instances, the first target antigen binding domain is specific for a tumor antigen, such as EGFR, and the second target antigen binding domain is specific for CD3. The binding of the first target antigen

binding domain to its target, e.g., a tumor antigen such as EGFR, is masked by the binding moiety of the first embodiment, via its non-CDR loops. One exemplary conditionally active protein, comprising a binding moiety according to the first embodiment, is shown in FIG. 3.

In some cases, the non-CDR loops within the binding moiety of the second embodiment provide a binding site for a non-immunoglobulin binding molecule.

In some cases, the binding moieties comprise a binding site for a bulk serum protein. In some embodiments, the CDRs within the binding moieties provide a binding site for the bulk serum protein. The bulk serum protein is, in some examples, a globulin, albumin, transferrin, IgG1, IgG2, IgG4, IgG3, IgA monomer, Factor XIII, Fibrinogen, IgE, or pentameric IgM. In some embodiments, the binding moieties comprise a binding site for an immunoglobulin light chain. In some embodiments, the CDRs provide a binding site for the immunoglobulin light chain. The immunoglobulin light chain is, in some examples, an IgK free light chain or an Ig, free light chain.

In some examples, the binding moieties comprise any type of binding domain, including but not limited to, domains from a monoclonal antibody, a polyclonal antibody, a recombinant antibody, a human antibody, a humanized antibody. In some embodiments, the binding moiety is a single chain variable fragment (scFv), a soluble TCR fragment, a single-domain antibody such as a heavy chain variable domain (VH), a light chain variable domain (VL) and a variable domain (VHH) of camelid derived nanobody. In other embodiments, the binding moieties are non-Ig binding domains, i.e., antibody mimetic, such as anticalins, affilins, affibody molecules, affimers, affitins, alphabodies, avimers, DARPins, fynomers, kunitz domain peptides, and monobodies.

TABLE 1

Exemplary Sequences for Masking Sequences within the Binding Moieties of this Disclosure are Provided In SEQ ID Nos. 50, 259-301 And 795.			
MaskID	Sequence	Alt. Names	SEQ ID No.
MASK001	APGKG	WT	SEQ ID No. 795
MASK002	GGQDGNEEMGGG	CC12	SEQ ID No. 259
MASK004	GGQDGNEEGG	CC10	SEQ ID No. 260
MASK006	GGGGQDGNEEMGGGGG	CC16	SEQ ID No. 261
MASK007	APFGSEM		SEQ ID No. 262
MASK008	AWNPGPYE		SEQ ID No. 263
MASK009	AQDNGDTKTG		SEQ ID No. 264
MASK010	AEAKETQG		SEQ ID No. 265
MASK011	ATRREQVEG		SEQ ID No. 266
MASK012	AQAPSQP		SEQ ID No. 267
MASK013	ATRSRTRNDG		SEQ ID No. 268
MASK014	ADVDPDGLG		SEQ ID No. 269
MASK015	AADISDPGG		SEQ ID No. 270
MASK016	ALSVDPSG		SEQ ID No. 271
MASK017	ARLSVDPG		SEQ ID No. 272

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TABLE 1-continued

Exemplary Sequences for Masking Sequences within the Binding Moieties of this Disclosure are Provided In SEQ ID Nos. 50, 259-301 And 795.			
MaskID	Sequence	Alt. Names	SEQ ID No.
MASK018	AVEAADRG		SEQ ID No. 273
MASK020	GGPDGNEEMGGG	CC12-Q1P	SEQ ID No. 274
MASK021	GGFDGNEEMGGG	CC12-Q1F	SEQ ID No. 275
MASK022	GGGDGNEEMGGG	CC12-Q1G	SEQ ID No. 276
MASK023	GGEMDGEGQNGG	CC12-scramble	SEQ ID No. 277
MASK024	GGGGPDGNEEPGG		SEQ ID No. 278
MASK025	GGGSLDGNEEPGG		SEQ ID No. 279
MASK026	GGGGALDGNEEPGG		SEQ ID No. 280
MASK027	GGGGGLDGNEEPGG		SEQ ID No. 281
MASK028	GGGALDGNEEPGG		SEQ ID No. 282
MASK029	GGGGPDGNEEPGGG		SEQ ID No. 283
MASK030	GGSGALDGNEEPGG		SEQ ID No. 284
MASK031	GGSGSLDGNEEPGG		SEQ ID No. 285
MASK038	GGSGPDGNEEPGG		SEQ ID No. 286
MASK039	GGVRDGPDGNEEPGG		SEQ ID No. 287
MASK040	GGSGPDGNEEPGGGG		SEQ ID No. 288
MASK041	GGGRPDGNEEPGG		SEQ ID No. 289
MASK042	GGSGGLDGNEEPGG		SEQ ID No. 290
MASK043	GGGVPGDGNEEPGG		SEQ ID No. 291
MASK044	GGGEGPDGNEEPGG		SEQ ID No. 292
MASK046	GGVALDGNEEPGG		SEQ ID No. 293
MASK047	GGGRALDGNEEPGG		SEQ ID No. 294
MASK048	GGYAGLDGNEEPGG		SEQ ID No. 295
MASK049	GGAGPDGNEEPGG		SEQ ID No. 296
MASK051	GGRGGPDGNEEPGG		SEQ ID No. 297
MASK052	GGGGPDGNEEPGGGG		SEQ ID No. 298
MASK053	GGGEALDGNEEPGG		SEQ ID No. 299
MASK054	GGDASLDGNEEPGG		SEQ ID No. 300
MASK055	GGRDAPDGNEEGG		SEQ ID No. 301

It is contemplated herein that in some embodiments of this disclosure the binding moieties described herein comprise at least one cleavable linker. In one aspect, the cleavable linker comprises a polypeptide having a sequence recognized and cleaved in a sequence-specific manner. The cleavage, in certain examples, is enzymatic, based on pH sensitivity of the cleavable linker, or by chemical degradation. The conditionally active binding proteins contemplated herein, in some cases, comprise a protease cleavable linker recognized in a sequence-specific manner by a matrix metalloprotease (MMP), for example a MMP9. In some cases, the protease cleavable linker is recognized by a MMP9

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comprises a polypeptide having an amino acid sequence PR(S/T)(L/I)(S/T) (SEQ ID NO: 3). In some cases, the protease cleavable linker recognized by a MMP9 comprises a polypeptide having an amino acid sequence LEATA (SEQ ID NO: 4). In some cases, the protease cleavable linker is recognized in a sequence-specific manner by a MMP11. In some cases, the protease cleavable linker recognized by a MMP11 comprises a polypeptide having an amino acid sequence GGAANLVRGG (SEQ ID NO: 5). In some cases, the protease cleavable linker is recognized by a protease disclosed in Table 3. In some cases, the protease cleavable linker is recognized by a protease disclosed in Table 3 comprises a polypeptide having an amino acid sequence selected from a sequence disclosed in Table 3 (SEQ ID NOS: 1-42, 53, and 58-62). In some cases, the cleavable linker has an amino acid sequence as set forth in SEQ ID No. 59. In some cases, the cleavable linker is recognized by MMP9, matriptase, Urokinase plasminogen activator (uPA) and has an amino acid sequence as set forth in SEQ ID No. 59.

In some embodiments of this disclosure the binding moieties described herein comprise at least one non-cleavable linker. The non-cleavable linker comprises, in some examples, a sequence as set forth in SEQ ID No. 51, SEQ ID No. 302, SEQ ID No. 303, SEQ ID No. 304, or SEQ ID No. 305.

TABLE 2

Exemplary Non-cleavable Linker Sequences		
Linker ID	Sequence	SEQ ID No.
L002	SGGGSGGVV	SEQ ID No. 302
L016	SGGGSGGGSGGGGS	SEQ ID No. 303
L017	SGGGSGGGSGGGGS	SEQ ID No. 304
L046	SGGGSGGGGS	SEQ ID No. 305

Proteases are proteins that cleave proteins, in some cases, in a sequence-specific manner. Proteases include but are not limited to serine proteases, cysteine proteases, aspartate proteases, threonine proteases, glutamic acid proteases, metalloproteases, asparagine peptide lyases, serum proteases, cathepsins, Cathepsin B, Cathepsin C, Cathepsin D, Cathepsin E, Cathepsin K, Cathepsin L, kallikreins, hK1, hK10, hK15, plasmin, collagenase, Type IV collagenase, stromelysin, Factor Xa, chymotrypsin-like protease, trypsin-like protease, elastase-like protease, subtilisin-like protease, actinidain, bromelain, calpain, caspases, caspase-3, Mir1-CP, papain, HIV-1 protease, HSV protease, CMV protease, chymosin, renin, pepsin, matriptase, legumain, plasmepsin, nepenthesin, metalloexopeptidases, metal loendopeptidases, matrix metalloproteases (MMP), MMP1, MMP2, MMP3, MMP7, MMP8, MMP9, MMP13, MMP11, MMP14, urokinase plasminogen activator (uPA), enterokinase, prostate-specific antigen (PSA, hK3), interleukin-1 β converting enzyme, thrombin, FAP (FAP- α), dipeptidyl peptidase, type II transmembrane serine proteases (TTSP), neutrophil serine protease, cathepsin G, proteinase 3, neutrophil serine protease 4, mast cell chymase, and mast cell tryptases.

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TABLE 3

Exemplary Proteases and Protease Recognition Sequences			SEQ ID NO:
Protease	Cleavage Domain Sequence		
MMP7	KRALGLPG	1	5
MMP7	(DE) ₈ RPLALWRS (DR) ₈	2	
MMP9	PR (S/T) (L/I) (S/T)	3	10
MMP9	LEATA	4	
MMP11	GGAANLVRGG	5	15
MMP14	SGRIGFLRTA	6	
MMP	PLGLAG	7	20
MMP	PLGLAX	8	
MMP	PLGC (me) AG	9	25
MMP	ESPAYYTA	10	
MMP	RLQLKL	11	30
MMP	RLQLKAC	12	
MMP2, MMP9, MMP14	EP (Cit) G (Hof) YL	13	35
Urokinase plasminogen activator (uPA)	SGRSA	14	
Urokinase plasminogen activator (uPA)	DAFK	15	40
Urokinase plasminogen activator (uPA)	GGGRR	16	
Lysosomal Enzyme	GFLG	17	45
Lysosomal Enzyme	ALAL	18	
Lysosomal Enzyme	FK	19	50
Cathepsin B	NLL	20	
Cathepsin D	PIC (Et) FF	21	55
Cathepsin K	GGPRGLPG	22	
Prostate Specific Antigen	HSSKLQ	23	60
Prostate Specific Antigen	HSSKLQL	24	
Prostate Specific Antigen	HSSKLQEDA	25	65
Herpes Simplex Virus Protease	LVLASSSGFY	26	
HIV Protease	GVSQNYPIVG	27	70
CMV Protease	GVVQASCRLA	28	
Thrombin	F (Pip) RS	29	75
Thrombin	DPRSFL	30	
Thrombin	PPRSFL	31	80
Caspase-3	DEV D	32	
Caspase-3	DEVDP	33	85
Caspase-3	KSGDVEG	34	

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TABLE 3-continued

Exemplary Proteases and Protease Recognition Sequences			SEQ ID NO:
Protease	Cleavage Domain Sequence		
Interleukin 1 β converting enzyme	GWEHDG	35	90
Enterokinase	EDDDDKA	36	
FAP	KQEONPGST	37	95
Kallikrein 2	GKAFRR	38	
Plasmin	DAFK	39	100
Plasmin	DVLK	40	
Plasmin	DAFK	41	105
TOP	ALLLALL	42	
MMP9 + matriptase	KPLGLQARVV	58	110
MMP9 + matriptase + uPA	PQASTGRSGG	59	
MMP9 + matriptase + uPA	PQGSTGRAAG	60	115
Matriptase + uPA	PPASSGRAGG	61	
MMP9 + matriptase	PIPVQGRAH	62	120
MMP9 + matriptase	PQGSTARSAG	909	

Proteases are known to be secreted by some diseased cells and tissues, for example tumor or cancer cells, creating a microenvironment that is rich in proteases or a protease-rich microenvironment. In some case, the blood of a subject is rich in proteases. In some cases, cells surrounding the tumor secrete proteases into the tumor microenvironment. Cells surrounding the tumor secreting proteases include but are not limited to the tumor stromal cells, myofibroblasts, blood cells, mast cells, B cells, NK cells, regulatory T cells, macrophages, cytotoxic T lymphocytes, dendritic cells, mesenchymal stem cells, polymorphonuclear cells, and other cells. In some cases, proteases are present in the blood of a subject, for example proteases that target amino acid sequences found in microbial peptides. This feature allows for targeted therapeutics such as antigen binding proteins to have additional specificity because T cells will not be bound by the antigen binding protein except in the protease rich microenvironment of the targeted cells or tissue.

The binding moiety comprising the cleavable linker thus masks the binding of a first or a second target antigen binding domain to their respective targets. In some embodiments, the binding moiety is bound to a first target antigen binding domain, which is further bound to a second target antigen binding domain, in the following order: binding moiety (M); cleavable linker (L); first target antigen binding domain (T1); second antigen binding domain (T2). In other examples, the domains are organized in any one of the following orders: M:L:T2:T1; T2:T1:L:M, T1:T2:L:M. The binding moiety is further bound to a half-life extending protein, such as albumin or any other of its targets as described below. In some instances, the binding moiety is albumin or comprises a binding site for albumin. In some instances the binding moiety comprises a binding site for IgE. In some embodiments, the binding moiety comprises a binding site for Igk free light chain.

TABLE 4

Exemplary sequences for the binding moieties comprising a cleavable linker are provided in SEQ ID Nos. 796-800.	
Sequence	
EVOLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGGGGLDGNE EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNLSRPE DTAVYYCTIGGSLSVSSQGTGLTVTVSSGGGGKPLGLQARVVGGGGT	SEQ ID No. 796
EVOLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGGGGLDGNE EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNLSRPE DTAVYYCTIGGSLSVSSQGTGLTVTVSSGGGGPQASTGRSGGGGGGT	SEQ ID No. 797
EVOLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGGGGLDGNE EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNLSRPE DTAVYYCTIGGSLSVSSQGTGLTVTVSSGGGGPQGSTGRAAGGGGGT	SEQ ID No. 798
EVOLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGGGGLDGNE EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNLSRPE DTAVYYCTIGGSLSVSSQGTGLTVTVSSGGGGPPASSGRAGGGGGT	SEQ ID No. 799
EVOLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGGGGLDGNE EPGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNLSRPE DTAVYYCTIGGSLSVSSQGTGLTVTVSSGGGGPIPVQGRAHGGGGT	SEQ ID No. 800

Targets of Conditionally Active Binding Proteins

The conditionally active binding proteins described herein are activated by cleavage of the at least one cleavable linker attached to the binding moieties within said conditionally active proteins. It is contemplated that in some cases the activated binding protein binds to a target antigen involved in and/or associated with a disease, disorder or condition. In particular, target antigens associated with a proliferative disease, a tumorous disease, an inflammatory disease, an immunological disorder, an autoimmune disease, an infectious disease, a viral disease, an allergic reaction, a parasitic reaction, a graft-versus-host disease or a host-versus-graft disease are contemplated to be the target for the activated binding proteins disclosed herein.

In some embodiments, the target antigen is a tumor antigen expressed on a tumor cell. Tumor antigens are well known in the art and include, for example, EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, and CEA. 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, ROR1, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1.

In some embodiments, the target antigen is an immune checkpoint protein. Examples of immune checkpoint proteins include but are not limited to CD27, CD137, 2B4, TIGIT, CD155, ICOS, HVEM, CD40L, LIGHT, OX40, DNAM-1, PD-L1, PD1, PD-L2, CTLA-4, CD8, CD40, CEACAM1, CD48, CD70, A2AR, CD39, CD73, B7-H3, B7-H4, BTLA, IDO1, IDO2, TDO, KIR, LAG-3, TIM-3, or VISTA.

In some embodiments, a target antigen is a cell surface molecule such as a protein, lipid or polysaccharide. In some embodiments, a target antigen is a on a tumor cell, virally

infected cell, bacterially infected cell, damaged red blood cell, arterial plaque cell, inflamed or fibrotic tissue cell.

In some embodiments, the target antigen comprises an immune response modulator that is not a cytokine.

5 Examples of immune response modulator include but are not limited to B7-1 (CD80), B7-2 (CD86), CD3, or GITR.

In some embodiments, the first target antigen binding domain or the second target antigen binding domain comprises an anti-EGFR domain, an anti-EpCAM domain, an anti-DLL3 domain, an anti-MSLN domain, an anti-PSMA domain, an anti-BDMA domain, or any combinations thereof. In some embodiments, the first target antigen binding domain or the second target antigen binding domain comprises an anti-EGFR sdAb, an anti-EpCAM sdAb, an anti-DLL3 sdAb, an anti-MSLN sdAb, an anti-PSMA sdAb, an anti-BDMA sdAb, or any combinations thereof.

In some embodiments, an anti-EGFR domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 55, and 737-785. In some embodiments, an anti-PSMA domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 57-73. In some embodiments, an anti-BCMA domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 91-214. In some embodiments, an anti-MSLN domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 215-258. In some embodiments, an anti-DLL3 domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 306-736. In some embodiments, an anti-EpCAM domain of this disclosure comprises an amino acid selected from the group consisting of SEQ ID Nos. 804-841.

In some embodiments, the first target antigen binding domain or the second target antigen binding domain comprises an anti-CD3 domain. In some embodiments, the anti-CD3 domain comprises an anti-CD3 scFv. In some embodiments, the anti-CD3 scFv comprises an amino acid sequence selected from the group consisting of: SEQ ID Nos. 74-90, and 794.

Binding Protein Variants

As used herein, the term "binding protein variants" refers to variants and derivatives of the conditionally active target-binding proteins described herein, containing a binding moiety as described above, comprising non-CDR loops that bind to an immunoglobulin binding molecule, such as a first or a second target antigen binding domain, a non-immunoglobulin binding molecule. In certain embodiments, amino acid sequence variants of the conditionally active target-binding proteins described herein are contemplated. For example, in certain embodiments amino acid sequence variants of the conditionally active target-binding proteins described herein are contemplated to improve the binding affinity and/or other biological properties of the binding proteins. Exemplary method for preparing amino acid variants include, but are not limited to, introducing appropriate modifications into the nucleotide sequence encoding the antibody, or by peptide synthesis. Such modifications include, for example, deletions from, and/or insertions into and/or substitutions of residues within the amino acid sequences of the antibody.

Any combination of deletion, insertion, and substitution can be made to the various domains to arrive at the final construct, provided that the final construct possesses the desired characteristics, e.g., antigen-binding. In certain embodiments, binding protein variants having one or more amino acid substitutions are provided. Sites of interest for substitution mutagenesis include the CDRs and framework

regions. Amino acid substitutions may be introduced into the variable domains of a conditionally active protein of interest and the products screened for a desired activity, e.g., retained/improved antigen binding, decreased immunogenicity, or improved antibody-dependent cell mediated cytotoxicity (ADCC) or complement dependent cytotoxicity (CDC). Both conservative and non-conservative amino acid substitutions are contemplated for preparing the antibody variants.

In another example of a substitution to create a variant conditionally active antibody, one or more hypervariable region residues of a parent antibody are substituted. In general, variants are then selected based on improvements in desired properties compared to a parent antibody, for example, increased affinity, reduced affinity, reduced immunogenicity, increased pH dependence of binding. For example, an affinity matured variant antibody can be generated, e.g., using phage display-based affinity maturation techniques such as those described herein and known in the field.

In another example, substitutions are made in hypervariable regions (HVR) of a parent conditionally active antibody to generate variants and variants are then selected based on binding affinity, i.e., by affinity maturation. In some embodiments of affinity maturation, diversity is introduced into the variable genes chosen for maturation by any of a variety of methods (e.g., error-prone PCR, chain shuffling, or oligo-nucleotide-directed mutagenesis). A secondary library is then created. The library is then screened to identify any antibody variants with the desired affinity. Another method to introduce diversity involves HVR-directed approaches, in which several HVR residues (e.g., 4-6 residues at a time) are randomized. HVR residues involved in antigen binding may be specifically identified, e.g., using alanine scanning mutagenesis or modeling. Substitutions can be in one, two, three, four, or more sites within a parent antibody sequence.

In some embodiments, a conditionally active binding protein, as described herein comprises a VL domain, or a VH domain, or both, with amino acid sequences corresponding to the amino acid sequence of a naturally occurring VL or VH domain, respectively, but that has been "humanized", i.e., by replacing one or more amino acid residues in the amino acid sequence of said naturally occurring VL or VH domains (and in particular in the framework sequences) by one or more of the amino acid residues that occur at the corresponding position(s) in a VL or VH domain from a conventional 4-chain antibody from a human being (e.g., as indicated above). This can be performed in a manner known in the field, which will be clear to the skilled person, for example on the basis of the further description herein. Again, it should be noted that such humanized conditionally active target-binding antibodies of the disclosure are obtained in any suitable manner known per se and thus are not strictly limited to polypeptides that have been obtained using a polypeptide that comprises a naturally occurring VL and/or VH domain as a starting material. In some additional embodiments, an conditionally active target-binding antibody, as described herein, comprises a VL and a VH domain with amino acid sequences corresponding to the amino acid sequence of a naturally occurring VL or VH domain, respectively, but that has been "camelized", i.e., by replacing one or more amino acid residues in the amino acid sequence of a naturally occurring VL or VH domain from a conventional 4-chain antibody by one or more of the amino acid residues that occur at the corresponding position(s) in a VL or a VH domain of a heavy chain antibody. Such "camelizing" substitutions are preferably inserted at amino acid positions that

form and/or are present at the VH-VL interface, and/or at the so-called Camelidae hallmark residues (see for example WO 94/04678 and Davies and Riechmann (1994 and 1996)). Preferably, the VH sequence that is used as a starting material or starting point for generating or designing the camelized single domain is preferably a VH sequence from a mammal, more preferably the VH sequence of a human being, such as a VH3 sequence. However, it should be noted that such camelized conditionally active antibodies of the disclosure, in certain embodiments, are obtained in any suitable manner known in the field and thus are not strictly limited to polypeptides that have been obtained using a polypeptide that comprises a naturally occurring VL and/or VH domain as a starting material. For example, both "humanization" and "camelization" is performed by providing a nucleotide sequence that encodes a naturally occurring VL and/or VH domain, respectively, and then changing, one or more codons in said nucleotide sequence in such a way that the new nucleotide sequence encodes a "humanized" or "camelized" conditionally active antibody, respectively. This nucleic acid can then be expressed, so as to provide the desired target-antigen binding capability. Alternatively, in other embodiments, a "humanized" or "camelized" conditionally active antibody is synthesized de novo using known peptide synthesis technique from the amino acid sequence of a naturally occurring antibody comprising a VL and/or VH domain. In some embodiments, a "humanized" or "camelized" conditionally active antibody is synthesized de novo using known peptide synthesis technique from the amino acid sequence or nucleotide sequence of a naturally occurring antibody comprising a VL and/or VH domain, respectively, a nucleotide sequence encoding the desired humanized or camelized conditionally active domain antibody of the disclosure, respectively, is designed and then synthesized de novo using known techniques for nucleic acid synthesis, after which the nucleic acid thus obtained is expressed in using known expression techniques, so as to provide the desired conditionally active antibody of the disclosure.

Other suitable methods and techniques for obtaining the conditionally active binding protein of the disclosure and/or nucleic acids encoding the same, starting from naturally occurring sequences for VL or VH domains for example comprises combining one or more parts of one or more naturally occurring VL or VH sequences (such as one or more framework (FR) sequences and/or complementarity determining region (CDR) sequences), and/or one or more synthetic or semi-synthetic sequences, and/or a naturally occurring sequence for a CH2 domain, and a naturally occurring sequence for a CH3 domain comprising amino acid substitutions that favor formation of heterodimer over homodimer, in a suitable manner, so as to provide a conditionally active binding protein of the disclosure or a nucleotide sequence or nucleic acid encoding the same.

Affinity Maturation

In designing conditionally active binding proteins for therapeutic applications, it is desirable to create proteins that, for example, modulate a functional activity of a target, and/or improved binding proteins such as binding proteins with higher specificity and/or affinity and/or binding proteins that are more bioavailable, or stable or soluble in particular cellular or tissue environments.

The conditionally active binding proteins described in the present disclosure exhibit improved the binding affinities towards the target, for example a tumor antigen expressed on a cell surface. In some embodiments, the conditionally active binding protein of the present disclosure is affinity matured to increase its binding affinity to the target, using

any known technique for affinity-maturation (e.g., mutagenesis, chain shuffling, CDR amino acid substitution). Amino acid substitutions may be conservative or semi-conservative. For example, the amino acids glycine, alanine, valine, leucine and isoleucine can often be substituted for one another (amino acids having aliphatic side chains). Of these possible substitutions, typically glycine and alanine are used to substitute for one another since they have relatively short side chains and valine, leucine and isoleucine are used to substitute for one another since they have larger aliphatic side chains which are hydrophobic. Other amino acids which may often be substituted for one another include but are not limited to: phenylalanine, tyrosine and tryptophan (amino acids having aromatic side chains); lysine, arginine and histidine (amino acids having basic side chains); aspartate and glutamate (amino acids having acidic side chains); asparagine and glutamine (amino acids having amide side chains); and cysteine and methionine (amino acids having sulphur-containing side chains). In some embodiments, the conditionally active target-binding proteins are isolated by screening combinatorial libraries, for example, by generating phage display libraries and screening such libraries for antibodies possessing the desired binding characteristics towards a target antigen, such as a tumor antigen expressed on a cell surface.

Conditionally Active Binding Protein Modifications

The conditionally active binding proteins described herein encompass derivatives or analogs in which (i) an amino acid is substituted with an amino acid residue that is not one encoded by the genetic code, (ii) the mature polypeptide is fused with another compound such as polyethylene glycol, or (iii) additional amino acids are fused to the protein, such as a leader or secretory sequence or a sequence to block an immunogenic domain and/or for purification of the protein.

Typical modifications include, but are not limited to, acetylation, acylation, ADP-ribosylation, amidation, covalent attachment of flavin, covalent attachment of a heme moiety, covalent attachment of a nucleotide or nucleotide derivative, covalent attachment of a lipid or lipid derivative, covalent attachment of phosphatidylinositol, cross-linking, cyclization, disulfide bond formation, demethylation, formation of covalent crosslinks, formation of cystine, formation of pyroglutamate, formylation, gamma carboxylation, glycosylation, GPI anchor formation, hydroxylation, iodination, methylation, myristylation, oxidation, proteolytic processing, phosphorylation, prenylation, racemization, selenoylation, sulfation, transfer-RNA mediated addition of amino acids to proteins such as arginylation, and ubiquitination.

Modifications are made anywhere in the conditionally active binding proteins described herein, including the peptide backbone, the amino acid side-chains, and the amino or carboxyl termini. Certain common peptide modifications that are useful for modification of the conditionally active binding proteins include glycosylation, lipid attachment, sulfation, gamma-carboxylation of glutamic acid residues, hydroxylation, blockage of the amino or carboxyl group in a polypeptide, or both, by a covalent modification, and ADP-ribosylation.

In some embodiments, the conditionally active binding proteins of the disclosure are conjugated with drugs to form antibody-drug conjugates (ADCs). In general, ADCs are used in oncology applications, where the use of antibody-drug conjugates for the local delivery of cytotoxic or cyto-

static agents allows for the targeted delivery of the drug moiety to tumors, which can allow higher efficacy, lower toxicity, etc.

Polynucleotides Encoding the Binding Moieties or the Conditionally Active Binding Proteins

Also provided, in some embodiments, are polynucleotide molecules encoding the binding moieties as described herein. In some embodiments, the polynucleotide molecules are provided as a DNA construct. In other embodiments, the polynucleotide molecules are provided as a messenger RNA transcript.

Also provided, in some embodiments, are polynucleotide molecules encoding the conditionally active binding proteins as described herein. In some embodiments, the polynucleotide molecules are provided as a DNA construct. In other embodiments, the polynucleotide molecules are provided as a messenger RNA transcript.

The polynucleotide molecules are constructed by known methods such as by combining the genes encoding the various domains (e.g. binding moiety, target antigen binding domain, etc.) either separated by peptide linkers or, in other embodiments, directly linked by a peptide bond, into a single genetic construct operably linked to a suitable promoter, and optionally a suitable transcription terminator, and expressing it in bacteria or other appropriate expression system such as, for example CHO cells. Depending on the vector system and host utilized, any number of suitable transcription and translation elements, including constitutive and conditionally active promoters, may be used. The promoter is selected such that it drives the expression of the polynucleotide in the respective host cell.

In some embodiments, the polynucleotides described herein are inserted into vectors, such as expression vectors, which represent further embodiments. This recombinant vector can be constructed according to known methods. Vectors of particular interest include plasmids, phagemids, phage derivatives, virii (e.g., retroviruses, adenoviruses, adeno-associated viruses, herpes viruses, lentiviruses, and the like), and cosmids.

A variety of expression vector/host systems may be utilized to contain and express the polynucleotide encoding the polypeptide of the described conditionally active binding protein. Examples of expression vectors for expression in *E. coli* are pSKK (Le Gall et al., J Immunol Methods. (2004) 285(1):111-27) or pcDNA5 (Invitrogen) for expression in mammalian cells.

Thus, the binding moieties or the conditionally active binding proteins comprising the binding moieties as described herein, in some embodiments, are produced by introducing vectors encoding the binding moieties or the binding proteins as described above into host cells and culturing said host cells under conditions whereby the binding moieties or the binding proteins, or domains thereof are expressed.

Pharmaceutical Compositions

Also provided, in some embodiments, are pharmaceutical compositions comprising a therapeutically effective amount of a conditionally active binding protein of the present disclosure, and at least one pharmaceutically acceptable carrier. The term "pharmaceutically acceptable carrier" includes, but is not limited to, any carrier that does not interfere with the effectiveness of the biological activity of the ingredients and that is not toxic to the patient to whom it is administered. Examples of suitable pharmaceutical carriers are well known in the art and include phosphate buffered saline solutions, water, emulsions, such as oil/water emulsions, various types of wetting agents, sterile solutions

etc. Such carriers can be formulated by conventional methods and can be administered to the subject at a suitable dose. Preferably, the compositions are sterile. These compositions may also contain adjuvants such as preservative, emulsifying agents and dispersing agents. Prevention of the action of microorganisms is, in some cases, ensured by the inclusion of various antibacterial and antifungal agents.

The conditionally active binding proteins described herein are contemplated for use as medicaments. Administration is effected by different ways, e.g., by intravenous, intraperitoneal, subcutaneous, intramuscular, topical or intradermal administration. In some embodiments, the route of administration depends on the kind of therapy and the kind of compound contained in the pharmaceutical composition. The dosage regimen will be determined by the attending physician and other clinical factors. Dosages for any one patient depends on many factors, including the patient's size, body surface area, age, sex, the particular compound to be administered, time and route of administration, the kind of therapy, general health and other drugs being administered concurrently. An "effective dose" refers to amounts of the active ingredient that are sufficient to affect the course and the severity of the disease, leading to the reduction or remission of such pathology and may be determined using known methods.

Methods of Treatment

Also provided herein, in some embodiments, are methods and uses for stimulating the immune system of an individual in need thereof comprising administration of a conditionally active binding protein as described herein. In some instances, administration induces and/or sustains cytotoxicity towards a cell expressing a target antigen. In some instances, the cell expressing a target antigen is a cancer or tumor cell, a virally infected cell, a bacterially infected cell, an autoreactive T or B cell, damaged red blood cells, arterial plaques, or fibrotic tissue. In some embodiments, the target antigen is an immune checkpoint protein.

Also provided herein are methods and uses for a treatment of a disease, disorder or condition associated with a target antigen comprising administering to an individual in need thereof a conditionally active binding protein as described herein. Diseases, disorders or conditions associated with a target antigen include, but are not limited to, viral infection, bacterial infection, auto-immune disease, transplant rejection, atherosclerosis, or fibrosis. In other embodiments, the disease, disorder or condition associated with a target antigen is a proliferative disease, a tumorous disease, an inflammatory disease, an immunological disorder, an autoimmune disease, an infectious disease, a viral disease, an allergic reaction, a parasitic reaction, a graft-versus-host disease or a host-versus-graft disease. In one embodiment, the disease, disorder or condition associated with a target antigen is cancer. In one instance, the cancer is a hematological cancer. In another instance, the cancer is a melanoma. In a further instance, the cancer is non-small cell lung cancer. In yet further instance, the cancer is breast cancer.

As used herein, in some embodiments, "treatment" or "treating" or "treated" refers to therapeutic treatment wherein the object is to slow (lessen) an undesired physiological condition, disorder or disease, or to obtain beneficial or desired clinical results. For the purposes described herein, beneficial or desired clinical results include, but are not limited to, alleviation of symptoms; diminishment of the extent of the condition, disorder or disease; stabilization (i.e., not worsening) of the state of the condition, disorder or disease; delay in onset or slowing of the progression of the condition, disorder or disease; amelioration of the condition,

disorder or disease state; and remission (whether partial or total), whether detectable or undetectable, or enhancement or improvement of the condition, disorder or disease. Treatment includes eliciting a clinically significant response without excessive levels of side effects. Treatment also includes prolonging survival as compared to expected survival if not receiving treatment. In other embodiments, "treatment" or "treating" or "treated" refers to prophylactic measures, wherein the object is to delay onset of or reduce severity of an undesired physiological condition, disorder or disease, such as, for example is a person who is predisposed to a disease (e.g., an individual who carries a genetic marker for a disease such as breast cancer).

In some embodiments of the methods described herein, the conditionally active binding proteins described herein are administered in combination with an agent for treatment of the particular disease, disorder or condition. Agents include but are not limited to, therapies involving antibodies, small molecules (e.g., chemotherapeutics), hormones (steroidal, peptide, and the like), radiotherapies (gamma-rays, X-rays, and/or the directed delivery of radioisotopes, microwaves, UV radiation and the like), gene therapies (e.g., antisense, retroviral therapy and the like) and other immunotherapies. In some embodiments, the conditionally active binding proteins described herein are administered in combination with anti-diarrheal agents, anti-emetic agents, analgesics, opioids and/or non-steroidal anti-inflammatory agents. In some embodiments, the conditionally active binding proteins described herein is administered before, during, or after surgery.

EXAMPLES

The examples below further illustrate the described embodiments without limiting the scope of the disclosure.

Example 1: Construction of an Exemplary Binding Moiety which Binds to Albumin and a Target Antigen Binding Domain Whose Target is EGFR

The sequence of an engineered protein scaffold comprising CDR loops capable of binding albumin and non-CDR loops is obtained. Overlapping PCR is used to introduce random mutations in the non-CDR loop regions, thereby generating a library. The resultant sequences are cloned into a phage display vector, thereby generating a phage display library. *Escherichia coli* cells are transformed with the library and used to construct a phage display library. ELISA is performed using an immobilized target antigen binding domain with specificity for EGFR. A clone with high specificity for EGFR is selected. Affinity maturation is performed by re-randomizing residues in the non-CDR loop regions as before.

Sequence alignment of non-CDR loop regions of the resultant proteins is performed to determine sequence conservation between proteins with high affinity for the EGFR binding target antigen binding domain. Site directed mutagenesis of one or more amino acids within these regions of sequence conservation is performed to generate additional proteins. Binding of the resultant proteins to an immobilized target antigen binding domain whose target is EGFR is measured in an ELISA. A protein with the highest affinity for the target antigen binding domain is selected.

The sequence of this binding moiety is cloned into a vector comprising a sequence for a cleavable linker, and sequences for a second target antigen binding domain that binds to a second target antigen, e.g., CD3. The resultant

vector is expressed in a heterologous expression system to obtain a conditionally active target binding protein that comprises a binding moiety comprising a cleavable linker and non-CDR loops which provide a binding site specific for the target antigen binding domain whose target is EGFR, and CDR loops which are specific for albumin.

Example 2: Construction of an Exemplary Binding Moiety which Binds to Albumin and a Target Antigen Binding Domain Whose Target is CD3

The sequence of an engineered protein scaffold comprising CDR loops capable of binding albumin and non-CDR loops is obtained. Overlapping PCR is used to introduce random mutations in the non-CDR loop regions, thereby generating a library. The resultant sequences are cloned into a phage display vector, thereby generating a phage display library. *Escherichia coli* cells are transformed with the library and used to construct a phage display library. ELISA is performed using an immobilized target antigen binding domain with specificity for CD3. A clone with high specificity for CD3 is selected. Affinity maturation is performed by re-randomizing residues in the non-CDR loop regions as before.

Sequence alignment of non-CDR loop regions of the resultant proteins is performed to determine sequence conservation between proteins with high affinity for the EGFR binding target antigen binding domain. Site directed mutagenesis of one or more amino acids within these regions of sequence conservation is performed to generate additional proteins. Binding of the resultant proteins to an immobilized target antigen binding domain whose target is CD3 is measured in an ELISA. A protein with the highest affinity for the target antigen binding domain is selected.

The sequence of this binding moiety is cloned into a vector comprising a sequence for a cleavable linker, and sequences for a second target antigen binding domain that binds to a second target antigen, e.g., EGFR. The resultant vector is expressed in a heterologous expression system to obtain a conditionally active target binding protein that comprises a binding moiety comprising a cleavable linker and non-CDR loops which provide a binding site specific for the target antigen binding domain whose target is CD3, and CDR loops which are specific for albumin.

Example 3: A Conditionally Active Binding Protein of this Disclosure Exhibit Reduced Specificity Toward Cell Line which Overexpresses EGFR but is Protease Deficient

Cells overexpressing EGFR and exhibiting low expression of a matrix metalloprotease are separately incubated with an exemplary conditionally active binding protein according to the present disclosure and a non-conditionally active control binding protein. Cells expressing normal levels of EGFR and proteases are also incubated with a conditionally active binding protein according to the present disclosure and a non-conditionally active control binding protein. Both proteins comprise a target antigen binding domain with specificity toward PSMA.

Results indicate that in the absence of protease secretion, the conditionally active binding protein of the present disclosure interacts with the protease expressing cells but does not interact with the EGFR expressed on the surface of the protease deficient cells. In contrast, the non-conditionally active control binding protein lacks the ability to selectively bind the protease expressing cells over the protease deficient

ones. Thus, the exemplary conditionally active binding protein receptor of the present disclosure is advantageous, for example, in terms of reducing off-tumor toxicity.

Example 4: Treatment with an Exemplary Conditionally Active Binding Protein of the Present Disclosure Inhibits In Vivo Tumor Growth

Murine tumor line CT26 is implanted subcutaneously in Balb/c mice and on day 7 post-implantation the average size of the tumor is measured. Test mice are treated with an exemplary conditionally active binding protein which has a target antigen binding domain specific for CTLA4 and another target antigen binding domain specific for CD3, wherein either the CTLA4 or the CD3 specific domain is bound to a binding moiety via its non-CDR loops, the binding moiety comprises a cleavable linker, and is bound to albumin. Control mice are treated with binding protein that contains CD3/CTLA4 specific domains but do not contain the binding moiety or the cleavable linker, and are not conditionally active. Results show that treatment with the exemplary conditionally active binding protein of the present disclosure inhibits tumor more efficiently than the comparator binding protein which does not contain the moiety with non-CDR loops.

Example 5: Exemplary Conditionally Active Binding Protein Exhibits Reduced Specificity Toward Cell Line which Overexpresses Antigen but is Protease Deficient

Cells overexpressing CTLA-4 and exhibiting low expression of a matrix metalloprotease are separately incubated with an exemplary CTLA4 specific conditionally active binding protein of this disclosure, containing a binding moiety that binds to a CTLA4 binding domain via its non-CDR loops and albumin via its CDRs; or a control CTLA-4 binding antibody which does not contain the binding moiety which binds to a CTLA4 binding domain via its non-CDR loops and to albumin via its CDRs. Cells expressing normal levels of antigens and proteases are also incubated with the exemplary CTLA4 specific conditionally active binding protein, or the control CTLA4 binding antibody.

Results indicate that in the absence of protease secretion, the conditionally active binding protein of the present disclosure binds the protease expressing cells but does not bind the protease-deficient antigen expressing cells. In contrast, the control antibody lacks the ability to selectively bind the protease expressing cells over the protease deficient ones. Thus, the exemplary conditionally active binding protein of the present disclosure is advantageous, for example, in terms of reducing off-tumor toxicity.

Example 6: Purification, Pharmacokinetic Analysis, and Potency Assays for Exemplary ProTriTAC Molecules

A) Expression, Purification and Stability of Exemplary ProTriTAC Molecules
Protein Production

Sequences of exemplary ProTriTAC (also referred to as Pro-trispecific) molecules were cloned into mammalian expression vector pcDNA 3.4 (Invitrogen) preceded by a leader sequence and followed by a 6× Histidine Tag (SEQ ID NO: 927). Expi293F cells (Life Technologies A14527) were maintained in suspension in Optimum Growth Flasks

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(Thomson) between 0.2 to 8×10^6 cells/ml in Expi 293 media. Purified plasmid DNA was transfected into Expi293 cells in accordance with Expi293 Expression System Kit (Life Technologies, A14635) protocols, and maintained for 4-6 days post transfection. Alternatively, sequences of trispecific molecules were cloned into mammalian expression vector pDEF38 (CMC ICOS) transfected into CHO-DG44 dhfr-cells, stable pools generated, and cultured in production media for up to 12 days prior to purification. The amount of the exemplary trispecific proteins in conditioned media was quantitated using an Octet RED 96 instrument with Protein A tips (ForteBio/Pall) using a control trispecific protein for a standard curve. Conditioned media from either host cell was filtered and partially purified by affinity and desalting chromatography. Trispecific proteins were subsequently polished by ion exchange and upon fraction pooling formulated in a neutral buffer containing excipients. Final purity was assessed by SDS-PAGE and analytical SEC using an Acquity BEH SEC 200 1.7u 4.6x150 mm column (Waters Corporation) resolved in an aqueous/organic mobile phase with excipients at neutral pH on a 1290 LC system and peaks integrated with Chemstation CDS software (Agilent). Trispecific proteins purified from CHO host cells were analyzed by running an SDS-PAGE, as shown in FIG. 5.

Stability Assessment

Purified Protrispecific proteins in two formulations were sub-aliquoted into sterile tubes and stressed by five freeze-thaw cycles each comprising greater than 1 hour at -80°C . and room temperature or by incubation at 37°C . for 1 week. Stressed samples were evaluated for concentration and turbidity by UV spectrometry using UV transparent 96 well plates (Corning 3635) with a SpectraMax M2 and SoftMax-Pro Software (Molecular Devices), SDS-PAGE, and analytical SEC and compared to the same analysis of control non-stressed samples. An overlay of chromatograms from analytical SEC of control and stressed samples for a single exemplary trispecific ProTriTAC molecule purified from 293 host cells is depicted in FIG. 6.

B) ProTriTAC Exhibits Potent, Protease-Dependent, Anti-Tumor Activity in a Rodent Tumor Xenograft Model

An exemplary ProTriTAC molecule (SEQ ID NO: 46) containing an EGFR binding domain as the target binding domain, a CD3 binding domain and an albumin binding domain comprising a masking moiety (SEQ ID NO: 50) and a cleavable linker (SEQ ID NO: 53) was evaluated for anti-tumor activity in vivo in an HCT116 subcutaneous xenograft tumor admixed with expanded human T cells in immunocompromised NCG mice. A non-cleavable EGFR targeting ProTriTAC molecule (SEQ ID NO: 47) and a GFP targeting ProTriTAC molecule (SEQ ID NO: 49) were also used in the study. Specifically, 5×10^6 HCT116 cells were admixed with 2.5×10^6 expanded T cells per mouse on day 0. Dosing of the test molecules (EGFR targeting ProTriTAC, non-cleavable EGFR targeting Pro-TriTAC, and GFP targeting ProTriTAC) were performed starting on the following day with a q.d.x10 (single daily dose for 10 days) schedule via intraperitoneal injection, at a dose of 0.03 mg/kg. Tumor volumes were determined using caliper measurements and calculated using the formula $V = (\text{length} \times \text{width} \times \text{width})/2$, at the indicated times. Results shown in FIG. 7 indicate that following final dose of test molecules, on day 10, tumor growth was arrested in mice administered the activatable EGFR targeting ProTriTAC molecule. Whereas, administering the GFP targeting ProTriTAC molecule was not effective in inhibiting tumor growth and the administration of the

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EGFR targeting non-cleavable ProTriTAC molecule was not as potent as the activatable ProTriTAC, in arresting tumor growth.

C) Demonstration of Functional Masking and Stability of ProTriTAC In Vivo in a Three-Week Cynomolgus Monkey Pharmacokinetic Study

Single doses of PSMA targeting ProTriTAC (SEQ ID NO: 43) containing a PSMA binding domain as the target binding domain, a CD3 binding domain, and an albumin binding domain comprising a masking moiety (SEQ ID NO: 50) and a cleavable linker (SEQ ID NO: 53), non-cleavable PSMA targeting ProTriTAC (SEQ ID NO: 44), non-masked/non-cleavable TriTAC (SEQ ID NO: 52), and active drug mimicking protease-activated PSMA targeting ProTriTAC (SEQ ID NO: 45) were dosed into cynomolgus monkeys at 0.1 mg/kg via intravenous injection. Plasma samples were collected at the time points indicated in FIG. 9. The designs of the above described test molecules are shown in FIG. 8. Concentrations of the various test molecules, as described above, were determined using ligand binding assays with biotinylated recombinant human PSMA (R&D systems) and sulfo-tagged anti-CD3 idiotype antibody cloned 11D3 in a MSD assay (Meso Scale Diagnostic, LLC). Pharmacokinetic parameters were estimated using Phoenix WinNonlin pharmacokinetic software using a non-compartmental approach consistent with the intravenous bolus route of administration.

To calculate the rate of in vivo conversion of the test molecules (i.e., conversion of PSMA targeting ProTriTAC, non-cleavable PSMA targeting ProTriTAC, non-masked/non-cleavable PSMA targeting ProTriTAC) the concentration of active drug in circulation was estimated by solving the following system of differential equations where P is the concentration of prodrug, A is the concentration of active drug, k_a is the rate of prodrug activation in circulation, $k_{c,P}$ is the clearance rate of the prodrug, and $k_{c,A}$ is the clearance rate of the active drug.

$$\frac{dP}{dt} = -k_{c,P}P$$

$$\frac{dA}{dt} = k_aP - k_{c,A}A$$

The clearance rates of the prodrug, active drug, non-masked non-cleavable prodrug control, and a non-cleavable prodrug control ($k_{c,NCLV}$) were determined empirically in cynomolgus monkeys. To estimate the rate of prodrug activation in circulation, it was assumed that the difference between the clearance rate of cleavable prodrug and the non-cleavable prodrug arose solely from non-specific activation in circulation. Therefore, the rate of prodrug conversion to active drug in circulation was estimated by subtracting the clearance rate of the cleavable prodrug from the non-cleavable prodrug.

$$k_a = k_{c,NCLV} - k_{c,P}$$

The initial concentration of prodrug in circulation was determined empirically and the initial concentration of active drug was assumed to be zero. Further calculations showed that the ProTriTAC comprising the protease cleavable linker was sufficiently stable in circulation, with 50% non-tumor mediated conversion every 194 hours and the $t_{1/2}$ of the molecule was determined, empirically, to be around 211 hours. This indicated that ProTriTAC molecules are sufficiently stable and protected against off-tumor effects. In contrast, the $t_{1/2}$ of the active drug fragment mimicking the

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activated ProTriTAC molecule was determined, empirically, to be 0.97 hours. Thus, active drug was rapidly cleared from circulation. Results are shown in FIG. 10.

FIG. 11 shows that because the active drug mimicking the activated ProTriTAC molecule was rapidly cleared from circulation and that the ProTriTAC and the control non-masked non-cleavable ProTriTAC had longer half-lives, there is a significant >200-fold differential between the ProTriTAC and the activated ProTriTAC circulating exposure. It was also observed that the masked but non-cleavable ProTriTAC control molecule was present in circulation for a longer time than the non-masked non-cleavable ProTriTAC control, thereby indicating that masking plays a role in increased circulation half-life ($t_{1/2}$) and limited peripheral T cell binding. The combination of the functional masking that renders ProTriTAC inert outside the tumor environment and the half-life differential that renders any aberrantly activated ProTriTAC in circulation to be rapidly cleared in vivo works together to ensure on-target, off-tumor activity of ProTriTAC is minimized. Supporting pharmacokinetic parameters are shown in Table 5.

TABLE 5

Pharmacokinetics of ProTriTAC, Activated ProTriTAC and Control Molecules				
Test Article	Terminal $t_{1/2}$ (hr)	C_{max} (nM)	AUC, 0-last (hr*nM)	Clearance (mL/hr/kg)
Control #1 (No mask, non-cleavable)	118	48.2	2490	0.735
Control #2 (Masked, non-cleavable)	211	58.0	7000	0.238
ProTriTAC	101	42.7	2670	0.686
Activated ProTriTAC	0.969	66.6	41.5	58.5

D) Protease Activation of ProTriTAC Molecule Leads to Significantly Enhanced Activity In Vitro

The aim of this study was to assess the relative potency of protease activatable ProTriTAC molecules, non-cleavable ProTriTAC molecules and recombinant active drug fragment mimicking the protease-activated ProTriTAC molecule, in CD3 binding and T cell mediated cell killing. The active drug fragment mimicking the protease activated ProTriTAC molecule contained the CD3 binding domain and the target antigen binding domain but lacked the albumin binding domain. Whereas the protease activatable ProTriTAC molecule contained the albumin binding domain comprising a masking domain and a protease cleavable site, the CD3 binding domain, and the target antigen binding domain. The non-cleavable ProTriTAC molecule lacked the protease cleavable site but was otherwise identical to the protease activatable ProTriTAC molecule.

Purified ProTriTAC (labeled as prodrug in FIGS. 12-14), non-cleavable ProTriTAC (labeled as prodrug (non-cleavable) in FIGS. 12-14), and recombinant active drug fragment mimicking the protease-activated ProTriTAC (labeled as active drug in FIGS. 12-14) were tested for binding to recombinant human CD3 in an ELISA assay (FIG. 12), binding to purified human primary T cells in a flow cytometry assay (FIG. 13), and functional potency in a T cell-dependent cellular cytotoxicity assay (FIG. 14).

For ELISA, soluble test molecules (i.e., active drug, prodrug, and prodrug (non-cleavable)) at the indicated concentrations were incubated, in multi-well plates, with immobilized recombinant human CD3e (R&D Systems) for 1 hour at room temperature in PBS supplemented with 15 mg/mL human serum albumin. Plates were blocked using

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SuperBlock (Thermo Fisher), washed using PBS with 0.05% Tween-20, and detected using a non-competitive anti-CD3 idiotype monoclonal antibody 11D3 followed by peroxidase-labeled secondary antibody and TMB-ELISA substrate solution (Thermo Fisher). Results shown in FIG. 12 demonstrate that the active drug fragment mimicking the protease-activated ProTriTAC molecule was about 250 times more potent in binding CD3 as compared to prodrug non-cleavable. The EC_{50} values are provided in Table 6. The masking ratio is the ratio between the prodrug EC_{50} over the active drug EC_{50} ; the higher the number, the higher the fold-shift between prodrug and active drug and thus greater functional masking.

TABLE 6

CD3 binding potential		
	EC_{50} (nM)	Masking Ratio
Active Drug	0.16	—
Prodrug	11.91	74
Prodrug (non-cleavable)	39.44	247

For binding to human primary T cells, determined by flow cytometry, soluble test molecules (i.e., active drug, prodrug, and prodrug (non-cleavable)) at the indicated concentrations (shown in FIG. 13) were incubated, in multi-well plates, with purified human primary T cells for 1 h at 4° C. in the presence of PBS with 2% fetal bovine serum and 15 mg/ml human serum albumin. Plates were washed with PBS with 2% fetal bovine serum, detected using AlexaFluor 647-labeled non-competitive anti-CD3 idiotype monoclonal antibody 11D3, and data was analyzed using FlowJo 10 (FlowJo, LLC). Results shown in FIG. 13 demonstrate that the active drug fragment mimicking the protease-activated ProTriTAC molecule was greater than 1000 times more potent in binding human primary T cells as compared to prodrug non-cleavable. The EC_{50} values are provided in Table 7.

TABLE 7

Human primary T cell binding potential		
	EC_{50} (nM)	Masking Ratio
Active Drug	1.19	—
Prodrug	>1000	n/a
Prodrug (non-cleavable)	>1000	n/a

For functional potency in a T cell-dependent cellular cytotoxicity assays, soluble test molecules (i.e., active drug, prodrug, and prodrug (non-cleavable)) at the indicated concentrations, shown in FIG. 14, were incubated, in multi-well plates, with purified resting human T cells (effector cell) and HCT116 cancer cell (target cell) at 10:1 effector:target cell ratio for 48 h at 37° C. The HCT116 target cell line had been stably transfected with a luciferase reporter gene to allow specific T cell-mediated cell killing measurement by ONE-Glo (Promega). Results shown in FIG. 14 demonstrate that the active drug fragment mimicking the protease-activated ProTriTAC molecule was about 500 times more potent in T cell mediated killing of cancer cells, as compared to prodrug non-cleavable. The EC_{50} values are provided in Table 8.

TABLE 8

T cell mediated cell killing potential		
	EC ₅₀ (nM)	Masking Ratio
Active Drug	0.004	—
Prodrug	0.485	121
Prodrug (non-cleavable)	2.197	549

Example 7: Anti-Tumor Activity of Exemplary ProTriTAC Molecules Containing Various Exemplary Linkers, in an Admixed Mouse Tumor Model

The aim of this study was to explore the anti-tumor activity of ProTriTAC molecules containing different linkers. NSG female mice, 7 weeks old, were used for this study. At the commencement of the study, on day 0, the NSG female mice were injected with 2.5×10^6 expanded human T cells, and 5×10^6 HCT116 (human colorectal carcinoma) tumor cells. The following day, on day 1, the mice were divided into groups and each group was treated with at least one of the ProTriTAC molecules listed in Table 9 (SEQ ID Nos. 786-790), or with a control GFP TriTAC molecule (SEQ ID No. 792), or with a ProTriTAC molecule that contains a non cleavable linker (NCLV) (SEQ ID No. 791).

The ProTriTAC molecules and the ProTriTAC NCLV molecule used in the following examples were targeted to EGFR and had the following orientation of the individual domains: (anti-albumin binding domain (sdAb): anti-CD3 domain (scFv): anti-EGFR domain (sdAb)). The only differences between the ProTriTAC molecules listed in Table 6 were in the linker sequences. The ProTriTAC molecules, ProTriTAC NCLV molecule, or the GFP TriTAC molecule (the GFP TriTAC molecule had the following orientation of individual domains: anti-GFP sdAb: anti-Alb sdAb: anti-CD3 scFv) were administered daily for a period of 10 days (i.e., final dose was administered on day 10 following injection of tumor cells and expanded cells to the animals) and tumor volumes were measured at regular intervals, beginning a few days prior to the administration of the final dose at day 10.

TABLE 9

ProTriTAC sequences and linkers			
ProTriTAC molecule sequence	Linker in the ProTriTAC molecule	Linker Sequence	Cleavability/ Recognition of the linker sequence by enzymes
SEQ ID No. 786	L001 (SEQ ID NO: 58)	KPLGLQARVV	MMP9 + matriptase
SEQ ID No. 787	L040 (SEQ ID NO: 59)	PQASTGRSGG	MMP9 + matriptase + uPA
SEQ ID No. 788	L041 (SEQ ID NO: 60)	PQGSGRAAG	MMP9 + matriptase + uPA
SEQ ID No. 789	L042 (SEQ ID NO: 61)	PPASSGRAGG	matriptase + uPA
SEQ ID No. 790	L045 (SEQ ID NO: 62)	PIPVQGRAH	MMP9 + matriptase
	L043 (SEQ ID NO: 909)	PQGSTARSAG	MMP9 + matriptase

As shown in FIG. 18, the ProTriTAC molecules containing linker sequences L001, L045, L040, and L041 demonstrated more potent anti-tumor activity as compared to the GFP TriTAC control of the ProTriTAC NCLV molecule. The statistical significance of the data was determined by repeated measures one-way ANOVA-Dunnett post test. The mean tumor volume of each group of mice were compared to the mean tumor volume of the mice group that received the NCLV molecule.

The pharmacokinetics following administration of the various molecules, as described above, were also assessed and the data is shown in FIG. 19. The control GFP TriTAC molecule was rapidly cleared from circulation, following administration, whereas the NCLV molecule remained in circulation for the longest time. The pharmacokinetic clearance profile of the test ProTriTAC molecules were in between the GFP TriTAC and NCLV, except for the ProTriTAC containing the linker sequence L001, which was cleared almost as rapidly as the control GFP TriTAC.

Example 8: Individual Tumor Volumes of Admixed Xenograft Tumors, Following Treatment with Exemplary ProTriTAC Molecules Containing Various Exemplary Linkers

The ProTriTAC molecules listed in Table 9, the control GFP TriTAC molecule, and the ProTriTAC NCLV molecule were evaluated in an admixed xenograft model, in order to determine the efficacy of the ProTriTAC molecules containing different linkers, in vivo. As described in previous example (Example 7), the xenograft tumor model was generated by injecting 7 week old NSG mice with 2.5×10^6 expanded human T cells, and 5×10^6 HCT116 (human colorectal carcinoma) tumor cells. The mice were divided into groups and each group was treated with at least one of the ProTriTAC molecules listed in Table 9, with the control GFP TriTAC molecule, or with the ProTriTAC NCLV molecule. Tumor volumes were measured at regular intervals, starting from day 10 post injection of tumor cells and expanded T cells.

It was observed that in animals treated with the exemplary ProTriTAC molecules containing linker L040 there was a statistically significant delay in tumor growth as compared to the mice group which was treated with the control GFP TriTAC molecule, or the mice group that was treated with the ProTriTAC NCLV molecule. Similar observation was also made for the ProTriTAC molecules containing linker sequences L001, L041, and L045. The data is shown in FIG. 20.

It is also possible to carry out a similar study with xenograft models using other cell lines, such as A549 (non-small cell lung carcinoma) cells, DU-145 (prostate) cells, MCF-7 (breast) cells, Colo 205 (colon) cells, 3T3/J GF-IR (mouse fibroblast) cells, NCI H441 cells, HEP G2 (hepatoma) cells, MDA MB 231 (breast) cells, HT-29 (colon) cells, MDA-MB-435s (breast) cells, U266 cells, SH-SY5Y cells, Sk-Mel-2 cells, NCI-H929, RPM18226, and A431 cells.

Example 9: Demonstration of Reduced Cytokine Levels in Cynomolgus Monkeys, Correlated with Masking of TriTAC Molecules

In this study, cynomolgus monkeys were treated with three different concentrations (30 μ g/kg; 300 μ g/kg; and 1000 μ g/kg) of an exemplary EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC

(NCLV), or with three different concentrations (10 µg/kg; 30 µg/kg; and 100 µg/kg) of an exemplary EGFR targeting TriTAC molecule (SEQ ID No. 793).

As shown in FIG. 21, after 4 hours following administering the ProTriTAC (NCLV) molecule, IFN-gamma IL-6 and IL-10 levels were significantly lower in comparison with administering the EGFR targeting TriTAC molecule.

Example 10: Demonstration of Improved Tolerability in Mouse, Conferred by an Exemplary EGFR Targeting ProTriTAC Molecule

In this study, the tolerability of an exemplary EGFR targeting ProTriTAC molecule was assessed. Seven weeks old NSG female tumor free mice were intraperitoneally injected with 2×10^7 expanded human T cells at the commencement of the study, i.e., at day 0. On day 2, treatment was started by dividing the mice into various groups and administering to them varying concentrations of the exemplary EGFR targeting ProTriTAC molecule, containing the linker sequence L001, an EGFR targeting TriTAC molecule, and an EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)). The molecules were administered once daily for 10 days, at the following dosages: 30 µg/kg, 100 µg/kg, 300 µg/kg. Starting from day 2, body weight of the animals were recorded daily.

As shown in FIG. 22, the EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) and a GFP TriTAC (used as a negative control) were very well tolerated in mice even at the highest dose of 1000 µg/kg. The EGFR targeted ProTriTAC molecule containing the linker sequence of L001 was well tolerated at the dosage of 100 µg/kg, whereas the EGFR targeted TriTAC was well tolerated at 30 µg/kg. It was thus observed that the ProTriTAC containing the L001 linker sequence conferred about 3 fold increase in tolerability and the ProTriTAC (NCLV) conferred about a 30 fold increase in tolerability, in mouse. The mouse maximum tolerated dose for the ProTriTAC (NCLV) and TriTAC molecules was consistent with what was observed in cynomolgus monkeys.

To further explore the role of the linker in tolerability of the EGFR targeting ProTriTAC molecule in mouse, the linker sequence was changed from L001 to that of L040. In this experiment, seven weeks old NSG female tumor free mice were subcutaneously injected with 5×10^6 HCT116 tumor cells, at the commencement of the study, i.e., at day 0. At day 7 following the tumor cell injection, when the tumor volumes were about 180-200 mm³ (e.g., 183 mm³), the mice were injected intraperitoneally with 2×10^7 expanded human T cells. Treatment was started on day 9, by dividing the mice into various groups and each group was administered an EGFR targeting TriTAC molecule, an EGFR targeting ProTriTAC molecule with linker sequence L040 (ProTriTAC(L040)), and a ProTriTAC molecule containing a non-cleavable linker (ProTriTAC(NCLV)). The molecules were administered once daily for 10 days, at the following dosages: 300 µg/kg and 1000 µg/kg. Starting from day 2, body weight of the animals were recorded daily. The results shown in FIG. 23 provide that the EGFR targeting ProTriTAC molecule with the linker sequence L040 conferred better tolerability than when the linker sequence L001 was used. The ProTriTAC (L040) was well-tolerated at 300 µg/kg and at 1000 µg/kg, with body weight percentage change comparable to that with the ProTriTAC (NCLV) molecule. It was thus observed that compared to the TriTAC molecule, the ProTriTAC containing the L040 linker sequence conferred about a 30 fold increase in tolerability,

in mouse, similar to the 30 fold increase in tolerability observed with the ProTriTAC (NCLV) molecule.

Example 11: Clinical Pathology Studies in Mice and Cynomolgus Monkeys

In this study, mice were treated with various concentrations of an EFGR targeting TriTAC molecule, an EGFR targeting ProTriTAC molecule containing the linker sequence L001 (ProTriTAC (L001)), and an EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC(NCLV)). Tolerability was assessed by measuring serum concentration of ALT (alanine aminotransferase) and AST (aspartate aminotransferase). Results are shown in FIGS. 24A, 24B, and 24C and FIGS. 25A, 25B, and 25C. It was observed that serum concentrations of AST and ALT were not elevated following administering the ProTriTAC (L001) at dosages up to 0.3 mg/kg, and that the serum concentrations of AST and ALT were not elevated following administering the ProTriTAC(NCLV) at dosages up to 1 mg/kg. In contrast, serum concentration of AST and ALT were not elevated following administering the TriTAC molecule at a dosage of 0.3 mg/kg.

In another study, cynomolgus monkeys were treated with various concentrations of an EFGR targeting TriTAC molecule, and an EGFR targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC(NCLV)). Tolerability was assessed by measuring serum concentration of ALT (alanine aminotransferase) and AST (aspartate aminotransferase). Results are shown in FIG. 26. It was observed that serum concentrations of AST and ALT were not elevated following administering the ProTriTAC (NCLV) at dosages up to 1000 µg/kg. In contrast, serum concentration of AST and ALT were not elevated following administering the TriTAC molecule at a dosage of 10 µg/kg.

Example 12: Demonstration of Therapeutic Window Expansion with an Exemplary ProTriTAC Molecule of this Disclosure in a Tumor-Bearing Mouse Model

The aim of this study was to evaluate the expansion of therapeutic window by measuring anti-tumor activity and observable on-target toxicity in the same tumor-bearing mice. NSG female mice, 7 weeks old, were used for this study.

At the commencement of the study, on day 0, the NSG female mice were injected with 2.5×10^6 expanded human T cells, and 5×10^6 HCT116 (human colorectal carcinoma) tumor cells. The following day, on day 1, the mice were divided into groups and each group was treated with either GFP TriTAC molecule (SEQ ID No. 792), EGFR TriTAC molecule (SEQ ID No. 793), or an EGFR targeting ProTriTAC molecule containing linker L040, (SEQ ID No. 787) at the indicated dose levels in FIGS. 27A-297D (for GFP TriTAC the dosage was 300 µg/kg; for EGFR TriTAC dosages were 10 µg/kg, 30 µg/kg, 100 µg/kg, and 300 µg/kg; for EGFR ProTriTAC dosages were 30 µg/kg, 100 µg/kg, 300 µg/kg, and 1000 µg/kg) and administered daily for a period of 10 days (i.e., final dose was administered on day 10 following injection of tumor cells and expanded cells to the animals) and tumor volumes were measured at regular intervals, beginning a few days prior to the administration of the final dose at day 10. Results shown in FIGS. 27A-27D.

On-target EGFR-related toxicity was determined by measuring the radius of the red scarring skin lesion above the original tumor implantation site with a caliper and applying

the equation $\text{Area} = \pi \times (\text{radius of lesion})^2$ on day 14. Results provided in FIG. 28 show that an exemplary EGFR ProTriTAC has 30× better tolerability compared to an EGFR TriTAC in the same HCT116 tumor-bearing mice as measured by the onset of red scarring skin lesions above the original tumor implantation site. Therapeutic window is defined as the difference between the minimal dose level required for anti-tumor activity and the highest skin lesion-free dose level.

Results (from FIGS. 27 and 28) show that the exemplary protease-cleavable EGFR ProTriTAC is 3× less potent but 30× more tolerated (i.e., has a 10× improved therapeutic window) than the EGFR TriTAC when efficacy and toxicity are measured on the same tumor-bearing mice, as summarized in below Table. This would make it possible to dose the ProTriTAC at a dose that is about 3× higher than the TriTAC, to get at least the same efficiency and better tolerability.

	Min efficacious dose	Max scar-free dose	Therapeutic Window
TriTAC	10 µg/kg	10 µg/kg	1
ProTriTAC	30 µg/kg	300 µg/kg	10

**Example 13: An Exemplary ProTriTAC Molecule
Containing a Binding Moiety with Extended
Non-CDR Loop into which a Human CD3ε
Epitope is Grafted**

The sequence of a binding moiety comprising non-CDR loops (AB, EF, C'D, and CC') was obtained. A portion of the human CD3ε sequence was grafted into the CC' loop of the non-CDR loops within the binding moiety, along with glycine residues to further extend the CC' loop. FIG. 29 illustrates three different variants comprising 10, 12, or 16 amino acid extensions to the CC' loop. In case of the variant CC10, the portion of human CD38 sequence grafted into the CC' loop, to replace the wild type sequence of APGKG (SEQ ID NO: 795) was QDGNEE (SEQ ID NO: 801), and in addition 4 glycine residues were inserted to extend the CC' loop. In case of the variant CC12, the portion of human CD3ε sequence grafted into the CC' loop, to replace the wild type sequence of APGKG (SEQ ID NO: 795) was QDGNEEMGG (SEQ ID No. 802), and in addition 3 glycine residues were inserted to extend the CC' loop. In case of the variant CC16, the portion of human CD3ε sequence grafted into the CC' loop, to replace the wild type sequence of APGKG (SEQ ID NO: 795) was QDGNEEMGG (SEQ ID No. 803), and in addition 7 glycine residues were inserted to extend the CC' loop. The binding moiety comprising extended non-CDR loops comprising the CD38 sequences, as described above, were cloned into a vector further comprising coding sequences for a protease cleavable linker, a scFv containing a CD3 binding domain, and an EGFR binding domain, to express a ProTriTAC molecule. The ProTriTAC molecule contained an exemplary binding moiety of this disclosure, a CD3 binding scFv, and an EGFR binding domain. The ProTriTAC molecule was subsequently exposed to a tumor associated protease, matriptase, to assay activation of the molecule upon cleavage of the protease cleavable linker, which separates the binding moiety (depicted as aALB in FIGS. 29 and 30) comprising the cleavable linker from the rest of the molecule, i.e., the scFv containing the CD3 binding domain (depicted as aCD3 in FIGS. 29 and 30) and the EGFR

binding domain (aEGFR). FIG. 30 shows the activation of the ProTriTAC molecules, containing the CC10, CC12, or CC16 variants of CC' non-CDR loop, CD3 binding scFv in a VH-VL (left panel) or a VL-VH (right panel) format, upon treatment with matriptase. A ProTriTAC molecule containing a wild-type CC' loop in the binding moiety was used as a control for the protease activation assay. In addition, a TriTAC molecule that is not in the "pro" form, i.e., a molecule that includes the same domains as the ProTriTAC molecule, except that it has a half-life extension domain, such as albumin, instead of a binding moiety, was also treated with matriptase and used as a control. Results indicated that the ProTriTAC molecules were activated upon cleavage, to generate a free albumin binding domain (depicted as Free aALB in FIG. 30) whereas the albumin domain did not separate from the TriTAC molecules. Thus, the ProTriTAC molecules containing a binding moiety of this disclosure was able to readily dissociate from half-life extending domain upon cleavage in a tumor microenvironment, unlike the TriTAC versions, and thereby were amenable to rapid clearance from the systemic circulation upon activation.

Further studies were carried out to assay the binding of the binding moiety containing the human CD38 to CD3. It was observed that, in the presence of human serum albumin, the activated forms of ProTriTAC molecules that contained the binding moiety containing the human CD3ε were about 20 times potent in binding CD3 than their activated forms which did not contain the binding moiety. Results are shown in FIG. 31.

Cell killing potential of a ProTriTAC molecule that contained a binding moiety as described herein was also assayed in a study where CaOV4 cell line was treated with the ProTriTAC molecule or its activated form in the presence of human serum albumin. As shown in FIG. 32, the ProTriTAC molecule containing the CC16 variant of CC' non-CDR loop was about 50 times more potent in killing cancer cells compared to its activated form, which was separated from the albumin binding domain. The total observed masking is a combination of steric masking due to binding to human serum albumin and specific masking from the CC16 mask in the CC' non-CDR loop.

**Example 14: Soft Library Mutagenesis Identified
CC' Loop as Most Amenable to Modification**

To identify locations within the non-CDR loops (AB, CC', C'D, and EF) that were most amenable to modification, for creating a masking capability, libraries were assembled and generated using four groups of overlapping DNA oligos containing randomized degenerate "NNK" codons and with different loop lengths, as indicated in the schematic below:

AB loop oligos:

WT: LVQPGN (20%) (SEQ ID NO: 903)

AB0: XXXXXX (20%)

AB1: XXXXXX (20%)

AB2: XXXXXXXX (20%)

AB3: XXXXXXXXXX (20%)

CC' loop oligos:

WT: APGKG (20%) (SEQ ID NO: 795)

CC0: XXXXX (20%)

CC1: XXXXXX (20%)

CC2: XXXXXXXX (20%)

CC3: XXXXXXXXXX (20%)

C'D loop oligos:

WT: DSVKGR (20%) (SEQ ID NO: 904)

CD0: XXXXXX (20%)

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CD1: XXXXXXXX (20%)
 CD2: XXXXXXXX (20%)
 CD3: XXXXXXXX (20%)
 EF loop oligos:
 WT: SLRPED (20%) (SEQ ID NO: 905)
 EF0: XXXXXXXX (20%)
 EF1: XXXXXXXX (20%)
 EF2: XXXXXXXX (20%)
 EF3: XXXXXXXXXX (20%)

Note: "X" denotes a randomized residue ("NNK" codon) that could be any of the 20 natural amino acids as well as stop codon. The goal was to have approximately 20% of each non-CDR loop be wild-type. These wild-type oligos served as internal benchmarks to gauge the tolerance of each loop to modification (sequence composition and/or length changes). A loop that was less tolerable to change could easily revert to wild-type; in contrast, a loop that was highly amenable to change would maintain the diverse sequence repertoire. To this end, 24 clones were sequenced from the naive library to verify the randomization of non-CDR loops prior to panning with HAS, as shown in FIG. 33. After two rounds of phage panning against HSA, 30 clones were sequenced to examine the non-CDR loop composition. The results showed that 3 out of 4 non-CDR loops (AB, C'D, and EF) mostly utilized the 20% wild-type oligo and reverted back to the wild-type, suggesting that they may be less preferred than the wild-type sequence. However, the CC' loop kept the diverse sequence repertoire (both sequence and length), suggesting that the CC' loop may be the loop most tolerable to randomization that we could exploit for specific masking of adjacent domains, as shown in FIG. 34.

Example 15: Screening of Phage Display Library for Identification of EpCAM Binding Domains

Llamas were immunized with purified EpCAM protein expressed in Expi293 cells. A phage display library for expression of heavy variable antibody domains was constructed from circulating B cells. See van der Linden, de Geus, Stok, Bos, van Wassenaar, Verrips, and Frenken. 2000. J Immunol Methods 240:185-195. Phage clones were screened for binding to EpCAM by expressing anti-EpCAM proteins in *E. coli*, preparing periplasmic extracts, and proteins were screened for human and cynomolgus EpCAM binding activity using a colorimetric ELISA. Thirty-eight unique heavy chain only sequences were identified (SEQ ID Nos. 804-841) that produced a signal in the ELISA screening relative to the control with human and/or cynomolgus EpCAM proteins (as shown in Table 10).

TABLE 10

Binding of Llama anti-Human EpCAM heavy chain only single domain antibodies to Human and Cynomolgus EpCAM, as demonstrated by signal in an ELISA Assay (absorbance readings in a colorimetric ELISA assay), relative to control heavy chain only single domain antibodies					
Sequence name	ELISA Human EpCAM	ELISA Cynomolgus EpCAM	ELISA Control	Human EpCAM/ Control	Cynomolgus EpCAM/ Control
EPL90	1.6	1.7	0.2	10	10
EPL118	3.4	2.9	0.7	5	4
EPL138	3.1	2.7	0.9	3	3
EPL145	0.6	0.4	0.2	3	2
EPL164	0.6	2.8	0.1	7	34

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TABLE 10-continued

Binding of Llama anti-Human EpCAM heavy chain only single domain antibodies to Human and Cynomolgus EpCAM, as demonstrated by signal in an ELISA Assay (absorbance readings in a colorimetric ELISA assay), relative to control heavy chain only single domain antibodies					
Sequence name	ELISA Human EpCAM	ELISA Cynomolgus EpCAM	ELISA Control	Human EpCAM/ Control	Cynomolgus EpCAM/ Control
EPL31	0.6	0.5	0.1	7	6
EPL55	0.4	0.6	0.1	5	7
EPL57	1.7	3.4	0.1	14	27
EPL136	0.9	1.3	0.1	9	13
EPL15	0.7	3.2	0.1	8	35
EPL34	0.9	3.1	0.1	10	36
EPL86	0.8	3.0	0.1	8	31
EPL153	3.1	2.2	0.1	31	21
EPL20	2.9	2.2	0.5	6	4
EPL70	3.2	2.2	0.1	24	16
EPL125	3.5	3.5	0.5	8	8
EPL13	2.9	4.0	0.2	17	23
EPL129	2.0	3.1	0.1	20	29
EPL159	0.4	0.2	0.1	3	2
EPL120	3.2	2.3	0.6	5	4
EPL126	3.4	2.8	0.7	5	4
EPL60	1.1	3.5	0.1	10	33
EPL156	3.6	4.0	0.2	16	17
EPL2	2.2	3.6	0.1	15	24
EPL43	1.8	3.8	0.1	13	27
EPL10	2.7	2.0	0.2	12	9
EPL49	1.1	0.6	0.1	8	4
EPL58	1.1	0.5	0.1	8	4
EPL74	1.4	0.7	0.1	10	5
EPL78	3.1	2.0	0.2	17	11
EPL82	0.9	1.1	0.2	4	5
EPL83	2.0	1.1	0.6	3	2
EPL97	2.5	1.7	0.3	8	5
EPL109	0.4	0.1	0.1	5	2
EPL117	0.9	0.6	0.2	4	3
EPL127	0.4	0.9	0.1	5	11
EPL152	3.4	2.8	0.8	4	4
EPL189	1.3	0.9	0.1	12	8

Example 16: Incorporation of EpCAM Binding Heavy Chain Only Single Domain Antibodies into Fusion Proteins and T Cell Dependent Cellular Cytotoxicity Assays

Selected anti-EpCAM heavy chain only single domain antibodies from Example 15 were cloned into DNA constructs for expression of recombinant proteins. These expression constructs all encoded a signal peptide. One set of anti-EpCAM constructs (SEQ ID Nos. 842 to 868) was designed to express a fusion protein with a humanized anti-CD3 scFv domain on the N-terminus of the mature secreted fusion protein followed by a llama anti-EpCAM domain, with the two domains linked by the sequence GGGGSGGGS (SEQ ID NO: 928), and with a HHHHHH (SEQ ID NO: 927) on the C-terminus. One second of anti-EpCAM constructs (SEQ ID Nos. 869 to 895) was designed to express a fusion protein with a llama anti-EpCAM domain on the N-terminus of the mature secreted fusion protein followed a humanized anti-CD3-scFv domain, with the two domains linked by the sequence GGGGSGGGS (SEQ ID NO: 928), and with a HHHHHH (SEQ ID NO: 927) on the C-terminus.

These anti-EpCAM/anti-CD3 (from N-terminus to C-terminus) or anti-CD3/anti-EpCAM (from N terminus to C terminus) fusion protein constructs were transfected into

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Expi293 cells. The amount of anti-EpCAM/anti-CD3 fusion protein in the conditioned media from the transfected Expi293 cells was quantitated using by using an Octet instrument with streptavidin and loaded with biotinylated CD3-Fc fusion protein using an anti-CD3 fusion protein of similar molecular weight to the anti-EPCAM/anti-CD3 proteins as a standard.

The conditioned media were tested in a T-cell dependent cellular cytotoxicity assay. See Nazarian A A, Archibeque I L, Nguyen Y H, Wang P, Sinclair A M, Powers D A. 2015. J Biomol Screen. 20:519-27. In this assay, luciferase labelled NCI-H508 cells, which express EpCAM, were combined with purified human T cells and a titration of the anti-EpCAM/anti-CD3 fusion protein or the anti-CD3/anti-EpCAM. It was hypothesized that if the fusion protein directs T cells to kill the NCI-H508 cells, the signal in a luciferase assay performed at 48 hours after starting the experiment should decrease. FIGS. 36-39 provide the TDCC data in graphical format. EC₅₀ values from the TDCC assays are listed in Table 11 (lists EC₅₀ data for SEQ ID Nos. 842 to 868) and Table 12 (lists EC₅₀ data for SEQ ID Nos. 869-895). The most potent molecule (EPL13) had an EC₅ value of about 1.6 pM. Some of the anti-EpCAM binding proteins were only active when present in an anti-CD3/anti-EpCAM configuration. One anti-EpCAM sequence, EPL34, was only active in the anti-EpCAM/anti-CD3 configuration. A negative control for the TDCC assays was anti-GFP/anti-CD3 protein, and this protein did not direct the T cells to kill the NCI-H508 cells (data not shown).

TABLE 11

EC ₅₀ Values for Redirected T Cell Killing of NCI-H508 Cells by Anti-CD3/Anti-EPCAM Proteins Containing Llama Anti-EPCAM Sequences (n/a = insufficient activity to calculate an EC ₅₀ using the protein concentrations tested)	
Anti-EpCAM Sequence	NCI-H508 Cell Killing EC ₅₀ (M)
EPL10	n/a
EPL109	2.2E-09
EPL117	n/a
EPL120	5.8E-010
EPL125	1.3E-09
EPL127	n/a
EPL13	1.6E-12
EPL136	6.1E-12
EPL138	7.9E-12
EPL145	n/a
EPL152	n/a
EPL153	1.4E-10
EPL156	n/a
EPL164	3.6E-10
EPL189	n/a
EPL2	n/a
EPL20	n/a
EPL34	n/a
EPL49	n/a
EPL58	n/a
EPL74	2.6E-09
EPL78	n/a
EPL82	n/a
EPL83	3.1E-10
EPL86	4.7E-10
EPL90	9.2E-12
EPL97	n/a

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TABLE 12

EC ₅₀ Values for Redirected T Cell Killing of NCI-H508 Cells by Anti-EpCAM/Anti-CD3 Proteins Containing Llama Anti-EpCAM Sequences (n/a = insufficient activity to calculate an EC ₅₀ using the protein concentrations tested)	
Anti-EPCAM Sequence	NCI-H508 Cell Killing EC ₅₀ (M)
EPL10	n/a
EPL109	n/a
EPL117	n/a
EPL120	n/a
EPL125	n/a
EPL127	n/a
EPL13	1.6E-11
EPL136	1.3E-10
EPL138	n/a
EPL145	n/a
EPL152	n/a
EPL153	not expressed
EPL156	n/a
EPL164	n/a
EPL189	n/a
EPL2	n/a
EPL20	n/a
EPL34	3.7E-10
EPL49	n/a
EPL58	n/a
EPL74	n/a
EPL78	n/a
EPL82	n/a
EPL83	2.1E-11
EPL86	2.7E-10
EPL90	n/a
EPL97	n/a

Using conditioned media with known concentrations of anti-EpCAM/anti-CD3 or anti-CD3/anti-EpCAM fusion proteins, the binding affinities of the fusion proteins for human and cynomolgus monkey EpCAM proteins were measured. An Octet instrument with streptavidin tips were loaded with biotinylated human or cynomolgus EpCAM protein, and K_D values were calculated by measuring the on rate and off rate of binding of the anti-EPCAM/anti-CD3 fusion or anti-CD3/anti-EpCAM fusion proteins to the biotinylated EpCAM proteins. The K_D measurements were made using a single 50 nM concentration of the anti-EPCAM/anti-CD3 or anti-CD3/anti-EpCAM fusion proteins, which allowed for rank ordering potency. The measured relative affinities are listed in Table 13. All of the fusion proteins bound to cynomolgus EpCAM, with K_D values ranging from 1.6 to 56 nM. Most, but not all of the fusion proteins were measured binding to human EpCAM with K_D values ranging from 0.8 to 74 nM.

TABLE 13

Binding Affinities to Human and Cyno EpCAM of Anti-EPCAM/Anti-CD3 or Anti-CD3/Anti-EpCAM Fusion Proteins Containing Llama Anti-EpCAM Sequences					
		Anti-CD3/ Anti-EpCAM		Anti-EpCAM/ Anti-CD3	
	Humanized anti- EpCAM Binder	hu KD (nM)	cy KD (nM)	hu KD (nM)	cy KD (nM)
55	EPL13	29	13	14	7.6
	EPL136	74	56	32	31
	EPL138	2.2	2.2	0.8	1.3
	EPL153	n/q	3.5	No expression	
	EPL164	3.1	3.6	2.1	2.9
60	EPL34	n/q	14	n/q	6.4
	EPL83	1.1	3	1.6	3

TABLE 13-continued

Binding Affinities to Human and Cyno EpCAM of Anti-EPCAM/Anti-CD3 or Anti-CD3/Anti-EpCAM Fusion Proteins Containing Llama Anti-EpCAM Sequences				
Humanized anti- EpCAM Binder	Anti-CD3/ Anti-EpCAM		Anti-EpCAM/ Anti-CD3	
	hu KD (nM)	cy KD (nM)	hu KD (nM)	cy KD (nM)
EPL86	n/q	8.3	n/q	7
EPL90	1.9	1.6	0.8	1.2

Example 17: Humanization of EpCAM Binding Heavy Chain Only Single Domain Antibodies and T Cell Dependent Cellular Cytotoxicity Assays

Three of the llama anti-EpCAM antibodies sequences identified in Example 15 were humanized by grafting their CDR sequences onto human germline sequences, while retaining some llama framework sequences to ensure the antibodies did not lose activity (SEQ ID Nos. 896 to 898).

These sequences were cloned into expression constructs for expression of anti-EpCAM/anti-CD3 fusion proteins (SEQ ID Nos. 899 to 901) in Expi293 cells, as described in Example 16.

The amount of anti-EpCAM/anti-CD3 fusion proteins present in the conditioned medium was quantitated as described in Example 16. The affinities of these humanized proteins for human, cynomolgus, and mouse EpCAM were measured as described in Example 16. The relative K_D values calculated from these measurements are listed in Table 14. All three sequences bound to human and cynomolgus EpCAM, with relative K_D values ranging from about 0.3 to about 18 nM. Two of the sequences also bound to mouse EpCAM, with K_D values ranging from about 1.4 to about 1.8 nM.

TABLE 14

Binding Affinities to Human, Cynomolgus, and Mouse EpCAM of Anti-EpCAM/Anti-CD3 Fusion Proteins Containing Llama Anti-EpCAM Sequences (n/q = not quantifiable under the experimental conditions used)			
Humanized Anti-EpCAM Sequence	Human EpCAM (nM)	Cynomolgus EpCAM (nM)	Mouse EpCAM (nM)
H13	17	18	n/q
H90	0.3	1.3	1.4
H138	0.3	1.8	1.8

T cell killing potential of the anti-EpCAM/anti-CD3 fusion proteins present in the conditioned medium was assessed as described in Example 16. Results are provided in Table 15 and in FIG. 40.

TABLE 15

EC ₅₀ Values for Redirected T Cell Killing of NCI-H508 Cells by Purified Anti-CD3/Anti-EpCAM Proteins Containing Humanized Anti-EpCAM Sequences	
Humanized anti-EpCAM Binder Sequence	EC ₅₀ (pM)
H13	10
H90	10
H138	16

Example 18: Demonstration of Improved Tolerability in Mouse, Conferred by an Exemplary EpCAM Targeting ProTriTAC Molecule

In this study, the tolerability of an exemplary EpCAM targeting ProTriTAC molecule was assessed. Seven weeks old NSG female tumor free mice were intraperitoneally injected with 2×10^7 expanded human T cells at the commencement of the study, i.e., at day 0. On day 2, treatment was started by dividing the mice into various groups and administering to them varying concentrations of the exemplary EpCAM targeting ProTriTAC molecule, containing the linker sequence L040, an EpCAMR targeting TriTAC molecule, an EpCAM targeting ProTriTAC molecule containing a non-cleavable linker (EpCAM ProTriTAC (NCLV)), and a GFP TriTAC molecule (SEQ ID No. 792) as a control. The molecules were administered once daily for 10 days, at the following dosages: 0.03 mg/kg, 0.1 mg/kg, 0.3 mg/kg, and 1 mg/kg. Starting from day 2, body weight of the animals were recorded daily.

As shown in FIGS. 41A-41C, the EpCAM targeting ProTriTAC molecule containing a non-cleavable linker (ProTriTAC (NCLV)) (SEQ ID No. 908) and a GFP TriTAC (used as a negative control) were very well tolerated in mice even at the highest dose of 1 mg/kg. The EpCAM targeted ProTriTAC molecule containing the linker sequence of L040 (SEQ ID No. 907) was well tolerated at the highest tested dosage of 1 mg/kg, whereas the EpCAM targeted TriTAC (SEQ ID No. 906) was well tolerated 0.1 mg/kg. It was thus observed that the EpCAM targeting ProTriTAC containing the L040 linker sequence conferred at least about 10 times improved tolerability, in mouse, compared to the EpCAM targeting TriTAC.

Example 19: Xenograft Tumor Model

An EpCAM targeting fusion protein of this disclosure (e.g., a fusion protein which is a trispecific protein comprising an anti-EpCAM heavy chain only single domain antibody, an anti-CD3 scFv, and an anti-Albumin domain) is evaluated in a xenograft model. In order to determine efficacy of the exemplary EpCAM targeting fusion protein in vivo, multiple xenograft tumor models are used. Examples of common tumor cell lines for use in xenograft tumor studies include A549 (non-small cell lung carcinoma) cells, DU-145 (prostate) cells, MCF-7 (breast) cells, Colo 205 (colon) cells, 3T3 (mouse fibroblast) cells, NCI H441 cells, HEP G2 (hepatoma) cells, MDA MB 231 (breast) cells, HT-29 (colon) cells, MDA-MB-435s (breast) cells, U266 cells, SH-SY5Y cells, Sk-Mel-2 cells, NCI-H929, RPM18226, and A431 cells. Immune-deficient NOD/scid mice are sub-lethally irradiated (2 Gy) and subcutaneously inoculated with 1×10^6 tumor cells (e.g., NCI H441 cells) into their right dorsal flank. When tumors reach 100 to 200 mm³, animals are allocated into 3 treatment groups. Groups 2 and 3 are intraperitoneally injected with 1.5×10^7 activated human T-cells. Three days later, animals from Group 3 are subsequently treated with the exemplary EPCAM targeting trispecific antigen-binding protein of Groups 1 and 2 are only treated with vehicle. Body weight and tumor volume are determined for 30 days, beginning at least 5 days post treatment with the exemplary EPCAM targeting trispecific protein.

It is expected that animals treated with the exemplary EpCAM targeting trispecific protein have a statistically

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significant delay in tumor growth in comparison to the respective vehicle-treated control group.

Example 20: Proof-of-Concept Clinical Trial
Protocol for Administration of the EpCAM
Targeting Trispecific Antigen-Binding Protein of
Example 19 to Ovarian Cancer Patients

This is a Phase I/II clinical trial for studying an exemplary EpCAM targeting trispecific antigen-binding protein of this disclosure as a treatment for an epithelial ovarian cancer.

1. Study Outcomes:
2. Primary: Maximum tolerated dose of the exemplary EpCAM targeting trispecific protein.
3. Secondary: To determine whether in vitro response of the exemplary EpCAM targeting trispecific protein is associated with clinical response

Phase I

4. The maximum tolerated dose (MTD) will be determined in the phase I section of the trial.

1.1 The maximum tolerated dose (MTD) will be determined in the phase I section of the trial.

1.2 Patients who fulfill eligibility criteria will be entered into the trial to EPCAM targeting trispecific proteins of the previous examples.

1.3 The goal is to identify the highest dose of EpCAM targeting trispecific proteins of the previous examples that can be administered safely without severe or unmanageable side effects in participants. The dose given will depend on the number of participants who have been enrolled in the

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study prior and how well the dose was tolerated. Not all participants will receive the same dose.

Phase II

2.1 A subsequent phase II section will be treated at the MTD with a goal of determining if therapy with therapy of the exemplary EpCAM targeting trispecific protein results in at least a 20% response rate.

Primary Outcome for the Phase II—To determine if therapy of EPCAM targeting trispecific protein results in at least 20% of patients achieving a clinical response (blast response, minor response, partial response, or complete response)

Eligibility:

Histologically or cytologically confirmed epithelial ovarian cancer. May have Recurrent epithelial ovarian carcinoma or disease progression following failure of first-line, platinum-based chemotherapy with no more than one prior platinum based regimen therapy

Adequate laboratory values of bone marrow function, renal function, liver function, and echocardiogram tests

Phase III

5.3.1A subsequent phase III section will be carried out with the exemplary EpCAM targeting trispecific protein, wherein secondary endpoints such as response rate (RR), patient recorded outcomes (PRO), progression-free survival (PFS), duration of progression free survival, time to progression (TTP), overall survival, health-related quality of life assessment, number of participants with overall survival, duration of response, time to response, number of participants with response, and time to tumor growth etc. will be assessed.

SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
PSMA Prodrug C1872 (SEQ ID NO: 43)	EVQLVESGGGLVQPGLNSLRSLSCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLVWSSISGSGRDTLYADSVKGRFTISRDNK TTLYLQMNSLRPEDTAVYYCTIGGSLVSSQGTLVTVSSGGGGK PLGLQARVVGSGGT QTVVTQEPSTLVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLGGAALTLGVPQPEDEA EYYCTLWYSNRWVFGGKTTLVGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNINL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTSLCAASRFMISEYHMHWR QAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDNKNTLYLQ MNSLKPEDTAVYYCDSYGRGQGTQTVTSSHHHHH
PSMA Non-cleavable Prodrug C1873 (SEQ ID NO: 44)	EVQLVESGGGLVQPGLNSLRSLSCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLVWSSISGSGRDTLYADSVKGRFTISRDNK TTLYLQMNSLRPEDTAVYYCTIGGSLVSSQGTLVTVSSGGGGG GGGGSGGVVGSGGT QTVVTQEPSTLVSPGGTVTLTCASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLGGAALTLGVPQPEDEA EYYCTLWYSNRWVFGGKTTLVGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNINL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTSLCAASRFMISEYHMHWR QAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDNKNTLYLQ MNSLKPEDTAVYYCDSYGRGQGTQTVTSSHHHHH
PSMA Active Drug C1875 (SEQ ID NO: 45)	VVGGGGTQTVVTQEPSTLVSPGGTVTLTCASSTGAVTSGNYP NWVQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGAALTLG VQPEDEAEYYCTLWYSNRWVFGGKTTLVGGGSGGGGS GGGGSEVQLVESGGGLVQPGGSLKLSAASGFTFNKYAINWV RQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNT AYLQMNINLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLV TVSSGGGSGGGSEVQLVESGGGLVQPGGSLTSLCAASRFMIS EYHMHWRQAPGKGLEWVSTINPAGTTDYAESVKGRFTISR NAKNTLYLQMNSLKPEDTAVYYCDSYGRGQGTQTVTSSHH HHH

-continued

SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
EGFR (G8) Prodrug C1486 (SEQ ID NO: 46)	EVQLVESGGGLVQPGNSLRSLSCAASGFTFSKFGMSWVRQ GGG GGLDGN EE PGGL EWVSSISGSGRDTLYADSVKGRFTISRDNAK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLLTVTVSS GGGGK PLGLQARVVGGGGT QTVVTTQEPSTLTVSPGGTVTLTCASSTGAVTSGNYPNWWQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLGVPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNLL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLLTVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTSLCAASGRTFSSYAMGWFR QAPGKEREFVVAINWASGSTYYADSVKGRFTISRDNKNTLYL QMNSLRAEDTAVYYCAAGYQINSGNYNFKDYEYDYGQGTLL TVTSSHHHHHH
EGFR (G8) Non-cleavable Prodrug C1756 (SEQ ID NO: 47)	EVQLVESGGGLVQPGNSLRSLSCAASGFTFSKFGMSWVRQ GGG GGLDGN EE PGGL EWVSSISGSGRDTLYADSVKGRFTISRDNAK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLLTVTVSS GGGGG GGGGSGGVVGGGGT QTVVTTQEPSTLTVSPGGTVTLTCASSTGAVTSGNYPNWWQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLGVPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNLL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLLTVTVSSGGGGS GGGSEVQLVESGGGLVQPGGSLTSLCAASGRTFSSYAMGWFR QAPGKEREFVVAINWASGSTYYADSVKGRFTISRDNKNTLYL QMNSLRAEDTAVYYCAAGYQINSGNYNFKDYEYDYGQGTLL TVTSSHHHHHH
EGFR (G8) Active Drug C1300 (SEQ ID NO: 48)	VVGGGGT QTVVTTQEPSTLTVSPGGTVTLTCASSTGAVTSGNYP NWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGSGGGGS GGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWV RQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNT AYLQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLLV TVSSGGGSGGGSEVQLVESGGGLVQPGGSLTSLCAASGRTFS SYAMGWFRQAPGKEREFVVAINWASGSTYYADSVKGRFTISR DNSKNTLYLQMNSLRAEDTAVYYCAAGYQINSGNYNFKDYEY DYGQGTLLTVTSSHHHHHH
GFP TriTAC C646 (SEQ ID NO: 49)	QVQLVESGGALVQPGGSLRLSCAASGFPVNNRYSMRWYRQAP GKEREWVAGMSSAGDRSSYEDSVKGRFTISRDDARNTVYLQM NSLKPEDTAVYYCNVNVGFPEYWGQGTQVTVSSGGGSGGGGS EVQLVESGGGLVQPGNSLRSLSCAASGFTFSKFGMSWVRQAPG KGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSL RPEDTAVYYCTIGGSLSVSSQGTLLTVSSGGGSGGGSEVQLVE SGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLEWV ARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNLLKTE DTAVYYCVRHANFGNSYISYWAYWGQGTLLTVTVSSGGGSGG GSGGGGSGTVVTTQEPSTLTVSPGGTVTLTCASSTGAVTSGNYP NWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHHHH
Exemplary masking sequence (SEQ ID NO: 50)	GGGGGLDGN EE PGG
Exemplary non-cleavable linker sequence comprising protease cleavage site (SEQ ID NO: 51)	GGGGSGGGSGGVVGGGGT
PSMA Non-masked Non-cleavable Prodrug C1874 (SEQ ID NO: 52)	EVQLVESGGGLVQPGNSLRSLSCAASGFTFSKFGMSWVRQAPG KGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSL RPEDTAVYYCTIGGSLSVSSQGTLLTVSSGGGSGGGSGGGV GGGGTQTVVTTQEPSTLTVSPGGTVTLTCASSTGAVTSGNYPNW VQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLGVP QPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGSGGGSGG GSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQ APGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAY LQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLLTV SSGGGSGGGSEVQLVESGGGLVQPGGSLTSLCAASRFMISEY HMHWRQAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDN AKNTLYLQMNSLKPEDTAVYYCDSYGRGQGTQVTVSSHHHH HH

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary cleavable linker (SEQ ID NO: 53)	GGGGKPLGLQARVVGGGGT
Exemplary anti-albumin sdAb (SEQ ID NO: 54)	EVQLVESGGGLVQPGNSLRSLSCAASGFTFSKFGMSWVRQGGG GGLDGNEE PGGLEWVSSISGSGRDTLYADSVKGRFTISRDNAK TTLYLQMNSLRPEDTAVYYCTIGGSLVSSQGTLTIVSS GGGGK PLGLQARVVGGGGT
Exemplary anti-target (anti-EGFR) sdAb (SEQ ID NO: 55)	EVQLVESGGGLVQPGGSLTSLSCAASGRTFSSYAMGWFRQAPG KEREFVVAINWASGSTYYADSVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCAAGYQINSGNYNFKDYEYDWGQGTLTIVSS
Exemplary anti-CD3 scFv (SEQ ID NO: 56)	QTVVTQEPSLTVSPGGTVTLTLCASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 57)	EVQLVESGGGLVQPGGSLTSLSCAASRFMISEYHMHVVRQAPG KGLEWVSTINPAGTTDYAESVKGRFTISRDNKNTLYLQMNSL KPEDTAVYYCDSYGYRGQGTQTVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 63)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYHMHVVRQAP GKGLEWVSDINPAGTTDYAESVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 64)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYHMHVVRQAP GKGLEWVSTINPAGTTDYAESVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 65)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYSMHVVRQAPG KGLEWVSTINPAKTTDYAESVKGRFTISRDNKNTLYLQMNSLR AEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 66)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISPYSMHVVRQAPG KGLEWVSTINPAGTTDYAESVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCDGYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 67)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYSMHVVRQAPG KGLEWVSTINPAGQTDYAESVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCDGYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 68)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYSMHVVRQAPG KGLEWVSTINPAGTTDYAEYVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCDGYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 69)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYHMHVVRQAP GKGLEWVSDINPAKTTDYAESVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 70)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISPYHMHVVRQAP GKGLEWVSDINPAGTTDYAESVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 71)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYHMHVVRQAP GKGLEWVSDINPAGQTDYAESVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 72)	EVQLVESGGGLVQPGGSLRSLSCAASRFMISEYHMHVVRQAP GKGLEWVSDINPAGTTDYAEYVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCDSYGYRGQGTLTIVSS
Exemplary anti-target (anti-PSMA) sdAb (SEQ ID NO: 73)	EVQLVESGGGLVQPGGSLTSLSCAASRFMISEYHMHVVRQAPG KGLEWVSDINPAGTTDYAESVKGRFTISRDNKNTLYLQMNSL KPEDTAVYYCDSYGYRGQGTQTVSS
Exemplary anti-CD3 scFv (SEQ ID NO: 74)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPG KGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN NNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTLTIVSSGG GGGGGGGGGGSGTVVTQEPSLTVSPGGTVTLTLCASSTGAV TSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-CD3 scFv (SEQ ID NO: 75)	EVQLVESGGGLVQPGGSLKLSCAASGFENKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSFG AVTSGNYPNWVQKPGQAPRGLIGGTFKFLAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCVLWYDNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 76)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSHISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSSTG YVTSGNYPNWVQKPGQAPRGLIGGTSFLAPGTPARFSGSLLG GKAALTLSGVQPEDEAEYYCVLWYSNRWIFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 77)	EVQLVESGGGLVQPGGSLKLSCAASGFMPNKYAMNWVRQAP GKGLEWVARIRSKSNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSFG AVTSGNYPNWVQKPGQAPRGLIGGTKLLAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 78)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSSTG AVVSGNYPNWVQKPGQAPRGLIGGTEFLAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 79)	EVQLVESGGGLVQPGGSLKLSCAASGFTYNNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADEVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSKG AVTSGNYPNWVQKPGQAPRGLIGGTEKELAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 80)	EVQLVESGGGLVQPGGSLKLSCAASGNTFNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGLTVTVSSG GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSSTGA VTSGYYPNWVQKPGQAPRGLIGGTYFLAPGTPARFSGSLLGG KAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 81)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADAVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSQISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSSTG AVTDGNYPNWVQKPGQAPRGLIGGIKFLAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 82)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAVNWVRQAPG KGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQM NNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGLTVTVSSGG GGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGESTGAV TSGNYPNWVQKPGQAPRGLIGGTKILAPGTPARFSGSLLGGK AALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 83)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYPMNWVRQAP GKGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKNEDTAVYYCVRHGNFNNSYISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSSTG AVTKGNYPNWVQKPGQAPRGLIGGTMKLAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCALWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 84)	EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADEVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGLTVTVSS GGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCGSSSTG AVVSGNYPNWVQKPGQAPRGLIGGTEFLAPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 85)	EVQLVESGGGLVQPGGSLKLSCAASGNTFNKYAMNWVRQAP GKGLEWVARIRSKYNNKYATYYADSVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHGNFGDSYISYWAYWGQGLTVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	GGGGSGGGSGGGSGT VVTQEP SLTVSPGGTVTLT CGSSTG AVTHGNYPNWVQKPGQAPRGLIGG TKVLAPGT PARFSGSLL GGKAALTL SGVQPEDEAEYYCVLWYSNRWVFGG GTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 86)	EVQLVESGGGLVQPGGSLKLS CAASGFTFN NYAMNWVRQAP GKGLEWVARIRSGYNNYATYYADSVKDRFTISRDDS KNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGT LVTVSS GGGGSGGGSGGGSGT VVTQEP SLTVSPGGTVTLT CGSYTG AVTSGNYPNWVQKPGQAPRGLIGG TKFNAPGT PARFSGSLL GGKAALTL SGVQPEDEAEYYCVLWYANRWVFGG GTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 87)	EVQLVESGGGLVQPGGSLKLS CAASGFEPN KYAMNWVRQAP GKGLEWVARIRSKYNNYETYYADSVKDRFTISRDDS KNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSLI SYWAYWGQGT LVTVSS GGGGSGGGSGGGSGT VVTQEP SLTVSPGGTVTLT CGSSSG AVTSGNYPNWVQKPGQAPRGLIGG TKFGAPGT PARFSGSLL GGKAALTL SGVQPEDEAEYYCVLWYSNRWVFGG GTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 88)	EVQLVESGGGLVQPGGSLKLS CAASGFTFN KYAMNWVRQAP GKGLEWVARIRSKYNNYATYYADSVKDRFTISRDDS KNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSYI SYWAYWGQGT LVTVSS GGGGSGGGSGGGSGT VVTQEP SLTVSPGGTVTLT CGSSTG AVTSGNYPNWVQKPGQAPRGLIGG TKFLAPGT PARFSGSLL GGKAALTL SGVQPEDEAEYYCVLWYSNRWVFGG GTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 89)	EVQLVESGGGLVQPGGSLKLS CAASGFTFN KYALNWVRQAPG KGLEWVARIRSKYNNYATEYADSVKDRFTISRDDS KNTAYLQM NNLKTEDTAVYYCVRHGNFGNSPISYWAYWGQGT LVTVSSGG GGSGGGSGGGSGT VVTQEP SLTVSPGGTVTLT CGSSTGAV TSGNYPNWVQKPGQAPRGLIGG TNFLAPGT PERFSGSLGG KAALTL SGVQPEDEAEYYCVLWYSNRWAFGG GTKLTVL
Exemplary anti-CD3 scFv (SEQ ID NO: 90)	EVQLVESGGGLVQPGGSLKLS CAASGFTFN EYAMNWVRQAP GKGLEWVARIRSKYNNYATYYADDVKDRFTISRDDS KNTAYLQ MNNLKTEDTAVYYCVRHGNFGNSG ISYWAYWGQGT LVTVSS GGGGSGGGSGGGSGT VVTQEP SLTVSPGGTVTLT CGSSTG AVTVGNYPNWVQKPGQAPRGLIGG TEFLAPGT PARFSGSLL GGKAALTL SGVQPEDEAEYYCVLWYSNRWVFGG GTKLTVL
Exemplary anti-target (BCMA) (SEQ ID NO: 91)	EVQLVESGGGLVQGRSLTLSCAASTNIFSI SPMGWYRQAPGK QRELVAAIHGFTLYADSVKGRFTISRDN AKNSIYLQMN SLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 92)	EVQLVESGGGLVQGRSLTLSCAASTDIFS ISPMGWYRQAPGK QRELVAAIHGGSTLYADSVKGRFTISRDN AKNSIYLQMN SLRPE DTALYYCNKVPWGDYHPRNVAWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 93)	EVQLVESGGGLVQGRSLTLSCAASTNDFS ISPMGWYRQAPG KQRELVAAIHGGSTLYADSVKGRFTISRDN AKNSIYLQMN SLRP EDTALYYCNKVPWGDYHPRNVAWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 94)	EVQLVESGGGLVQGRSLTLSCAASTNIFSKSPMGWYRQAPGK QRELVAAIHGKSTLYADSVKGRFTISRDN AKNSIYLQMN SLRPE DTALYYCNKVPWGDYHPRNVVWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 95)	EVQLVESGGGLVQGRSLTLSCAASTNDFS ISPMGWYRQAPG KQRELVAAIHGKSTLYADSVKGRFTISRDN AKNSIYLQMN SLRP EDTALYYCNKVPWGDYHPRNVKVGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 96)	EVQLVESGGGLVQGRSLTLSCAASTNQFS ISPMGWYRQAPG KQRELVAAIHGKSTLYADSVKGRFTISRDN AKNSIYLQMN SLRP EDTALYYCNKVPWGDYHPRNVVWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 97)	EVQLVESGGGLVQGRSLTLSCAASTNIFSSSPMGWYRQAPGK QRELVAAIHGFTLYADSVKGRFTISRDN AKNSIYLQMN SLRPE DTALYYCNKVPWGDYHPRNVHWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 98)	EVQLVESGGGLVQGRSLTLSCAASTNIFSSSPMGWYRQAPGK QRELVAAIHGFTLYADSVKGRFTISRDN AKNSIYLQMN SLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVT VSS
Exemplary anti-target (BCMA) (SEQ ID NO: 99)	EVQLVESGGGLVQGRSLTLSCAASTNIFSTSPMGWYRQAPGK QRELVAAIHGFQTLYADSVKGRFTISRDN AKNSIYLQMN SLRPE DTALYYCNKVPWGDYHPRNVVWGQGTQVT VSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (BCMA) (SEQ ID NO: 100)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGFETLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVLWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 101)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSESPMGWYRQAPGK QRELVAAIHGFETLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 102)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSDSPMGWYRQAPG KQRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVWVGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 103)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSNSPMGWYRQAPG KQRELVAAIHGGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVHWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 104)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSNSPMGWYRQAPG KQRELVAAIHGRSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVWVGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 105)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSNSPMGWYRQAPG KQRELVAAIHGPSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 106)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSNSPMGWYRQAPG KQRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVRWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 107)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 108)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSKSPMGWYRQAPGK QRELVAAIHGQSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 109)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSSSPMGWYRQAPGK QRELVAAIHGHSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 110)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSSSPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRKVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 111)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGIYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 112)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSESPMGWYRQAPGK QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGTYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 113)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSVSPMGWYRQAPG KQRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGKYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 114)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSVSPMGWYRQAPG KQRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 115)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSKSPMGWYRQAPGK QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPREVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 116)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSDSPMGWYRQAPG KQRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 117)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGKYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 118)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSHSPMGWYRQAPG KQRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (BCMA) (SEQ ID NO: 119)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSTSPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 120)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSTSPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVQWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 121)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSLSPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 122)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSIISPMGWYRQAPGK QRELVAAIHGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 123)	EVQLVESGGGLVQPGRSLTSLSCAASTNHFSISPMGWYRQAPG KQRELVAAIHGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 124)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSASPMGWYRQAPG KQRELVAAIHGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPRNVNWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 125)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSASPMGWYRQAPG KQRELVAAIHGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 126)	EVQLVESGGGLVQPGRSLTSLSCAASTNISSISPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 127)	EVQLVESGGGLVQPGRSLTSLSCAASTNIESISPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 128)	EVQLVESGGGLVQPGRSLTSLSCAASTNIFSIISPYGWYRQAPGKQ RELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPEDT ALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 129)	EVQLVESGGGLVQPGRSLTSLSCAASTNIASISPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 130)	EVQLVESGGGLVQPGRSLTSLSCAASTNIASISPMGWYRQAPGK QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 131)	EVQLVESGGGLVQPGRSLTSLSCAASTNIASISPMGWYRQAPGK QRELVAAIHGSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 132)	EVQLVESGGGLVQPGRSLTSLSCAASTNITISPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 133)	EVQLVESGGGLVQPGRSLTSLSCAASTNIMISPMGWYRQAPG KQRELVAAIHGNTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 134)	EVQLVESGGGLVQPGRSLTSLSCAASTNITISPMGWYRQAPGK QRELVAAIHGNTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 135)	EVQLVESGGGLVQPGRSLTSLSCAASTNIVISPMGWYRQAPGK QRELVAAIHGHTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 136)	EVQLVESGGGLVQPGRSLTSLSCAASTNIVISPMGWYRQAPGK QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (BCMA) (SEQ ID NO: 137)	EVQLVESGGGLVQPGRSLTSLSCAASTNVVSI SPMGWYRQAPG KQRELVAAIHGKSTLYADSVKGRFTISRDN AKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 138)	EVQLVESGGGLVQPGRSLTSLSCAASTNII SISPMGWYRQAPGK QRELVAAIHGASTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 139)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SITPMGWYRQAPGK QRELVAAIHGASTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 140)	EVQLVESGGGLVQPGRSLTSLSCAASTNIT SISPMGWYRQAPGK QRELVAAIHGFETLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 141)	EVQLVESGGGLVQPGRSLTSLSCAASTNIQ SISPMGWYRQAPGK QRELVAAIHGFETLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 142)	EVQLVESGGGLVQPGRSLTSLSCAASTSD FISPMGWYRQAPGK QRELVAAIHGFETLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 143)	EVQLVESGGGLVQPGRSLTSLSCAASTNID SISPMGWYRQAPGK QRELVAAIHGFQTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 144)	EVQLVESGGGLVQPGRSLTSLSCAASTNIM SISPMGWYRQAPG KQRELVAAIHGFSTVYADSVKGRFTISRDN AKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 145)	EVQLVESGGGLVQPGRSLTSLSCAASTNIE SISPMGWYRQAPGK QRELVAAIHGFSTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 146)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SSSPMGWYRQAPGK QRELVAAIHGFKTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTARYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 147)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SSSPMGWYRQAPGK QRELVAAIHGFSTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 148)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SNSPMGWYRQAPG KQRELVAAIHGFSTLYADSVKGRFTISRDN AKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 149)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF STSPMGWYRQAPGK QRELVAAIHGFSTIYADSVKGRFTISRDN AKNSIYLQMNSLRPE TALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 150)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF STSPMGWYRQAPGK QRELVAAIHGFSTIYADSVKGRFTISRDN AKNSIYLQMNSLRPE TALYYCNKVPWGDYHPLNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 151)	EVQLVESGGGLVQPGRSLTSLSCVASTNIF STSPMGWYRQAPGK QRELVAAIHGFSTLYADSVKGRFTISRDN AKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 152)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SDSPMGWYRQAPG KQRELVAAIHGFSTFYADSVKGRFTISRDN AKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 153)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SQSPMGWYRQAPG KQRELVAAIHGDSLTYADSVKGRFTISRDN AKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGQGTQV TVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 154)	EVQLVESGGGLVQPGRSLTSLSCAASTNIF SQSPMGWYRQAPG KQRELVAAIHGKSTLYADSVKGRFTISRDN AKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPSNVYWGKGTQV TVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (BCMA) (SEQ ID NO: 155)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 156)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGISTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE TALYYCNKVPWGDYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 157)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSRSPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 158)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSGSPMGWYRQAPG KQRELVAAIHGNSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 159)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIYPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 160)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIYPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPKNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 161)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSKSPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 162)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSKSPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 163)	EVQLVESGGGLVQPGRSLTSLCAASTNEFSISPMGWYRQAPGK QRELVAAIHGLSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGAYHPRNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 164)	EVQLVESGGGLVQPGRSLTSLCAASTNEFSISPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPGNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 165)	EVQLVESGGGLVQPGRSLTSLCAASTNIPSISPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVWAGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 166)	EVQLVESGGGLVQPGRSLTSLCAASTNIPSISPMGWYRQAPGK QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVWAGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 167)	EVQLVESGGGLVQPGRSLTSLCAASTNIPSISPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 168)	EVQLVESGGGLVQPGRSLTSLCAASTNIPSISPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 169)	EVQLVESGGGLVQPGRSLTSLCAASTNITSISPMGWYRQAPGK QRELVAAIHGVSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVQWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 170)	EVQLVESGGGLVQPGRSLTSLCAASTNIPSISPMGWYRQAPGK QRELVAAIHQSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVQWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 171)	EVQLVESGGGLVQPGRSLTSLCAASTNIVSISPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVWGGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 172)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSISPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVWGGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (BCMA) (SEQ ID NO: 173)	EVQLVESGGGLVQPGRSLTSLSCAASNIDISIPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 174)	EVQLVESGGGLVQPGRSLTSLSCAASNIDISIPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVTWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 175)	EVQLVESGGGLVQPGRSLTSLSCAASNIRISIPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVVWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 176)	EVQLVESGGGLVQPGRSLTSLSCAASNITISIPMGWYRQAPGK QRELVAAIISGFSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE TALYYCNEVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 177)	EVQLVESGGGLVQPGRSLTSLSCAASNITISIPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 178)	EVQLVESGGGLVQPGRSLTSLSCAASNITISVSPMGWYRQAPG KQRELVAAIHGPFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGDYHPTNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 179)	EVQLVESGGGLVQPGRSLTSLSCAASNIGSIPMGWYRQAPGK QRELVAAIHQSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPQNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 180)	EVQLVESGGGLVQPGRSLTSLSCAASNIESIPMGWYRQAPGK QRELVAAIHGKSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRRVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 189)	EVQLVESGGGLVQPGRSLTSLSCAASNIVSIPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRRVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 190)	EVQLVESGGGLVQPGRSLTSLSCAASNIDISIPMGWYRQAPGK QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGDYHPRMVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 191)	EVQLVESGGGLVQPGRSLTSLSCAASNIFMISIPMGWYRQAPG KQRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRP EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 192)	EVQLVESGGGLVQPGRSLTSLSCAASNIFRISIPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 193)	EVQLVESGGGLVQPGRSLTSLSCAASNIFSIPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGEYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 194)	EVQLVESGGGLVQPGRSLTSLSCAASNIFSIPMGWYRQAPGK QRELVAAIHGDSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGKYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 195)	EVQLVESGGGLVQPGRSLTSLSCAASNIESIPMGWYRQAPGK QRELVAAIHGSSSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 196)	EVQLVESGGGLVQPGRSLTSLSCAASNIESIPMGWYRQAPGK QRELVAAIHGNSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 197)	EVQLVESGGGLVQPGRSLTSLSCAASNISIPMGWYRQAPGK QRELVAAIHGFSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 198)	EVQLVESGGGLVQPGRSLTSLSCAASNISIPMGWYRQAPGK QRELVAAIHGHSTLYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 199)	EVQLVESGGGLVQPGRSLTSLSCAASNISIPMGWYRQAPGK QRELVAAIHGFSTVYADSVKGRFTISRDNAKNSIYLQMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (BCMA) (SEQ ID NO: 200)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIIRPMGWYRQAPGK QRELVAAIHGFSTVYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 201)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIYPMGWYRQAPGK QRELVAAIHGFSTYYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGSYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 202)	EVQLVESGGGLVQPGRSLTSLCAASTNIFNISPMPGWYRQAPGK QRELVAAIHGFSTYYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 203)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSSSPMPGWYRQAPGK QRELVAAIHGFSTWYADSVKGRFTISRDNAKNSIYLMNSLRP EDTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 204)	EVQLVESGGGLVQPGRSLTSLCAASTNIISSIPMPGWYRQAPGK QRELVAAIHGFDTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 205)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIINPMGWYRQAPGK QRELVAAIHGFDTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGDYHPRNVSWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 206)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIITPMGWYRQAPGK QRELVAAIHGRSTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGSYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 207)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIITPMGWYRQAPGK QRELVAAIHGTSTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 208)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIITPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGRYHPRNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 209)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIITPMGWYRQAPGK QRELVAAIHGESTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGDYHPRDVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 210)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSTSPYGWYRQAPGK QRELVAAIHGFSTIYADSVKGRFTISRDNAKNSIYLMNSLRPED TALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 211)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSTSPGGWYRQAPGK QRELVAAIHGFSTIYADSVKGRFTISRDNAKNSIYLMNSLRPED TALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 212)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIITPYGWYRQAPGK QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 213)	EVQLVESGGGLVQPGRSLTSLCAASTNIFSIITPGGWYRQAPGK QRELVAAIHGASTLYADSVKGRFTISRDNAKNSIYLMNSLRPE DTALYYCNKVPWGDYHPGNVYWGQGTQVTVSS
Exemplary anti-target (BCMA) (SEQ ID NO: 214)	QVQLVESGGGLVQPGESLRLSCAASTNIFSIISPMGWYRQAPGK QRELVAAIHGFSTLYADSVKGRFTISRDNAKNTIYLMNSLKPE DTAVYYCNKVPWGDYHPRNVYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 215)	QVQLVESGGGLVQPGGSLRLSCAASGRFTSVRGMAWYRQAG NNRALVATMNPDPFPNYADAVKGRFTISWDIAENTVYLMN SLNSEDTTVYYCNSGPYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 216)	QVQLVESGGGLVQAGGSLRLSCAASGSIPISEQMFWYRQAPG KQRELVAALTSGGRANYADSVKGRFTISGDNVRNMVYLMNS LKPEDTAIYYCSAGRFGDYAQRSGMDYWGKGLTVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 217)	QVQLVESGGGLVQAGGSLRLSCAFSGTTTYTFDLMSWYRQAPG KQRTTVASISSDGRTSYADSVRGRFTISGNGKNTVYLMNSL KLEDTAVYYCLGQRSGVRAFWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (MSLN) (SEQ ID NO: 218)	QVQLVESGGGLVQAGGSLRLSCVASGSTSNINNMRYRQAP GKERELVAVITRGGYAIYLDVAVKGRFTISRDNANNAIYLEMNSL KPEDTAVYVCNADRVEGTSGGPQLRDYFGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 219)	QVQLVESGGGLVQAGGSLRLSCAASGSTFGINAMGWYRQAP GKQRELVAVISRGGSTNYADSVKGRFTISRDNANTVSLQMNT LKPEDTAVYFCNARTYTRHDYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 220)	QVRLVESGGGLVQAGGSLRLSCAASISAPRLMSVRWYRQDPSK QREWVATIDQLGRTNYADSVKGRFAISKDSTRNTVYLQMNML RPEDTAVYYCNAGGGPLGSRWLRGRHWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 221)	QVRLVESGGGLVQAGESLRLSCAASGRPFISINTMGWYRQAPG KQRELVASISSGDFTYTDSVKGRFTISRDNAKNTVYLQMNSLK PEDTAVYYCNARTYLPFRFGSWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 222)	QVQPVESGGGLVQPGGSLRLSCVVSIGSDFTEDAMAWYRQAS GKERESVAFVSKDGKRILYLDVSRGRFTISRDIKKTVYLQMDNL KPEDTGYYCNSAPGAARNYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 223)	QVQPVESGGGLVQPGGSLRLSCVVSIGSDFTEDAMAWYRQAS GKERESVAFVSKDGKRILYLDVSRGRFTISRDIKKTVYLQMDNL KPEDTGYYCNSAPGAARNVWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 224)	EVQLVESGGGLVQPGNSLRLSCAASGFTFSSPGMSWVRQAPG KGLEWVSSISGSGSDTLYADSVKGRFTISRDNAKTTLVYLQMNSL RPEDTAVYYCTIGGSLSRSSQGTTLTVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 225)	QVQIVESGGGLVQAGGSLRLSCVASGLTYSIVAVGWYRQAPGK EREMVADISPVGNTNYADSVKGRFTISKENAKNTVYLQMNSLK PEDTAVYYCHIVRGWLDERPFGPIVYWQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 226)	QVQLVESGGGLVQTGGSLRLSCAASGLTFGVYGMWFRQAP GKQREWVASHTSTGYVYYRDSVKGRFTISRDNASTVYLQMN SLKPEDTAIYYCKANRGSYEYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 227)	QVQLVESGGGLVQAGGSLRLSCAASTSSINSMSWYRQAQGK QREPVAVITDRGSTSYADSVKGRFTISRDNAKNTVYLQMNSLK PEDTAIYTCHVIAWRGYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 228)	QVQLVESGGGLVQAGGSLRLSCAASGRTLRYAMGWFRQAP GKERQFVAISRSGGTRYSDSVKGRFTISRDNANTFYLQMN NLRPDDTAVYYCNVRRRGWRTLEYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 229)	QVQLGESGGGLVQAGGSLRLSCAASGSIFSPNAMIWHRQAPG KQREPVASINSSGSTNYGDSVKGRFTVSRDIVKNTMYLQMNSL KPEDTAVYYCSYSDFRGTQYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 230)	QVQLVESGGGLVPSGGSLRLSCAASGATSAITNLGWYRRAPGQ VREMVARISVREDKEDYEDSVKGRFTISRDNQNLVYLQMNNL QPHDTAIYYCGAQRWGRPGTWTWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 231)	QVQLVESGGGLVQAGGSLRLSCAASGSTFRIRVMRYRQAPG TERDLVAVISGSSITYADSVKGRFTISRDNAKNTLYLQMNLP EDTAVYYCNADDSGIARDYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 232)	QVQLVESGGGLVQAGESRRLSCAVSGDTSKFKAVGWYRQAPG AQRELLAWINNSGVGNTAESVKGRFTISRDNAKNTVYLQMN LTPEDTDVYYCFYRRFGINKNYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 233)	QVQLVESGGGLVQAGGSLRLSCAASGSTFGNKPMGWYRQAP GKQRELVAVISSDGGSTRYAALVKGRFTISRDNAKNTVYLQME SLVAEDTAVYYCNALRTYYLNDPVVFSWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 234)	QVQLVESGGGLVQAGGSLRLSCAASGSTSINTMYWYRQAPG KERELVAFISGGSTNVDSVKGRFVSRSDAKNIVYLQMNSLT PEDTAVYYCNTYIPLRGLTHDYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 235)	QVQLVESGGGLVQAGGSLRLSCVASGRDTRITMGWYRQAP GKQRELVAVISNRGTSNYANSVKGRFTISRDNAKNTVYLQMNS LKPEDTAVYYCNARKWGRNYWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (MSLN) (SEQ ID NO: 236)	QVQLVESGGGLVQARGSLRLSCTASGRTIGINDMAWYRQAPG NQRELVATITKGGTTDYADSVDRFTISRDNKNTVYLQMNSL KPEDTAVYYCNTKRREWAKDFEYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 237)	QVQLVESGGGLVQAGGSLRLSCAASIGSINMSWYRQAPGK QREPVAVIDRGSYSYADSVKGRFTISRDNKNTVYLQMNSLK PEDTAIYTCVHIADWRGYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 238)	QVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWFRQAPG EERELVATINRGGSTNVRDSVKGRFSVSRDSAKNIVYLQMNRL KPEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 239)	QVQLVESGGGLVQAGGSLRLSCTTSTTFSINMSWYRQAPGN QREPVAVINRGTTSYADSVKGRFTISRDNARNTVYLQMDSLK PEDTAIYTCVHIADWRGYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 240)	QVQLVESGGGLVQAGGSLRLSCTASGSTFSIRAMRWYRQAPG TERDLVAVIYGSSTYYADAVKGRFTISRDNKNTLYLQMNMLPK EDTAVYYCNADTIGTARDYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 241)	QVQLVESGGGLVQAGGSLRLSCVASGRTSTIDTMWHRQAPG NERELVAVVTSRGTSNVADSVKGRFTISRDNKNTAYLQMNSL KPEDTAVYYCSVRTTSSYPVDFWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 242)	QVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWYRQAPG KERELVAFISSGGSTNVRDSVKGRFSVSRDSAKNIVYLQMNSLK PEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 243)	QVQLVESGGGLVQPGGSLRLSCAASGGDWSANFMYWYRQA PGKQRELVARISGRGVVDYVESVKGRFTISRDNKNTVYLQMN SLKPEDTAVYYCAVASYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 244)	EVQLVESGGGLVQPGGSLRLSCAASGGDWSANFMYWYRQAP GKQRELVARISGRGVVDYVESVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCAVASYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 245)	EVQLVESGGGLVQPGGSLRLSCAASGGDWSANFMYWVRQAP KGLEWVSRIISGRGVVDYVESVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCAVASYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 246)	EVQLVESGGGLVQAGGSLRLSCAASGSTSSINTMYWYRQAPG KERELVAFISSGGSTNVRDSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 247)	EVQLVESGGGLVQPGGSLRLSCAASGSTSSINTMYWYRQAPG KERELVAFISSGGSTNVRDSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 248)	EVQLVESGGGLVQPGGSLRLSCAASGSTSSINTMYWVRQAPG KGLEWVSFISGGSTNVRDSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCNTYIPYGGTLHDFWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 249)	QVQLVESGGGVVQAGGSLRLSCAASGSTFSIRAMRWYRQAPG TERDLVAVIYGSSTYYADAVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCNADTIGTARDYWGQGTQVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 250)	QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWYRQAPG KERELVAVIYGSSTYYADAVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCNADTIGTARDYWGQGTQVTVSSGG
Exemplary anti-target (MSLN) (SEQ ID NO: 251)	QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWVRQAPG KGLEWVSVIYGSSTYYADAVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCNADTIGTARDYWGQGTQVTVSSGG
Exemplary anti-target (MSLN) (SEQ ID NO: 252)	EVQLVESGGGLVQAGGSLRLSCVASGRTSTIDTMWHRQAPG NERELVAVVTSRGTSNVADSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCSVRTTSSYPVDFWGQGTQVTVSSGG
Exemplary anti-target (MSLN) (SEQ ID NO: 253)	EVQLVESGGGLVQPGGSLRLSCAASGRTSTIDTMWHRQAPG KERELVAVVTSRGTSNVADSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCSVRTTSSYPVDFWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (MSLN) (SEQ ID NO: 254)	EVQLVESGGGLVQPGGSLRLSCAASGRTSTIDTMWVRQAPG KGLEWVSIVTSRGTSNVADSVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCSVRTTSYPVDFWGGTLVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 255)	QVQLVESGGGVVQAGGSLTSLCAASGSTFSIRAMRWYRQAPG TERDLVAVIYGSTYYADAVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQTLVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 256)	QVQLVESGGGVVQAGGSLRLSCAASGSTFSIRAMRWYRQAPG TERDLVAVIYGSTYYADAVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQTLVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 257)	QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWYRQAPG KERELVAVIYGSTYYADAVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQTLVTVSS
Exemplary anti-target (MSLN) (SEQ ID NO: 258)	QVQLVESGGGVVQPGGSLRLSCAASGSTFSIRAMRWYRQAPG KGLEWVSIVYGSTYYADAVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCNADTIGTARDYWGQTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 306)	QVQLQESGGGLVQAGGSLRLSCAASGSIFSIA SMGWYRQAPG KQRELVAVITSPSSTNYADSVKGRFTISRDN AKNTVYLQMNSLK PEDTGVIYCNAVYFERTDWGGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 307)	QVQLQESGGGLVQAGGSLRLSCAAPGSIFSIA SMGWYRQAPG KQRELVAVITSPSSTNYADSVKGRFTISRDN AKNTVYLQMNSLK PEDTGVIYCNAVYFERTDWGGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 308)	QVQLQESGGGLVQAGGSLRLSCAASGSIFSIA SMWYRQAPG KQRELVAAITSPSSTNYADSVKGRFTISRDN AKNTVYLQMNSLK PEDTGVIYCNAVYFERTDWGGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 309)	QVQLQESGGGLVQAGGSLRLSCAASESIFSI NVMAWHRQAPG KQRELVARITSGGSTNYADSVKGRFTISRDN AKNTVYLQMNSL KPEDTGVIYCGAYQGLYAYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 310)	QVQLQESGGGLVQAGGSLRLSCVASGSSFSITSMWYRQAPG KQRDLVAAITSPGSTNYADSVKDRFTISRDN AKNTVYLQMNSL KPEDTAVYYCNGRVFDHVYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 311)	QVQLQESGGGLVQAGGSLKLSCAASSSIFSISMSWYRQAPGK QRELVAAITTFDYTNYADSVKGRFTISRDN AKNM MYLQMNSL KPEDTAVIYLCNARAFGRDYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 312)	QVQLQESGGGLVQAGGSLKLSCAASSSIFSISMSWYRQAPGK QRELVAAITSPGSTNYADSVKGRFTISRDN AKNM MYLQMNSL KPEDTAVYRCNARTMGRDYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 313)	QVQLQESGGGLVQPGGSLRLSCAASGSTLN IKIMAWHRQAPG KQRELVALTSGGNTNYADSVKGRFTISRDN AKNTVYLQMNSL QPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 314)	QVQLQESGGGLVQPGGSLRLSCAASGSTLN IKIMAWHRQAPG KQRELVALTSGGNTNYADSVKGRFTISRDN AKNTVYLQMNSL QPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 315)	QVQLQESGGGLVQAGGSLRLSCAASGSTFN IKTMAWHRQAP GNQRELVALTSGGNTNYADSVKGRFTISRDN AKNTVYLQMN SLKPEDTAVYYCGLWNGVGGAYWGRGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 316)	QVQLQDGGGLVQPGGSLRLSCAASGSTFN IKLMAWHRQAPG NQRELVALTSGGNTNYADSVKGRFTISRDN ASNIVYLQMNSL KPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 317)	QVQLQESGGGLVQAGGSLRLSCAASGSTFN IKMAWHRQAP GKQRELVASLTSEGLTNYRDSVKGRFTISRDN AKNTVYLQMN NLKPEDTAVYYCGLWDGVGGAYWGRGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 318)	QVQLQESGGGLVQPGGSLRLSCAASGFMFSSYSMSWYRQAP GKQRELVAAITTWGSTNYADSVKGRFTISRDN AKNTVVLQM NSLEPEDTAVYFCNARSWNINWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 319)	QVQLQESGGGLVQVGGSLRLSCAASGFMFSSYSMSWYRQAP GKQRELVAAITSYGSTNYADSVKGRFTISRDNAKNTVWLQMN SLKPEDTAVYFCNARSWNNYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 320)	QVQLQESGGGLVQPGGSLRLSCAASGFTFSSHSMSWYRQAPG KQRELVAAITTYGSTNYIDSVKGRFTISRDNKTNTVYLQMNSLK PEDTAVYFCNARSWNNYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 321)	QVQLQESGGGLVQAGGSLRLSCVASGSSFSHNTMGWYRQAP GKQRDLVARITTFGTTNYADSVKGRFTISRDNAKNTVYLQMNS LKPEDTAVYYCNGESFGRWYNWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 322)	QVQLQESGGGLVQAGASLRLTCTASGGRFSYATMGWSRQAP GKQREMVARITSSGFSTNYADSVKGRFTISRDNAKNAVYLMQ DSLKPEDTAVYYCNAQHFGTDSWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 323)	QVQLQESGGGLVQAGASLRLTCTASGSRFSYATMGWSRQAP GKQRELVARITSSGFSTNYADSVKGRFTISRDNAKNAVYLMQD SLKPEDTAVYYCNAQQFGTDSWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 324)	QVQLQESGGGLVQAGGSLRLSCAASGSTFTSNVMGWHRQAP GKQRELVANMHSGGSTNYADSVKGRFTISRDNAKNIVYLMQMN NLKIEDTAVYYCRWYGIQRAEGYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 325)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTVSRDNAKNTVYLQMNSL QPEDTAVYYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 326)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNTVYLQMNSL QPEDTAVYYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 327)	QVQLQESGGGLVVGGSRLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNTVYLQMNSL QPEDTAAYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 328)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTISRDAENTVYLQMNSL QPEDTAVYYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 329)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTISRDAENTVYLQMNSL QPEDTAVYYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 330)	QVQLQESGGGLVQPGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISTDGSTNYVDSVKGRFTISRDNAKNTVYLQMNSL QPEDTAVYYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 331)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISTDGTNTYVDSVKDRFTISRDNAKNTVYLQMNSL QPEDTAAYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 332)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSIWYRQAPGK KRELVAGISTDGTNTYVDSVKDRFTISRDNAKNTVYLQMNSLQ PEDTAAYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 333)	QVQLQESGGGLVQAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISTDGTNTYVDSVKDRFTISRDNAKNTVYLQMNSL QPEDTAAYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 334)	QVQLQESGGGLVQAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISTDGSTNYADSVKGRFTISEGNAKNTVDLQMNSL QPEDTAVYYCYAYRWVDRITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 335)	QVQLQESGGGLVAVAGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISTDGSTNYADSVKGRFTISEDNAKNTVDLQMNSL QPEDTAVYYCYAYRWIDRYTYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 336)	QVQLQESGGGLVQPGGSLRLSCAASGSSVSFLSMWYRQAPG KKRELVAGISTDGSTNYADSVKGRFTISEDNAKNTVDLQMNSL QPEDTAVYYCYAYRWVDRITYWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 337)	QVQLQESGGGLVVGAGSLRLSCAASGSSVSFLSMAYRQAPG KKRELVAGISSDGSTNYVDSVKGRFTISRDNKNIVFLQMNSLQ PQDTAVYYCYAYRWVGRDITYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 338)	QVQLQESGGGLVVGAGSLRLSCAASGSSVSFLSMAYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDSANNTMYLQMNSL QPEDTAVYYCYAYRWTRYTYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 339)	QVQLQESGGGLVQPGGSLRLSCAASGFTLDYYAIGWYRQAPG KKRELVAGISSDGSTHYVDSVKGRFAISRDNANTVYLQMNDL QPDDTAVYYCYAYRWGGYTYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 340)	QVQLQESGGGLVQAGGSLRLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKNYADSVKGRFTISRDNKNTVYLQMNSL KPEDTAVYYCYFYRTVAASSMQYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 341)	QVQLQESGGGLVQAGGSLRLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNKNTVYLQMNSL KPEDTAVFYCYFYRTVSGSSMRYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 342)	QVQLQESGGGLVQAGGSLRLSCAASGITSSVYSMGWYRQAPG KQRELVAGSSSDGSTHYVDSVGRFTISRDNKNTVYLQMSSL KPEDTAVYYCYANRFGAGAPSYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 343)	QVQLQESGGGLVQAGGSLRLSCAASGRTSMFNSMGWHRQA PGKQRELVAIIRSGGSSNYADTVKGRFTISRDNKNTVYLQMN DLKPEDTAVYYCFYYFQSSYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 344)	QVQLQESGGGLVQAGGSLRLSCAASGRTSMVNSMGWHRQA PGKQRELVALITSGGSSNYADTVKGRFTISRDNKNTVYLQMN DLKPEDTAVYYCFYYFQSSYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 910)	QVQLQESGGGLVQAGGSLRLSCAASGSSVMFNSMGWHRQP PGKQRELVAIITSGGSSNYADTVKGRFTISRDNKNTVYLQMN DLKPEDTAVYYCFYYFQSSYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 345)	QVQLQESGGGLVQAGGSLRLSCTASGSIFSIAVMGWYRQVPG KREWVATIFDGSYTNADSVKGRFTISRDNARNKVYLMNN LKPEDTAVYYCQTHWTQGSVPKESWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 346)	QVQLQESGGGLVQAGGSLRLSCVASSGIFSDMSMVYRQAP GKQRELVASITTFGSTNYADPVKGRFTISRDNKNTVYLQMNSL KPEDTAVYYCSGRSYSSDYWGRGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 347)	QVQLQESGGGLVQAGGSLRLSCVASGSISSIIVMGWSRQAPGK QRESVATITRDGTRNYADSLKGRFTISRDNKNTSYLQINSLKPE DTAVYSCYARYGDINWKGKTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 348)	QVQLQESGGGLVQAGGSLRLSCVASGSISSIIVMGWSRQAPGK QRESLATISRGGTRTYADSVKGRFTISRDNKNTSYLQMNSLKP EDTAVYSCYARYGDINWKGKTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 349)	QVQLQESGGGLVQAGGSLRLSCVASGSIFTTNSMGWHRQGP GKQRELVALIGSAGSTKYADSVKGRFTISRDNKNTVSLQMDSL KPEDTAVYYCFYYDSRSYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 350)	QVQLQESGGGMVQPGGSLRLSCAASGSREISTMGWHRQAPG KQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMNSLK SEDTAVYYCFAYDNINAYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 351)	QVQLQESGGGWVQAGGSLRLSCAASGSREISTMGWHRQAP GKQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMNSL KSEDTAVYYCFAYDNINAYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 352)	QVQLQESGGGWVQAGGSLRLSCTASGSREISTMGWHRQAPG KQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMDSLK SEDTAVYYCFAYDNINAYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 353)	QVQLQESGGGSVQAGRSLGLSCAASGSREISTMGWHRQAPG KQRELAARITSGGITKYADSVKGRFTISRDNAKKTVYLQMNSLK SEDTAVYYCFAYDNINAYWGQGTQVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 354)	QVQLQESGGGLVQAGGSLRLSCTASGSI FRGAAMYWHRQAP GKQRELVAAITTS GNTSYADSVKGRFTISRDNAKNTMYLQIISLK PEDTAVYYCAFWIAGKAYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 355)	QVQLQESGGGLVQPGGSLRLSCAASGSI SSFNFMSWHRQAPG KERELAGVITRGGATNYADSVKGRFTISRDNVKN TVYLQMNSL KPEDTAVYYCHGRS QLGS TWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 356)	QVQLQESGGGLVQAGGSLRLSCLASGTIFTASTMGWHRQPPG KQRELVASIAGDGRNTYAESTEGRFTISRDDAKNTMYLQMNSL KPEDTAVYYCYAYYLD TYAYWGQGTQVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 357)	EVQLVESGGGLVQPGGSLTLSCAASGSI FSIASMGWYRQAPGK QRELVAVITSFSSSTNYADSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCNARYFERTDWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 358)	EVQLVESGGGLVQPGGSLTLSCAASGRTSMFNSMGWHRQAP GKQRELVAIIRSGSSNYADTVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCFYFQSSYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 359)	EVQLVESGGGLVQPGGSLTLSCAASGSSVSFLSMAWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWVGRD TYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 360)	EVQLVESGGGLVQPGGSLTLSCTASGSI FRGAAMYWHRQAPG KQRELVAAITTS GNTSYADSVKGRFTISRDNAKNSMYLQMNSL RAEDTAVYYCAFWIAGKAYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 361)	EVQLVESGGGLVQPGGSLTLSCAASGSTFNIKTMAWHRQAPG NQRELVAITLTSGGNTNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCGLWNGVGGAYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 362)	EVQLVESGGGLVQPGGSLTLSCAASGSTLNKIMAWHRQAPGK QRELVAITLTSGGNTNYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCGLWDGVGGAYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 363)	EVQLVESGGGLVQPGGSLTLSCAASSI FSISSMSWYRQAPGK QRELVAAITTFDYTNADSVKGRFTISRDNAKNSMYLQMNSLR AEDTAVYYCNARAFGRDYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 364)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFFRTVAASSMQYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 365)	EVQLVESGGGLVQPGGSLTLSCAASGFMFSSYSMSWYRQAPG KQRELVAAITSYGSTNYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCNARSWNNYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 366)	EVQLVESGGGLVQPGGSLTLSCAASGSVMFNSMGWHRQPP GKQRELVAIITSGGSSNYADTVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCFYFQSSYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 367)	EVQLVESGGGLVQPGGSLTLSCTASGGRFSYATMGWSRQAPG KQREMVARITSSGFSTNYADSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCNAQHFGTDSWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 368)	EVQLVESGGGLVQPGGSLTLSCAASGSI SSFNFMSWHRQAPG KERELAGVITRGGATNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCHGRS QLGS TWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 369)	EVQLVESGGGLVQPGGSLTLSCAASESI FSINVMWHRQAPGK QRELVARITSGGSTNYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCGAYQGLYAYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 370)	EVQLVESGGGLVQPGGSLTLSCAASGSREISTMGWHRQAPGK QRELAARITSGGITKYADSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCFAYDNINAYWGQGT LVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 371)	EVQLVESGGGLVQPGGSLTLSCTASGSI FSIAMGWYRQVPGK RREWVATIFDGSYTNADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCQTHWTQGSVPKESWGQGT LVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 372)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 373)	EVQLVESGGGLVQPGGSLTSLSCVASSGIFSDMSMVYRQAPG KQRELVASITTFGSTNYADPVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCSGRSYSSDYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 374)	EVQLVESGGGLVQPGGSLTSLSCAASGFMFSSYSMSWYRQAPG KQRELVAAITTWGSTNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCNARSWNNYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 375)	EVQLVESGGGLVQPGGSLTSLCTASGSRFSYATMGWSRQAPG KQRELVARITSSGFSTNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCNAQQFGTDSWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 376)	EVQLVESGGGLVQPGGSLTSLSCAASGSTFTSNVMGWHRQAP GKQRELVANMHSGGSTNYADSVKGRFTISRDNAKNSVYLQM NSLRAEDTAVYYCRWYGIQRAEGYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 377)	EVQLVESGGGLVQPGGSLTSLSCAASGSSVSFLSMAWYRQAPG KKRELVAGISTDGSTNYVDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWGRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 378)	EVQLVESGGGLVQPGGSLTSLSCVASGSISSIIVMGWSRQAPGK QRESVATITRDGTRNYADSLKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYARYGDINYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 379)	EVQLVESGGGLVQPGGSLTSLSCAASGSSVSFLSMAWYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 380)	EVQLVESGGGLVQPGGSLTSLCLASGTIFTASTMGWHRQPPGK QRELVASIAGDGRNTYAESTEGRFTISRDNAKNSMYLQMNSLR AEDTAVYYCYAYLDTYAYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 381)	EVQLVESGGGLVQPGGSLTSLSCAASGSSVSFLSMAWYRQAPG KKRELVAGISVDGSTNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWEGRNTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 382)	EVQLVESGGGLVQPGGSLTSLSCVASGSSFSHNTMGWYRQAPG KQRDLVARITTFGTNYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCNGESFGRIWYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 383)	EVQLVESGGGLVQPGGSLTSLSCVASGSISSIIVMGWSRQAPGK QRESLATISRGGTRTYADSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYARYGDINYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 384)	EVQLVESGGGLVQPGGSLTSLSCVASGSSFSITSMWYRQAPGK QRDLVAAITSPGSTNYADSVKDRFTISRDNAKNSVYLQMNSLR AEDTAVYYCNGRVFDHVVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 385)	EVQLVESGGGLVQPGGSLTSLSCAASGRISMVNSMGWHRQAP GKQRELVALITSGSSNYADTVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCFYFQSSYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 386)	EVQLVESGGGLVQPGGSLTSLSCAASGSTFNFKIMAWHRQAPG KQRELVASLTSEGLTNYRDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCGLWDGVGGAYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 387)	EVQLVESGGGLVQPGGSLTSLSCAASGFTLDYYAIGWYRQAPGK KRELVAGISSDGSTHYVDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWGGYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 388)	EVQLVESGGGLVQPGGSLTSLSCVASGSIFTTNSMGWHRQGP KQRELVALIGSAGSTKYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCFYYDSRSYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 389)	EVQLVESGGGLVQPGGSLTSLSCAASGITSSVYSMGWYRQAPG KQRELVAGSSSDGSTHYVDSVGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYANRGFAGAPSYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 390)	EVQLVESGGGLVQPGGSLTSCAASSSIFSISSMSWYRQAPGK QRELVAAITSPGSTNYADSVKGRFTISRDNAKNSMYLQMNSLR AEDTAVYYCNARTMGRDYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 391)	EVQLVESGGGLVQPGGSLTSCVASGFTSSINAMGWYRRAPG KQRELVAGISSDGSFVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRHVS GSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 392)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRFTVSGSSSRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 393)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVSGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 394)	EVQLVESGGGLVQPGGSLTSCVASGSPSSINAMGWYRRAPG KQRELSAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVRGSSMYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 395)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMGWYRRAPG KQRELAAGISSDGSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVSGSSKRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 396)	EVQLVESGGGLVQPGGSLTSCVASGSISSINAMGWYRRAPGK QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFMVSGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 397)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKLYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVQSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 398)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAYGWYRRAPGK QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVYSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 399)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYFRFTVSGSSYRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 400)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVLGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 401)	EVQLVESGGGLVQPGGSLTSCVASGSTSIINAMGWYRRAPGK QRELAAGISSDGSKVIADSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRFRVSGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 402)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKIYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVSGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 403)	EVQLVESGGGLVQPGGSLTSCVASGKTSSINAMAWYRRAPG KQRELVAGISSDGSKVYTDVSKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRFTVSGSSARYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 404)	EVQLVESGGGLVQPGGSLTSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRIVRGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 405)	EVQLVESGGGLVQPGGSLTSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRVTVSGSSMRVWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 406)	EVQLVESGGGLVQPGGSLTSCVASGSGSSINAMGWYRRAPG KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRHVS GSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 407)	EVQLVESGGGLVQPGGSLTSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRFVSGSSMRVWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 408)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 409)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTKSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 410)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVYSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 411)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYRDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 412)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRSWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 413)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVGGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 414)	EVQLVESGGGLVQPGGSLTSLSCVASGNTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 415)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSHMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 416)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSIINAMGWYRRAPGK QRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRAVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 417)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 418)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELPAIGISSDGSKVYAVSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSPMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 419)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELVAGVSSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 420)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 421)	EVQLVESGGGLVQPGGSLTSLSCVASGITSSINAMGWYRRAPGK QRELVAGISSDGSKVYAGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 422)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSDINAMGWYRRAPG KQRELVAGISSDKSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 423)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSNGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRQVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 424)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVLADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRIVSGSSMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 425)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSKNAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGASMRYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 426)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVHGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 427)	EVQLVESGGGLVQPGGSLTSLSCVASGLTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRMVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 428)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYTDVSKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTISGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 429)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYTDVSKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 430)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGHSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 431)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSHINAMGWYRRAPG KQRELVAGISSDGSRVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGGSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 432)	EVQLVESGGGLVQPGGSLTSLSCVASGQTSSINAMGWYRRAPG KQRELVAGISSDGSQVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTKSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 433)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINGMGWYRRAPG KQRELPAGISDGSKAYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTASGTSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 434)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSVINAMAWYRRAPG KQRELAAGISSDGSKVYAKSAKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFNTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 435)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYNDVSKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVRGSSQRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 436)	EVQLVESGGGLVQPGGSLTSLSCVASGKTSSINAMGWYRRAPG KQRELVAGISSDGSKVIADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVLGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 437)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYTDVSKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 438)	EVQLVESGGGLVQPGGSLTSLSCVASGSVSSINAMGWYRRAPG KQRELVAGISSDGSKVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGLSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 439)	EVQLVESGGGLVQPGGSLTSLSCVASGNTSSINAMGWYRRAPG KQRELVAGISSDGSKVYYDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRGSSQRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 440)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSSTNAMGWYRRAPG KQRELVAGISSDGSKVYVDVSKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMVYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 441)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYGDVSKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSRSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 442)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELAAGISSDQSKVYADSAKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 443)	EVQLVESGGGLVQPGGSLTSLSCVASGGTSSINAMGWYRRAPG KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSARYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 444)	EVQLVESGGGLVQPGGSLTLSCVASGSTRSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFHTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 445)	EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVIADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVLGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 446)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVDADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 447)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRNVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 448)	EVQLVESGGGLVQPGGSLTLSCVASGNTSSINAMGWYRRAPG KQRELVAGISSNGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVTGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 449)	EVQLVESGGGLVQPGGSLTLSCVASGSTRINAMGWYRRAPG KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 450)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVKGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 451)	EVQLVESGGGLVQPGGSLTLSCVASGLTSSINAMGWYRRAPG KQRELVAGISSDGSKVYQDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTNSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 452)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYAESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGASMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 453)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVLADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVNLSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 454)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKYIADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVTGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 455)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVYAVSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRKVS GSARYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 456)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVVDADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTYSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 457)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSKSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 458)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFKTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 459)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMAWYRRAPG KQRELAAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTTRSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 460)	EVQLVESGGGLVQPGGSLTLSCVASGSKSSINAMGWYRRAPG KQRELAAGISSDGSKVYAQSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTSSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 461)	EVQLVESGGGLVQPGGSLTLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRFLSGSSMRYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 462)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAFGWYRRAPGK QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 463)	EVQLVESGGGLVQPGGSLTSLSCVASGSTFSINAMGWYRRAPG KQRELVAGISSDGSKVLADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRLVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 464)	EVQLVESGGGLVQPGGSLTSLSCVASGSTRSINAMGWYRRAPG KQRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRFWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 465)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTQSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 466)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 467)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVVDVSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTLSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 468)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVYGDVSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSAMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 469)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYTDVSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTTSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 470)	EVQLVESGGGLVQPGGSLTSLSCVASGRTSSINAMGWYRRAPG KQRELVAGISSDGSKVYNDVSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGTSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 471)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSRNAMGWYRRAPG KQRELVAGISSDGSKVTADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTTRSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 472)	EVQLVESGGGLVQPGGSLTSLSCVASGSTKSINAMGWYRRAPG KQRELVAGISSDGSKVYRDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTSSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 473)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSRNAMGWYRRAPG KQRELVAGISSNGSKVYDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 474)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 475)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSLINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRHVSAGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 476)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTKSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 477)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFTTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 478)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGTKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFHTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 479)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAFGWYRRAPGK QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRGSSMRYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 480)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSRNAMGWYRRAPG KQRELVAGISSDGSGLYDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVLGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 481)	EVQLVESGGGLVQPGGSLTSLSCVASGNTSSINAMGWYRRAPG KQRELVAGISSDGSRVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRSWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 482)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVRGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 483)	EVQLVESGGGLVQPGGSLTSLSCVASGSTASINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 484)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAVGWYRRAPGK QRELVAGISSDGSKVYDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVYGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 485)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSRNAMGWYRRAPG KQRELVAGISSDGSGLYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVLGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 486)	EVQLVESGGGLVQPGGSLTSLSCVASGSTNSINAMGWYRRAPG KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 487)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRSVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 488)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELVAGISSDGSKVYQDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 489)	EVQLVESGGGLVQPGGSLTSLSCVPSGSTSNINAMGWYRRAPG KQRELPAIGSSDGTKIYADSAKVPFTITRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGTSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 490)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSKINAMGWYRRAPG KQRELVAGISSDRSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVAGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 491)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINALGWYRRAPGK QRELVAGISSDGSGLVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRIVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 911)	EVQLVESGGGLVQPGGSLTSLSCVASGKTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGVSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 912)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAVGWYRRAPGK QRELVAGISSDGSKVYRDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVQGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 492)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTASGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 493)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 494)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAVGWYRRAPGK QRELVAGISSDGTKVYRDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVQGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 495)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSINAMGWYRRAPG KQRELAAGISSDGSKVYNDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVRGSSMRYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 496)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTKSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 497)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVWGSMSRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 498)	EVQLVESGGGLVQPGGSLTSLSCVASGKTSSINAMGWYRRAPG KQRELVAGISSDGSKVYTDVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTKSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 499)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPFK QGELPAGISPDGTKAYADSAKVRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFHTVCGTSMGYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 500)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSAINAMGWYRRAPG KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSQRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 501)	EVQLVESGGGLVQPGGSLTSLSCVASGSPSSINAYGWYRRAPGK QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYFRTVSGSSMSYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 502)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYASSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVRGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 503)	EVQLVESGGGLVQPGGSLTSLSCVASGSRSSINAMGWYRRAPG KQRELVAGISADGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTQSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 504)	EVQLVESGGGLVQPGGSLTSLSCVASGSSINAMGWYRRAPG KQRELVAGISSDGSKVYASSAKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTLGSSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 505)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFHTVSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 506)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRRVSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 507)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRLVSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 508)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAVGWYRRAPGK QRELVAGISSDGSKVYAGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSYMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 509)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELAAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVGGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 510)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAYGWYRRAPGK QRELVAGISSDGSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTKSGSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 511)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAVGWYRRAPGK QRELVAGISSDGSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSTSSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 512)	EVQLVESGGGLVQPGGSLTSLSCVASGSKSSINAMGWYRRAPG KQRELPAAGISSNGTKVYADSAKVRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVLGTSMRYWGQGLTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 513)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRYWGQGLTLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 514)	EVQLVESGGGLVQPGGSLTSLSCVASGSVSSINAMGWYRRAPG KQRELVAGISSDGSKVYKDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 515)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRWQGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 516)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGSLVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRILSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 517)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVQGS SMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 518)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSYINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGQSMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 913)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGVSSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSARYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 914)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELPAGISRDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRYVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 915)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELAAGISSDGSKLYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSMRWYGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 916)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSRINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 917)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELAAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFHVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 918)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMAWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRQVSGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 919)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDTSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSYMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 920)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSTINAMGWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTASGSSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 519)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAVGWYRRAPGK QRELVAGISSDGSSTVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGHSMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 520)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAMGWYRRAPG KQRELAAGISKDGSKVYADSAKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVSGSSRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 521)	EVQLVESGGGLVQPGGSLTSLSCVASGSPSSINAYGWYRRAPGK QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTVSGSSSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 522)	EVQLVESGGGLVQPGGSLTSLSCVASGSPSSINAYGWYRRAPGK QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTVSGSSQSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 523)	EVQLVESGGGLVQPGGSLTSLSCAASGSSVKFLSMAWYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 524)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSLAWYRQAPGK KRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 525)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFSLMAWYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 526)	EVQLVESGGGLVQPGGSLTSLCAASGSQVSFSLMAWYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 527)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFSLMAWYRQAPG KKRELVAGISEAGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 528)	EVQLVESGGGLVQPGGSLTLRCAASGSKVSFSLMAWYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR RAEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 529)	EVQLVESGGGLVQPGGSLTSLCAASGSSVGFSLMAWYRQAPG KKRELVAGISADGSTDYIRSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 530)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFSLMAWYRQAPG KKRELVAGISADGSVDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 531)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFSLMAWYRQAPG KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 532)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSLAWYRQAPGK KRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 533)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFSLMAWYRQAPG KKRELVAGISGDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 534)	EVQLVESGGGLVQPGGSLTSLCAASGSSVQFSLMAWYRQAPG KKRELVAGISADGSTDYINSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 535)	EVQLVESGGGLVQPGGSLTSLCAASGSNVSFSLMAWYRQAPG KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYHWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 536)	EVQLVESGGGLVQPGGSLTSLSCVAGSSVKFSLMAWYRQAPG KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 537)	EVQLVESGGGLVQPGGSLTSLCAASGKSVSFSLMAWYRQAPG KKRELVAGISKDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 538)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFSLMAWYRQAPG KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 539)	EVQLVESGGGLVQPGGSLTSLCAASGSHVSFSLMAWYRQAPG KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 540)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFSLMAWYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 541)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFSLMAWYRQAPG KKRELVAGISADGSADYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 542)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISAHGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 543)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 544)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISRDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTGYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 545)	EVQLVESGGGLVQPGGSLTSLCAASGSHVSFLSMAWYRQAPG KKRELVAGISADGPTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWDTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 546)	EVQLVESGGGLVQPGGSLTSLSCVASGTSVFLSMAWYRQAPG KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRTTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 547)	EVQLVESGGGLVQPGGSLTSLCAASGTSVSFLSIWYRQAPGK KRELVAGISADGSTDYIASVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 548)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAWYRQAPG KKRELVAGISLDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRGYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 549)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAWYRQAPG KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 550)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISAHGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 551)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAWYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWITRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 552)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWITRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 553)	EVQLVESGGGLVQPGGSLTSLCAASGSSVFLSMAWYRQAPG KKRELVAGISADGSMIDYIDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 554)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 555)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYSWTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 556)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 557)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRLSMAWYRQAPG KKRELVAGISANGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 558)	EVQLVESGGGLVQPGGSLTSLCAASGSSKFLSMAWYRQAPG KKRELVAGISADGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 559)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRLSMAWYRQAPG KKRELVAGISADGSRDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 560)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGISADGSTMYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWHTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 561)	EVQLVESGGGLVQPGGSLTSLCAASGSGVRFLSMAYRQAPG KKRELVAGISPDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 562)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWMTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 563)	EVQLVESGGGLVQPGGSLTSLCAASGSSVHFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 564)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 565)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTYFWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 566)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 567)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAYRQAPG KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 568)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWATRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 569)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWHTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 570)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWGTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 571)	EVQLVESGGGLVQPGGSLTSLCAASYSSVSRLSMAYRQAPG KKRELVAGISADGSTVIYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRNTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 572)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 573)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTYAYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 574)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISADGRTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 575)	EVQLVESGGGLVQPGGSLTSLCVASGTSVSFLSMAYRQAPG KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWTRRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 576)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 577)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTSRYTYWGQGLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 578)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISKDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRVTYWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 579)	EVQLVESGGGLVQPGGSLTSLCAASGSSSVLSMAWYRQAPG KKRELVAGISADGSTDYIGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRTTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 580)	EVQLVESGGGLVQPGGSLTSLCAASGSSSVSFLSMAYRQAPG KKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 581)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTGYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWATRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 582)	EVQLVESGGGLVQPGGSLTSLSCVAGSSVKFLSMAYRQAPG KKRELVAGISDGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 583)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYALRWTTTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 584)	EVQLVESGGGLVQPGGSLTSLCAASGSSVQLSMAYRQAPG KKRELVAGISADGSTDYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRGTYWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 585)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 586)	EVQLVESGGGLVQPGGSLTSLCAASKSSVSFLSMAYRQAPGK KRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWTTTRATYWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 587)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 588)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWPTTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 589)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISQGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 590)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISNDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWKTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 591)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 592)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSLAWYRQAPGK KRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWKTRRTYWGGTLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 593)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 594)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAYRQAPG KKRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRYRYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 595)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISADGSTNYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTWGGTLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 596)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRYKYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 597)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAYRQAPG KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWKTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 598)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGISADGSTDYIGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRVTYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 599)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRFTYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 600)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRFTYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 601)	EVQLVESGGGLVQPGGSLTSLCAASGSSVFLSMAYRQAPG KKRELVAGVSDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 602)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGHTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRYTHWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 603)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTDYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 604)	EVQLVESGGGLVQPGGSLTSLCAASGSSVGFSLMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 605)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFMSMAYRQAP GKKRELVAGISADGSTDYIASVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWTTSTRTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 606)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISADGSTDYISSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYSWTTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 607)	EVQLVESGGGLVQPGGSLTSLCAASGSSVTFLSMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRGTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 608)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWKTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 609)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAYRQAPG KKRELVAGISADGSTTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRFTYWQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 610)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFMSMAYRQAP GKKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 611)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSNLSMAYRQAPG KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTTRTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 612)	EVQLVESGGGLVQPGGSLTSLCAASNSSVSKLSMAYRQAPG KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 613)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAYRQAPG KKRELVAGISADGSKDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTTRLTYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 614)	EVQLVESGGGLVQPGGSLTSLSCVASGSQVSFLSMAYRQAPG KKRELVAGISADGSTDYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 615)	EVQLVESGGGLVQPGGSLTSLSCAASGSKVSFMSMAWYRQAP GKKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR RAEDTAVYYCYAYRWTRRLTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 616)	EVQLVESGGGLVQPGGSLTSLSCAASGSRVSFMSMAWYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 617)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVRFLSMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 618)	EVQLVESGGGLVQPGGSLTSLSCAASGSKVSFLSMAYRQAPG KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYQWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 619)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVRFLSMAYRQAPG KKRELVAGISATGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 620)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVSFLSIAYRQAPGK KRELVAGISKDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWTRRMTYWQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 621)	EVQLVESGGGLVQPGGSLTSLSCAASGSSSFLSMAYRQAPGK KRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWTRRTRYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 622)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVKFLSMAYRQAPG KKRELVAGISPDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYRYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 623)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVNFLSMAYRQAPG KKRELVAGISADGSTHYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWLTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 624)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVKFLSMAYRQAPG KKRELVAGISADGSTDYIHSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYEWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 625)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVSFLSMAYRQAPG KKRELVAGISADGSTDYIHSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 626)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVRFLSMAYRQAPG KKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 627)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVSFLSAWYRQAPGK KRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWTRRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 628)	EVQLVESGGGLVQPGGSLTSLSCAASGSQVSFLSMAYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 629)	EVQLVESGGGLVQPGGSLTSLSCAASGS TSVFLSMAYRQAPG KKRELVAGISADGSTDYIRSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRLTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 630)	EVQLVESGGGLVQPGGSLTSLSCAASGS RVSFMSMAWYRQAPG KKRELVAGISADGSTMYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRLTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 631)	EVQLVESGGGLVQPGGSLTSLSCAASGS SVKFLSMAYRQAPG KKRELVAGISDGDSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYKWTRRYTYWGQGLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 632)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSSAWYRQAPGK KRELVAGISADGSTLYIDSVKGRFTISRDNAKNSVYLQMNSLR EDTAVYYCYAYRWTTRSTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 633)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAWYRQAPG KKRELVAGISADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 634)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAWYRQAPG KKRELVAGISATGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 635)	EVQLVESGGGLVQPGGSLTSLCAASGSTVSFLSMAWYRQAPG KKRELVAGISHDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 636)	EVQLVESGGGLVQPGGSLTSLCAASGSSVQFLSMAWYRQAPG KKRELVAGISYDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 637)	EVQLVESGGGLVQPGGSLTSLCAASRSSVSFLSMAWYRQAPG KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWLTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 638)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAWYRQAPG KKRELVAGISADGSTAYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 639)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAWYRQAPG KKRELVAGISADGSTDYIESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 640)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAWYRQAPG KKRELVAGISIDGSTDYIKSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 641)	EVQLVESGGGLVQPGGSLTSLCAASGSKVSFLSMAWYRQAPG KKRELVAGISADGSKDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 642)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISADGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWPTTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 643)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAWYRQAPG KKRELVAGISRDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRHTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 644)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAWYRQAPG KKRELVAGISADGSTDYIHSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 645)	EVQLVESGGGLVQPGGSLTSLCAASGSSVILSMAWYRQAPGK KRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRAE DTAVYYCYAYRWHTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 646)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSVAWYRQAPGK KRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWTRNRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 647)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAWYRQAPG KKRELVAGISTDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 648)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAWYRQAPG KKRELVAGISYDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRTRYTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 649)	EVQLVESGGGLVQPGGSLTSLCAASGHSVSFLSMAWYRQAPG KKRELVAGISADGSTDYIASVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 650)	EVQLVESGGGLVQPGGSLTSLCAASGSSVRFLSMAYRQAPG KKRELVAGISADGSTDYIGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 651)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSFLSMAYRQAPG KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWRTTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 652)	EVQLVESGGGLVQPGGSLTSLCAASGSRVSFLSMAYRQAPG KKRELVAGISADGSTYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 653)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGVSADGSTDYIDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYEWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 654)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSRLSMAYRQAPG KKRELVAGISARGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRSTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 655)	EVQLVESGGGLVQPGGSLTSLCAASGRSVSFLSMAYRQAPG KKRELVAGISADGSTIYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 656)	EVQLVESGGGLVQPGGSLTSLCAASGRSVSFLSMAYRQAPG KKRELVAGISANGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWSTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 657)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSMAYRQAPG KKRELVAGISADGSTDYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWSTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 658)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSKLSMAYRQAPG KKRELVAGISADGSTDYRDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 659)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSRLSMAYRQAPG KKRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTRRLTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 660)	EVQLVESGGGLVQPGGSLTSLCAASGSSVKFLSLAWYRQAPGK KRELVAGISADGSTDYILSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYEWTRTRYTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 661)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSRLSLAWYRQAPGK KRELVAGISVDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWTRRLTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 662)	EVQLVESGGGLVQPGGSLTSLCVASGTSSSINAMGWYRRAPG KQRELVAGISSDGSKVFNESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFPAAGSPMRWYGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 663)	EVQLVESGGGLVQPGGSLTSLCVASGTTSSINAIGWYRRAPGK QRELVAGISSDGSEVYTDVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTVDGSPLRWYGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 664)	EVQLVESGGGLVQPGGSLTSLCVASGSTSSINAMAWYRRAPG KQRELVAGISSDDSNVYVESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSKRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 665)	EVQLVESGGGLVQPGGSLTSLCVASGQTYRVNFAFWYRRAPG KQRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFSAAGSGTEMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 666)	EVQLVESGGGLVQPGGSLTSLCVASGSTSSINAMAWYRRAPG KQRELVAGISSDESTLYVDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFGLSGSSTTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 667)	EVQLVESGGGLVQPGGSLTSLCVASGSASLTNATGWYRRAPGK QRELVAGISSDDSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFPGSVSGSWTRYWGQGLTVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 668)	EVQLVESGGGLVQPGGSLTLSCVASGYPSLNNAMGWYRRAPG KQRELVAGISSDGSQVYGASVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRLVSGSSMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 669)	EVQLVESGGGLVQPGGSLTLSCVASGSSSTINAI GWYRRAPGK QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTSGGTSKSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 670)	EVQLVESGGGLVQPGGSLTLSCVASGSTSYINAMGWYRRAPG KQRELVAGISSDGSNMYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFSPNMSGTTRRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 671)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSVNALGWYRRAPGK QRELVAGISSDGSKVYTD SVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVPGSAMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 672)	EVQLVESGGGLVQPGGSLTLSCVASGSTSLNNAV GWYRRAPGK QRELVAGISSDGSKVAESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFFRAESGSSMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 673)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSTNAI GWYRRAPGK QRELVAGISSDGSKVYDSDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTLYGSSRSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 674)	EVQLVESGGGLVQPGGSLTLSCVASGLTSTINAMGWYRRAPG KQRELVAGISSDGSKVYDSDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFSPFSGSDTGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 675)	EVQLVESGGGLVQPGGSLTLSCVASGVSPSKNAI GWYRRAPGK QRELVAGISSDGS AVYVGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFSTFGSSISYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 676)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAV GWYRRAPGK QRELVAGISSDGSYVYSESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTLASSEMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 677)	EVQLVESGGGLVQPGGSLTLSCVASGSTTMMNAMAWYRRAP GKQRELVAGISSDSSHVYADSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYFRTVSGSGVRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 678)	EVQLVESGGGLVQPGGSLTLSCVASGSTSKINAI GWYRRAPGK QRELVAGISSDSSIVYTD SVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFFRPGAGHSNSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 679)	EVQLVESGGGLVQPGGSLTLSCVASGQTALNAMGWYRRAP GKQRELVAGISSDGS EVNTDSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYFFRASGTAMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 680)	EVQLVESGGGLVQPGGSLTLSCVASGATSSINAI GWYRRAPGK QRELVAGISSDGSKLSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFTSASGTDLSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 681)	EVQLVESGGGLVQPGGSLTLSCVASGSTSTINAMGWYRRAPG KQRELVAGISSDNSKVYADSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFFRSANGSSKRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 682)	EVQLVESGGGLVQPGGSLTLSCVASGSTSSINAMGWYRRAPG KQRELVAGISSDGS RVYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFKTIAGAMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 683)	EVQLVESGGGLVQPGGSLTLSCVASGSTSLVNAMGWYRRAPG KQRELVAGISSDGS LVYAESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFFRYGSSLSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 684)	EVQLVESGGGLVQPGGSLTLSCVASGSTSLNNAI GWYRRAPGK QRELVAGISSDGSVVYVDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVPGASKYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 685)	EVQLVESGGGLVQPGGSLTLSCVASGSTSPVNAMAWYRRAPG KQRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVDGSAISYWGQGLTVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 686)	EVQLVESGGGLVQPGGSLTSLSCVASGTTSSMNAIGWYRRAPG KQRELVAGISSDGSGLYDESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVKSGGSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 687)	EVQLVESGGGLVQPGGSLTSLSCVASGETSSINAMAWYRRAPG KQRELVAGISSDYSKLYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGSSRGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 688)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSTINAIGWYRRAPGK QRELVAGISSDSSKVYTESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRRPGPSQMAYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 689)	EVQLVESGGGLVQPGGSLTSLSCVASGSTYSMNAMGWYRRAP GKQRELVAGISSDGSQVYVDSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYFRTVAGSASGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 690)	EVQLVESGGGLVQPGGSLTSLSCVASGSPSSINAYGWYRRAPGK QRELVAGISSDGSKVYSDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTVSGSSSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 691)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSTINAIGWYRRAPGK QRELVAGISSDGSKVYVDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRLNKGSSMAYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 692)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDGSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRMVTGSYGGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 693)	EVQLVESGGGLVQPGGSLTSLSCVASGSISSINAMGWYRRAPGK QRELVAGISSDGSVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFVKSSYGLPMRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 694)	EVQLVESGGGLVQPGGSLTSLSCVASGSTQVNNAMAWYRRAP GKQRELVAGISSDGSQVYYGSKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYFRTVSGQSLRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 695)	EVQLVESGGGLVQPGGSLTSLSCVASGSTASFNAMAWYRRAPG KQRELVAGISSDGSKVYTDVSKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVTGRAARYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 696)	EVQLVESGGGLVQPGGSLTSLSCVASGSPLSINAIGWYRRAPGK QRELVAGISSDGSKVSADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFGPAIGASRTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 697)	EVQLVESGGGLVQPGGSLTSLSCVASGSTTFINAIGWYRRAPGK QRELVAGISSDGSKVYEDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFRTVSGAPKSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 698)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSSINAIGWYRRAPGK QRELVAGISSDRSKVYADSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYFHTVSGSSMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 699)	EVQLVESGGGLVQPGGSLTSLSCVASGETDTINAVGWYRRAPGK QRELVAGISSDGSKVYAESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRLEGYSNRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 700)	EVQLVESGGGLVQPGGSLTSLSCVASGSTSPINAIGWYRRAPGK QRELVAGISSDGSVVTTESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYFRTGSGSSMGYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 701)	EVQLVESGGGLVQPGGSLTSLSCVASGSISSNAMGWYRRAPG KQRELVAGISSDGSVHVHQSVEVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYFRTVTGSSMSYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 702)	EVQLVESGGGLVQPGGSLTSLSCAASRYSVSNLSMAWYRQAPG KKRELVAGISADGSTVYVESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCAYYWTERRPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 703)	EVQLVESGGGLVQPGDSLTLSCAASMSTSVLSMAWYRQAPG KKRELVAGISSDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAIYYCAYSWDDAHPYWGQGLTVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 704)	EVQLVESGGGLVQPGGSLTSLCAASDSYVSLLSMAWYRQAPG KKRELVAGISVDGSTHYVASVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWMTRLTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 705)	EVQLVESGGGLVQPGGSLTSLCAASDSAVSVLSIAWYRQAPGK KRELVAGISTDGSKHYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYDWADAQPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 706)	EVQLVESGGGLVQPGGSLTSLCAASHSSVTSLSLAWYRQAPGK KRELVAGISYDGSKYAESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWTDRLPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 707)	EVQLVESGGGLVQPGGSLTSLCAASDSVVKFLSMAWYRQAPG KKRELVAGISANGSRTYMESVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYRWATRLPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 708)	EVQLVESGGGLVQPGGSLTSLCAASDPSVWNLSMAWYRQAP GKKRELVAGISPDGSTDYVDSVKGRFTISRDNAKNSVYLQMNS LRAEDTAVYYCYAYKWSNRLPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 709)	EVQLVESGGGLVQPGGSLTSLCAASGTSVMLLSLAWYRQAPG KKRELVAGISPNGSAVYTESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYGWKTRQPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 710)	EVQLVESGGGLVQPGGSLTSLCAASSSPVSNLSLAWYRQAPGK KRELVAGISPDGSTAYMESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWPNNRRGYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 711)	EVQLVESGGGLVQPGGSLTSLCAASWRSVLLSVAWYRQAPG KKRELVAGISNDGSTDYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYDWTTTRQRYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 712)	EVQLVESGGGLVQPGGSLTSLCAASSSSVQYLSMAWYRQAPG KKRELVAGISTDGSVAVYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYNWSYAQPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 713)	EVQLVESGGGLVQPGGSLTSLCAASGTSVSLLSLAWYRQAPGK KRELVAGISTGGSTHYIESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYNWTDSLQYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 714)	EVQLVESGGGLVQPGGSLTSLCAASLSSVSNLSIAWYRQAPGKK RELVAGISTDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRAE DTAVYYCYAYSWTTSLPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 715)	EVQLVESGGGLVQPGGSLTSLCAASMYSVSFLSMAWYRQAPG KKRELVAGISNEGSTYIMDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYKWSRSSTYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 716)	EVQLVESGGGLVQPGGSLTSLCAASKSSVSHLSLAWYRQAPGK KRELVAGISADGSHVYTNSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYSQTTRDPYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 717)	EVQLVESGGGLVQPGGSLTSLCAASYTSVLDLSIAWYRQAPGK KRELVAGISDDGSRYYTDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYRWARTDYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 718)	EVQLVESGGGLVQPGGSLTSLCAASMSDVSFLSMAWYRQAPG KKRELVAGISAEGSTLYMESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWTSRRLSYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 719)	EVQLVESGGGLVQPGGSLTSLCAASESSVSFLSSAWYRQAPGK KRELVAGISTDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWTTRSRYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 720)	EVQLVESGGGLVQPGGSLTSLCAASGDSVSLLSMAWYRQAPG KKRELVAGISANGSTSYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYNWTSSRYRYWGQGLVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 721)	EVQLVESGGGLVQPGGSLTSLCAASGSDVWYLSLAWYRQAPG KKRELVAGISDDGSRHYIESVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYSWKTRFPYWGQGLVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (DLL3) (SEQ ID NO: 722)	EVQLVESGGGLVQPGGSLTSLCAASKSAVAFLSIAYWRQAPGK KRELVAGISPDGSTVYIESVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWTTRYPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 723)	EVQLVESGGGLVQPGGSLTSLCAASFSAVAYLSMAWYRQAPG KKRELVAGISDDGSTVYVDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYEWTNALPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 724)	EVQLVESGGGLVQPGGSLTSLCAASVYSVYDLSTAWYRQAPGK KRELVAGISDDGSTVYFDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWITRSPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 725)	EVQLVESGGGLVQPGGSLTSLCAASGDSVSFSLMAWYRQAPG KKRELVAGISDEGSTVYIGSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYSWTTRQYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 726)	EVQLVESGGGLVQPGGSLTSLCAASSSSVSLSLAWYRQAPGKK RELVAGISDDGSIVYMDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWITRSPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 727)	EVQLVESGGGLVQPGGSLTSLCAASADSVSFLSIAYWRQAPGK KRELVAGISDDGSKHYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWEESRQYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 728)	EVQLVESGGGLVQPGGSLTSLCAASASSVTLTSLIAYWRQAPGKK RELVAGISTDGSTDYLHVSVKGRFTISRDNAKNSVYLQMNSLRAE DTAVYYCYAYTWTTRLPYWGQGLTVTVTS
Exemplary anti-target (DLL3) (SEQ ID NO: 729)	EVQLVESGGGLVQPGGSLTSLCAASADSVSFLSIAYWRQAPGK KRELVAGISDDGSKHYFDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWEESRQYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 730)	EVQLVESGGGLVQPGGSLTSLCAASGTSVWLLSMAWYRQAP GKKRELVAGISYDGSTVYVESVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYSWTTRQPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 731)	EVQLVESGGGLVQPGGSLTSLCAASGSSVSILSIAYWRQAPGKK RELVAGISDDGSTVYIDSVKGRFTISRDNAKNSVYLQMNSLRAE DTAVYYCYAYVWGTRLPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 732)	EVQLVESGGGLVQPGGSLTSLCAASGTAVSNLSIAYWRQAPGK KRELVAGISDDGSTVYVDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYEWTNALPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 733)	EVQLVESGGGLVQPGGSLTSLCAASGSAVSMLSLAWYRQAPG KKRELVAGISDDGSQVYIDSVKGRFTISRDNAKNSVYLQMNSLR AEDTAVYYCYAYRWEDALTYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 734)	EVQLVESGGGLVQPGGSLTSLCAASGMTVFFLSMAWYRQAP GKKRELVAGISVDGSTVYSDSVKGRFTISRDNAKNSVYLQMNSL RAEDTAVYYCYAYSWTTRYPYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 735)	EVQLVESGGGLVQPGGSLTSLCAASQYSVTFLSVAWYRQAPGK KRELVAGISDDGSNVYIDSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWIDSLRYWGQGLTVTVSS
Exemplary anti-target (DLL3) (SEQ ID NO: 736)	EVQLVESGGGLVQPGGSLTSLCAASGETVSFSLAWYRQAPGK KRELVAGISTDGSTVYFVSVKGRFTISRDNAKNSVYLQMNSLRA EDTAVYYCYAYSWTTPRAYWGQGLTVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 737)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQRELVATSTRDGNVDYAESVKGRFTISRDNAKNTAYLQMNSL KPEDTAVYYCNADLRATAVDLIRANYWGQGTQTVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 738)	QVQLQESGGGLVQAGDSLRLSCVVSGRDTSWYVMGWFRQA PGKDREFVAGVSWSYGNTYYADSVKGRFTASRDNAKNTAYLQ MNSLNAEDTAVYYCAARVSREVIPTRWDLINYWGQGTQVTV SS
Exemplary anti-target (EGFR) (SEQ ID NO: 739)	QVQLQESGGGSVQPGGSLRVSCVVSRTIISINAMTWYHQAPG KRELVAIITSGGETNYADSVKGRFTISRDNAKNTAYLQMNNLK PEDTGYYCNVPPPLGSGWGQGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (EGFR) (SEQ ID NO: 740)	QVQLQESGGGRVQAGGSLRLSCSASARTLRLYAVGWFRQAPG KREFVAGIGRSERTYYTDSVKGRFTLSRDNAKNTVFLEMDLE PEDTAVYFCALTFQTDMVDVPTTQHEYDYWGRGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 741)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTHDGNTDYADSVKGRFTISRDNVKNVTYV LQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 742)	QVQLQESGGGLVQPGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTRDGNTDYADSVKGRFTISRDN AKDTYV LQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 743)	QVQLQESGGGLVQAGGSLTLSCAASGRYQMAWFRQAPEKER EFVGTISSGDSTWYTN SVKGRFAISRDSARNTVY LQMNDLKPE DTAIYYCAAALYYRDSRRADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 744)	QVQLQESGGGRVQAGESLRLSCSTSTRTLKLYAVGWFRQAPG KERDFVAGIGRSERIYYIDSVKGRFTLSRDNAKNTVFLEMDLEP EDTAVYFCAATFQTSNDVGVPTVQHEYDYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 745)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTHDGNTDYADSVKGRFTISRDN AKNTVY LQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 746)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATFTRDGNTDYADSVKGRFTISRDN AKNTVY LQMNSL KPEDTAVYYCNTDLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 747)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTHDGNTDYADSLKGRFTISRDN AKNTVY LQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 748)	QVQLQESGGGSVQAGGSLRLSCAASGRYQMAWFRQAPEKER EFVGTISSGDSTWYTN SLKGRFAISRDSARDTVY LQMNDLKPE DTAVYYCAAALYYRDSRRADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 749)	QVQLQESGGGLVQTGGSLRLSCAVSGSIVTINAMTWYRQAPG KRELVAIITSGGETNYADSVKGRFTISRDN AKNTAHLQMNSLN PEDTGYYCINVVPPLGSGWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 750)	QVQLQESGGGLVQTGGSLRLSCAVSRISVSIKSM TWYRQAPGK RRELVALITSGGETNYSDSVKGRFTISRDN AKNTVY LQMNSLKP EDTGYYCINVVPPLGSGWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 751)	QVQLQESGGGLVETGGSLRLSCAGSGSTFRHHAMAWFRQTP GKREFVSAINDHGDR TKYLD SVRGRFTISRDN TDMVY LQM TDLRPEDTANYSCAAGPLVDYLETTPLVYTYWGHGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 752)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTHDGNTDYADSVKGRFTISRDN AKNTVY LQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 753)	QVQLQESGGGSVQAGGSLTLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTHDGNTDYADSVKGRFTISRDN AKNTVY LQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 754)	QVQLQESGGGLVQPGGSLRLSCTASVSI FSVNAV DWYRQSPG KERELVAIMTSDGSTNYGDSVKGRFTISRDN AKNTVY LQMNNL KPEDTAVYYCNTVPPRYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 755)	QVQLQESGGGLVQPGGSLRLSCAASGSIAYIYTMDWYRQAPG KQREL VATSTRDGNIDYADSVKGRFTISRDS AKNTVY LQMSSLK PEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 756)	QVQLQESGGGLVQAGGSLTLSCAASGRYQLAWFRQAPEKVRE FVGTISSGDSTWYTN SVKGRFAISRDSARNTVY LQMNDLKPED TAVYYCAAALYYRDSRRADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 757)	QVQLQESGGGSVQAGGSLRLSCAASGRYHMAWFRQAPEKER EFVGTISSGDSTWYTN SVKGRFAISRDSARNTAY LQMNDLKPE DTAVYYCAAALYYGDSRRADYPYWGQGTQVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (EGFR) (SEQ ID NO: 758)	QVQLQESGGGSVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQRELVATSTHDGNTDYTDSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 759)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQTPG KQRELVATSTRDANTDYAGSVKGRFTISRDNAKDTVYLQMNSL KPEDTAVYYCHADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 760)	QVQLQESGGGLVQAGGSLRLSCAASGRYNMAWFRQAPEKER EFVGTITSADSTWYTNVSKGRFAITQDSARNTVYLMNDLKPE DTAVYYCAAALYYGDSRRADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 761)	QVQLQESGGGLVQPGGSLRLSCAASGRYQMAWFRQAPEKER EFVGTISSGDSWYTNVSKGRFAISRDSARTTVYLMNDLKPE DTAVYYCAAALYYRDSWRAADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 762)	QVQLQESGGGLVQPGESLRLSCAATGRYHLAWFRQAPEKERE FVGTITSADSTWYTNVSKGRFAITRDSARNTVYLMNDLKPED TAVYYCAAALYYGDSRRADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 763)	QVQLQESGGGLVQAGGSLKLSCADSGRSFSNYIMGWFRQAP GKEREVAGLGWSPGNTYYADSVKGRFTISRDNAKNMVYLQ MNSLNPEDTAVYYCAARRGDVIYTPWNVVYWGQGTQVTVS S
Exemplary anti-target (EGFR) (SEQ ID NO: 764)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTVPDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 765)	QVQLQESGGGSVQAGGSLRLSCAAPGRYQMAWFRQAPEKER EFVGTISSGDSWYTNVSKGRFAISRDSARNTVYLMNDLKSE DTAVYYCAAALYYRDSRRADYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 766)	QVQLQESGGGSVQAGGSLRLSCAASGLTFSSYAMAWFRQAP GKQRELVARITSGGTTDYADSVKGRFTISRDNAKNTMYLQMN SLKPEDTAVYYCAADLTyrnlllklPHYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 767)	QVQLQESGGGLVQAGGSLRLSCAASGNIAYIYTMDWYRQAPG KQRELVATSTHDGSTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNADLRTVPDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 768)	QVQLQESGGGLVQPGGSLRLSCAASGSIAYIYTMDWYRQAPG KQRELVATSTWDGNTDYADSVKGRFTISRDNAKNTVYLQMNS LKPEDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 769)	QVQLQESGGGLVQAGGSLRLSCAASGSIAYIYTMDWYRQAPG KQRELVATSTHDGNTDYADSVKGRFTISRDNAKNTVYLQMSSL KPDDTAVYYCNADLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 770)	QVQLQESGGGLVQAGESLSLSCAASGNDFVITDMHWYRQAP GKQREWVATITRFATTNYADSVKGRFTISRDNAKNTWYLQMN SLKPDDTAVYYCKAIGLRGVPDVRQFEVWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 771)	QVQLQESGGGLVQAGGSLRLSCAASGAIAYIYGMGWYRQAP GNQRELVAAISSGGSTDYADSVKGRFTISRDNAKNTVYLQMSS LKPEDTAVYYCNADVRSRNLVRSYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 772)	QVQLQESGGGLVQAGGSLRLSCAASGRYHTAWFRQAPEKERE FVGTISSGDSWYTNVSKGRFAISRDSARNTVYLMNDLKPED TAVYYCAAALYYGDSRRAGDYPYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 773)	QVQLQESGGGLVQAGGSLRLSCAASGNIAYIYTMNWYRQAPG KQRELVATSTHAGNTDYADSVKGRFTISRDNAKNTVYLQMNSL KPEDTAVYYCNVDLRTAVDLIRANYWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 774)	QVQLQESGGGLVQPGGSLRLSCAASGNIAYIYTMGWYRQAPG KQRELVATSTHDGNSDYADSVKGRFTISRDNAKNTVYLQMNTL KPDDTAVYYCNADLRTVPDRIRGNFWGQGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 775)	EVQLLESGGGLVQPGGSLTLSCAASGSIAYIYTMDWYRQAPGK QRELVATSTRDGNVDYADSVKGRFTISRDNKNTLYLQMNSLR AEDTAVYYCNADLRTAVDLIRANYWGLGTQVTVSS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-target (EGFR) (SEQ ID NO: 776)	EVQLLESGGGLVQPGGSLTSLCAASGRDTSWYVMGWFRQAP GKDREFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLYLQM NSLRAEDTAVYYCAARVSREVIPTRWDLNYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 777)	EVQLLESGGGLVQPGGSLTSLCAASARTLRLYAVGWFRQAPGK EREFVAGIGRSERTYYTDSVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCALTFQTTDMVDVPTTQHEYDYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 778)	EVQLLESGGGLVQPGGSLTSLCAASGSIVTINAMTWYRQAPGK RRELVAIIITSGGETNYADSVKGRFTISRDN SKNTLYLQMNSLRAE DTAVYYCNVVPPLGSWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 779)	EVQLLESGGGLVQPGGSLTSLCAASGRYHMAWFRQAPGKERE FVGTISSGDSTWYTN SVKGRFTISRDN SKNTLYLQMNSLRAEDT AVYYCAAALYYGDSRAADYPYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 780)	EVQLLESGGGLVQPGGSLTSLCAASGSTFRHHAMAWFRQTPG KREFVSAINDHGDRTKYLD SVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCAAGPLVDYLETTPLVYTYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 781)	EVQLLESGGGLVQPGGSLTSLCAASGRSFSNYIMGWFRQAPG KREFVAGLGWSPGNTYYADSVKGRFTISRDN SKNTLYLQMNS LRAEDTAVYYCAARRGDVIYTTFPWNVYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 782)	EVQLLESGGGLVQPGGSLTSLCAASGLTFSSYAMAWFRQAPGK QRELVARITSGGTTDYADSVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCAADLTyrnllkLPHYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 783)	EVQLLESGGGLVQPGGSLTSLCAASVSIFSVNAV DWYRQSPGK ERELVAIMTSDGSTNYDSDVKGRFTISRDN SKNTLYLQMNSLR AEDTAVYYCNTVPPRYWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 784)	EVQLLESGGGLVQPGGSLTSLCAASGND FVITDMHWYRQAPG KQREWVATITRFATTNYADSVKGRFTISRDN SKNTLYLQMNSL RAEDTAVYYCKAIGLRGVPD VNRQFEVWGLGTQVTVSS
Exemplary anti-target (EGFR) (SEQ ID NO: 785)	EVQLLESGGGLVQPGGSLTSLCAASGAIAIYIGMGWYRQAPGK QRELVAAISSGGSTDYADSVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCNADVRTSRNLVRSDYWGLGTQVTVSS
Exemplary EGFR ProTritAC containing linker sequence L001 (identified by residues in bold and italics) and masking sequence M027(identified by residues that are bold and underlined) (SEQ ID No. 786)	EVQLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLEWSSISGSGRDTLYADSVKGRFTISRDN AK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGT LVTVSS GGGGKPLGLQARVVGGGGT QTVVTQEP SLTVSPGGTVTLT CASSTGAVTSGNYPN WVQQKP GQAPRGLIGG TKFLVPGTPARFSGSLGGKAALT LSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWQG GTLVTVSSGGGGG GGGSEVQLLESGGGLVQPGGSLTSLCAASGRDTSWYVMGW F RQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLY LQMNSLRAEDTAVYYCAARVSREVIPTRWDLNYWGLGTQVT VSSHHHHHH
Exemplary EGFR ProTritAC containing linker sequence L040 (identified by residues in bold and italics) and masking sequence M027(identified by residues that are bold and underlined) (SEQ ID No. 787)	EVQLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLEWSSISGSGRDTLYADSVKGRFTISRDN AK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGT LVTVSSGGGGP QASTGRSGGGGGGT QTVVTQEP SLTVSPGGTVTLT CASSTGAVTSGNYPN WVQQKP GQAPRGLIGG TKFLVPGTPARFSGSLGGKAALT LSGVQPEDEA EYYCTLWYSNRWVFGGGTKLTVLGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWQG GTLVTVSSGGGGG GGGSEVQLLESGGGLVQPGGSLTSLCAASGRDTSWYVMGW F RQAPGKDREFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLY LQMNSLRAEDTAVYYCAARVSREVIPTRWDLNYWGLGTQVT VSSHHHHHH
Exemplary EGFR ProTritAC containing linker sequence L041 (identified by residues in bold and italics) and masking	EVQLVESGGGLVQPGNSLRSLCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLEWSSISGSGRDTLYADSVKGRFTISRDN AK TTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGT LVTVSS GGGGPQSGTGRAAGGGGT

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DESCRIPTION	SEQUENCE
sequence M027 (identified by residues that are bold and underlined) (SEQ ID No. 788)	QTVVTQEP SLTVSPGGTVTLT CASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGG TKLTVLGGGSGGGSGGGSGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN NL KTEDTAVYYCVRHANFGNSYISYWAYWGQGLTVTVSS GGGSGGGSEVQLLES GGGLVQPGGSLT LSCAASGR TDSWYV MGWFRQAPGKDRFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAARVSREVIPT RWDLYNYWGL GTQVTVSSHHHHH
Exemplary EGFR ProTriTAC containing linker sequence L042 (identified by residues in bold and italics) and masking sequence M027 (identified by residues that are bold and underlined) (SEQ ID No. 789)	EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLEWSSISGSGRDTLYADSVKGRFTISRDN AK TTLYLQMN SLRPEDTAVYYCTIGGSLSVSSQGLTVTVSSGGGGP PASSGRAGGGGT QTVVTQEP SLTVSPGGTVTLT CASSTGAVTSGNYPNWVQQKP GQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEA EYYCTLWYSNRWVFGG TKLTVLGGGSGGGSGGGSGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLE WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN NL KTEDTAVYYCVRHANFGNSYISYWAYWGQGLTVTVSS GGGSGGGSEVQLLES GGGLVQPGGSLT LSCAASGR TDSWYV MGWFRQAPGKDRFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAARVSREVIPT RWDLYNYWGL GTQVTVSSHHHHH
Exemplary EGFR ProTriTAC containing linker sequence L045 (identified by residues in bold and italics) and masking sequence M027 (identified by residues that are bold and underlined) (SEQ ID No. 790)	EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLEWSSISGSGRDTLYADSVKGRFTISRDN AK TTLYLQMN SLRPEDTAVYYCTIGGSLSVSSQGLTVTVSSGGGGP IPVQGRAHGGGGTQTVVTQEP SLTVSPGGTVTLT CASSTGAVT SGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGK AALTLSGVQPEDEAEYYCTLWYSNRWVFGG TKLTVLGGGSG GGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKY AINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR DDSKNTAYLQMN NLKTEDTAVYYCVRHANFGNSYISYWAYW GQGLTVTVSSGGGSGGGS EVQLLESGGGLVQPGGSLTSCAASGR TDSWYVMGWFRQAP GKDRFVAGVSWSYGNTYYADSVKGRFTISR DNSKNTLYLQMN NSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVTVSSH HHHHH
Exemplary EGFR ProTriTAC containing a non cleavable linker sequence (identified by residues in bold and italics) and masking sequence M027 (identified by residues that are bold and underlined) (SEQ ID No. 791)	EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQGGG GGLDGNEEPGGLEWSSISGSGRDTLYADSVKGRFTISRDN AK TTLYLQMN SLRPEDTAVYYCTIGGSLSVSSQGLTVTVSSGGGGSG GGGGSGGVVGGGGTQTVVTQEP SLTVSPGGTVTLT CASSTGA VTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLG GKAALTLSGVQPEDEAEYYCTLWYSNRWVFGG TKLTVLGGG SGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFN KYAINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR DDSKNTAYLQMN NLKTEDTAVYYCVRHANFGNSYISYWAYW WGQGLTVTVSS GGGSGGGSEVQLLES GGGLVQPGGSLT LSCAASGR TDSWYV MGWFRQAPGKDRFVAGVSWSYGNTYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAARVSREVIPT RWDLYNYWGL GTQVTVSSHHHHH
Exemplary GFP TriTAC Sequence (SEQ ID No. 792)	QVQLVESGGALVQPGGSLRLSCAASGFPVNRYSMRWYRQAP GKEREWVAGMSSAGDRSSYEDSVKGRFTISRDDARNTVYLQMN NSLKPEDTAVYYCNVNVGPEYWGQGTQVTVSSGGGGSGGGS EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQAPG KGLEWSSISGSGRDTLYADSVKGRFTISRDN AKTTLYLQMN SL RPEDTAVYYCTIGGSLSVSSQGLTVTVSSGGGGSGGGS EVQLVE SGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLEWV ARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN NLKTE DTAVYYCVRHANFGNSYISYWAYWGQGLTVTVSSGGGGSGG GGSGGGSGTQTVVTQEP SLTVSPGGTVTLT CASSTGAVTSGNYP NWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLS GVQPEDEAEYYCTLWYSNRWVFGG TKLTVLHHHHH
Exemplary EGFR TriTAC Sequence (SEQ ID No. 793)	EVQLLESGGGLVQPGGSLTSCAASGR TDSWYVMGWFRQAP GKDRFVAGVSWSYGNTYYADSVKGRFTISR DNSKNTLYLQMN NSLRAEDTAVYYCAARVSREVIPTRWDLYNYWGLGTQVTVSSG GGSGGGSEVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGM SWVRQAPGKGLEWSSISGSGRDTLYADSVKGRFTISRDN AKT TLYLQMN SLRPEDTAVYYCTIGGSLSVSSQGLTVTVSSGGGGSG

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	GGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQ APGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAY LQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVTV SSGGGGSGGGSGGGGSQTVVTTQEPSLTVSPGGTVTLTCASST GAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSL LGKKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGKTLTVLH HHHHH
Exemplary anti-CD3 scFv (SEQ ID NO: 794)	QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQK GQAPRGLIGGTFKFLVPGTPARFSGSLLGKKAALTLSGVQPEDE EYYCTLWYSNRWVFGGKTLTVLGGGGSGGGSGGGSEVQ LVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGL WVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN KTEDTAVYYCVRHANFGNSYISYWAYWGQGLVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 804) EPL90	QVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMAWYRQSPG NERELVASISSGFTNYADSVKARFTISRDNKNTVYLQMN PEDTAVYFCGATFLRSDGHHTINGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 805) EPL118	QVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMGWFRQSPG NERELVATVSSGFTNYADSVKGRFTISRDNKNTVYLQMN KPEDTAVYFCGATFVRSDGHHTIYGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 806) EPL138	QVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMDWYRQFP NERESIATISSGFTNYADSVKGRFTISRDNKNTVYLQMN PEDTAVYFCGATFLRSDGHHTINGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 807) EPL145	QVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMGWFRQSPG NERELVATVSSGFTNYADSVKGRFTISRDNKNTVYLQMN KPEDTAVYFCGATFVRSDGHHTIYGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 808) EPL164	QVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMDWYRQSPG TQPELVATISSGFTNYANSVKGRFTISRDNKNTVYLQMN PEDTAVYFCGATFLRSDGQHSIYGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 809) EPL31	QVQLQESGGGLVHTGGSLRLSCAASGDTFLRYAMGWFRQAP GKEREFVAAITWNGGNTDYAGSLKGRFTISRDNKNTVYLQ NSLKPEDTAVYYCAADLTFLGLASSHYQDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 810) EPL55	QVQLQESGGGLVQAGGSLRLSCAASGDTFLRYAMGWFRQAP GKEREFVAAITWNGGNTDYAGSLKGRFTISRDNKNTVYLQ NSLKPEDTAVYYCAADLTFLGLASSHYQDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 811) EPL57	QVQLQESGGGLVHTGGSLRLSCAFSGDTFLRYAMGWFRQAP GKEREFVAAITWNGGNTDYADSLKGRFTISRDNKNTVYLQ NSLRPEDTAVYYCAADLTFLGLASSHYQDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 812) EPL136	QVQLQESGGGLVQPGGSLRLSCAASGDTFLRYAMGWFRQAP GKEREFVAAITWNGGNTDYAGSLKGRFTISRDNKNTVYLQ NSLKPEDTAVYYCAADLTFLGLASSHYQDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 813) EPL15	QVQLQESGGGSLVLAGGSLRLSCAASGFTFSSYYMSWVRQAP KGLEWVSGIHYTGDTNYADSVKGRFTISRDNKNTVYLQ NLKPEDTAVYYCARGSDKGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 814) EPL34	QVQLQESGGGSLVQAGGSLRLSCAASGFTFSSYYMSWVRQAP GKGLEWVSGIHYTGDTNYADSVKGRFTISRDNKNTVYLQ NNLKPEDTAVYYCARGSDKGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 815) EPL86	QVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWVRQAP GKGLEWVSGIHYGDHTTHYADFVKGRFTISRDDAKNTVYLQ NSLKPEDTAVYYCARGSTKGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 816) EPL153	QVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWVRQAP GKGLEWVSSIHYGDHTTHYADFVKGRFTISRDDAKNTVYLQ SLKPEDTAVYYCEKGTTRGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 817) EPL20	QVQLQESGGGLVQAGGSLKLSCAASGNVFRAATMAWYRQAP EKQREMVATIASGGTTNYADFVKGRFTISRDNKNTVYLQ TLKPEDTAVYYCNAGYLTSLGPKNYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 818) EPL70	QVQLQESGGGLVQPGGSLRLSCAASGNVFRAATMAWYRQAP EKXREMVATIASGGTTNYADFVKGRFTISRDNKNTVYLQ LKPEDTAVYYCNAGYLTSLGPKNYWGQGTQVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (EpCAM) (SEQ ID NO: 819) EPL125	QVQLQESGGGLVQAGGSLRLSCAASGNVFRAATMAWYRQVP EKQREMVATIASGGTTNYADSVKGRFTISRDNAKNTVYLQMN TLKPEDTAVYYCNALYLTS LGPKSYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 820) EPL13	QVQLQESGGGLVQPGGSLRLSCAASGFAPGNHWMYWRQA PGRGRELVASISSGGSTNYVD SVKGRFTISRDNARNTVYLQMY LKPEDTAVYYCGTSDNWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 821) EPL129	QVQLQESGGGLVQAGGSLRLSCAASGFAPGNHWMYWRQA PGRGRELVASISSGGSTNYVD SVKGRFTISRDNARNTVYLQMY LRPEDTAVYYCGTSDNWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 822) EPL159	QVQLQESGGGLVQAGGSLRLSCAASGFAPGNHWMYWRQA PGRGRELVASISSGGSTNYVD SVKGRFTISRDNARNTVYLQMY LKPEDTAVYYCGTSDNWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 823) EPL120	QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASISWYRQSPGN ERELVATINSGGFTNYADSVLGRFTISRDNAKNTGYLQMN EDTAVYFCAATFLRSDGQPPIWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 824) EPL126	QVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMGWYRQSPG NERELVATINSGGFTNYADSVKGRFTISRDNAKNTGYLQMN KPEDTAVYFCAATFLRSDGQPPIWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 825) EPL60	QVQLQESGGGLVQAGGSLRLSCAASEYILSMYRMAWYRQAP GKVRELVADMSSGGTTNYADSVKGRFTISRDNDRNTVYLQMN RLQPEDTAVYYCNVAGRTGPPSYDAFNNWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 826) EPL156	QVQLQESGGGLVQPGASLRVSCAASEYILSMYRMAWYRQAP GKVRELVADMSSGGTTNYADSVKGRFTISRDNDRNTVYLQMN RLQPEDTAVYYCNVAGRTGPPSYDAFNNWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 827) EPL2	QVQLQESGGGLVQPGGSLRLSCAASESISFIAVGWYRQAPGK ERELVAGINRSGFTYYTDSVKGRFISRDNAKNTVLLQMTSLKP EDTAVYYCNAGGLYFSNAYTQGDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 828) EPL43	QVQLQESGGGLVQTGGSLRLSCAASESISFIAVGWYRQAPGK ERELVAGINRSGFTYYTDSVKGRFISRDNAKNTVLLQMTSLKP EDTAVYYCNAGGLYFSNAYTQGDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 829) EPL10	QVQLQESGGGLVQAGGSLRLSCAASGSVFRANVMGWYRQAP GKQHELVARIDPGGTTTADPVKGRFTISRDNAKNTVYLQMN LKPDDTAVYYCNAIILLSGGPKDYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 830) EPL49	QVQLQESGGGLVQAGGSLRLSCAPSGRTSSIFGMGWFRQAPG KEREFVASINWGGSTSYADSVKGRFTISRDNAKNEMYLQMN SLKFEDTAVYVCAAVLTNKP SWNFWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 831) EPL58	QVQLQESGGGLVQAGGSLRLSCAASGPIFSDTIRTMGWYRQA AGKQREL VATIASFPRTNYVD SVKGRFTISRDI AKNTVYLQMD SLKPEDTAVYYCNVDLASIPTKTYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 832) EPL74	QVQLQESGGGLVQAGGSLRLSCAASGSIFGINAMGWYRQAP GKQRESVAFITIGGNTNYLDSVKGRFTISRDNAKNTVYLQMN LKPEDTAVYYCNTNPPLILTAGGLYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 833) EPL78	QVQLQESGGGLVQPGGSLRLSCATSANRFNINVMGWYRQAP GQQR ELVATINIGGSTDYADSVKGRFTISRDNAKNTVYLQSDL KPEDTAVYYCNVLRVSGPTGPNVYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 834) EPL82	QVQLQESGGGLVQAGGSLKLSCTASGTILSTMAWYRQAPGKQ REL VATISRGGTTNYSDSVKGRFAISRSTKNTVYLQMN SLKPE DTAVYYCNTPLTDYGMGYNWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 835) EPL83	QVQLQESGGGLVQAGGSLRLSCAVSGSIFSLNTLAWYRQAPGR QRDLIARITGGGTTVYADSVKGRFTISRDNAKNTVYLQMN EDTAVYYCNLMVRHPSGSTYEYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 836) EPL97	QVQLQESGGGLVQAGGSLRLSCAASGIFRGTTMGWFRQAPG KQRESVASISPLGTTSYSGSVEGRFTVSRDNAKNTLFLQMN SLKSEDTAVYYCNAIQVTNVGPRVYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 837) EPL109	QVQLQESGGGLVQPGGSLRLSCASSGFTLDDYTI GWFRQAPG KEREGVSCISRRDDSTYYADSVKGRFTISRDNAKNTVDLQMI SLRPEDTAVYYCAATPRSYTLRCLGKFDFQGGTQVTVSS

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DESCRIPTION	SEQUENCE
Exemplary anti-target (EpCAM) (SEQ ID NO: 838) EPL117	QVQLQESGGGLVQAGGSLRLSCAASGNIVRMTNMAWYRQA PGKQREFVATISAGGSTTYVDSVKDRFTISRDNKNTVYLQMN YLKPEDTAVYYCATGSILTNRAIPGSWGHGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 839) EPL127	QVQLQESGGGLVQAGGSLRLSCAAPGFAPNDHAILWFRQAPG KREGVSEICRDGTTYTDSVKGRFTISSDNAKNTVYLQMNSVK TDDTAVYYCAVDRRRYYCSGNRAFSSDYIYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 840) EPL152	QVQLQESGGGLVQAGGSLRLSCVHSGSIFRASTMAWYRQAPG KQRELVAQIMSGGNTYAGSVKGRFTISRDNANNTVYLQMNS LKPEDTAVYYCNAAQITSWGPKVYWGQGTQVTVSS
Exemplary anti-target (EpCAM) (SEQ ID NO: 841) EPL189	QVQLQESGGGLVQPGGSLRLSCAASGRINSINTMGWYRQAPG NQRELVAEITRGGTTNYADSVQGRYAI SRDNAKNLVYLQMNSL KPEDTDVYYCNAQTPTFSRPTGLDYWGQGTQVTVSS
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 842) EPL10	VVGGGGTQTVVTTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGLKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTQVTVSS VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGSVFR ANVMGWYRQAPGKQHELVARIDPGGTTTYADPVKGRFTISR NAKNTVYLQMNLSKLPDDTAVYYCNAILLSGGPKDYWGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 843) EPL109	VVGGGGTQTVVTTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGLKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTQVTVSS VSSGGGGSGGGSQVQLQESGGGLVQPGGSLRLSCASSGFTLD DYTIWFRQAPGKEREVSCISRRDDSTYYADSVKGRFTISRDN AKNTVDLQMI SLRPEDTAVYYCAATPRSYTLRCLGKFD PQGQG TQVTVSSHHHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 844) EPL117	VVGGGGTQTVVTTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGLKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTQVTVSS VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGNIVR MTNMAWYRQAPGKQREFVATISAGGSTTYVDSVKDRFTISR DNKNTVYLQMNLYLKPEDTAVYYCATGSILTNRAIPGSWGHGT QVTVSSHHHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 845) EPL120	VVGGGGTQTVVTTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGLKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTQVTVSS VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR AASISWYRQSPGNERELVATINSGGFTNYADSVLGRFTISRDN KNTGYLQMNLSKPEDTAVYFCAATFLRSDGQPPIWGQGTQVT VSSHHHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 846) EPL125	VVGGGGTQTVVTTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGLKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTQVTVSS VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGNVFR AATMAWYRQVPEKQREMVATISAGGTTNYADPVKGRFTISR NAKNTVYLQMNLTLPEDTAVYYCNALYLTSLGPKSYWGQGTQ VTVSSHHHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 847) EPL127	VVGGGGTQTVVTTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGGLKLTVLGGGGSGGGSG

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAAPGFAN DHAILWFRQAPGKEREGVSEICRDGTTYTDSVKGRFTISSDNA KNTVYLMNSVKTDDTAVYYCAVDRRRYYCSGNRAFSSDY WGQGTQVTVSSHSHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 848) EPL13	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGKLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGFAG NHWMYWYRQAPGRGRELVASISSGGSTNYVDSVKGRFTISR NARNTVYLMYSLKPEDTAVYYCGTSDNWGQGTQVTVSSH HHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 849) EPL136	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGKLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGDTFL RYAMGWFRQAPGKEREFVAITWNGGNTDYAGSLKGRFTISR DNTKNTVYLMNSLKPEDTAVYYCAADLTFLGLASSHYQYDYW GQGTQVTVSSHSHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 850) EPL138	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGKLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR AASMDWYRQFPGNERESIATISSGGFTNYADSVKGRFTISRDN AKNTVYLMNSLKPEDTAVYFCGATFLRSDGHHTINGQGTQV TVSSHSHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 851) EPL145	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGKLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR AASMGWFRQSPGNERELVATVSSGGFTNYADSVKGRFTISR NAKNTVYLMNSLKPEDTAVYFCGATFVRSDGHHTIYGQGTQ VTVSSHSHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 852) EPL152	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGKLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCVHSGSIFR ASTMAWYRQAPGKQRELVAQIMSGGNTYAGSVKGRFTISR DNANNTVYLMNSLKPEDTAVYYCNAAQITSWGPKVYWQ GTQVTVSSHSHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 853) EPL153	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLG VQPEDEAEYYCTLWYSNRWVFGGKLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFS DWAMSWVRQAPGKGLEWSSIHYGDHTHYADFKGRFTISR RDDAKNTLYLMNSLKPEDTAVYYCEKGTTRGQGTQVTVSSH HHHH*

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 854) EPL156	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGLVQPGASLRVSCAASEYILS MYRMAWYRQAPGKVRELVADMSSGGTNYADPVKGRFTISR DNDRNTVYQLQMNRLQPEDTAAYYCNVAGRTGPPSYDAPNN WGQGTQTVTSSHHHHHH*
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 855) EPL164	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFI AASMDWYRQSPGTQPELVATISSTGFTNYANSVKGRFTISRDN AKNTVYQLQMNLSKPEDTAVYFCGATFLRSDGQHSIYGQGTQV TVSSHHHHHH*
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 856) EPL189	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGRINSI NTMGWYRQAPGNQRELVAEITRGGTNYADSVQGRYAI NAKNLVYQLQMNLSKPEDTDVYVCNAQTPTFTRPTGLDYWGQ GTQTVTSSHHHHHH*
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 857) EPL2	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASESISSFI AVGWYRQAPGKERELVAGINRSGFTYYTDSVKGRFSISRDN NTVLLQMTSLKPEDTAVYYCNAGGLYFSNAYTQGDYWGQGT QTVTSSHHHHHH*
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 858) EPL20	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGLVQAGGSLKLSCAASGNVFR AATMAWYRQAPKQREMVATIASGGTNYADPVKGRFTISR NAKNTVYQLQMNLTLPEDTAVYYCNAGYLTSLGPKNYWGQGT QTVTSSHHHHHH*
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 859) EPL34	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGSVQAGGSLRLSCAASGFTFS SYMSWVRQAPGKLEWVSGIHYTGDTNYADSVKGRFTISR DNAKNELYLEMNLTLPEDTAVYYCARGSDKGQGTQTVTSSHH HHHH*
Exemplary anti-CD3/anti-EpCAM fusion protein(SEQ ID NO: 860) EPL49	VVGGGGTQTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAPSGRTSSI FGMGWFRQAPGKEREFVASINWSGGSTSYADSVKGRFTISR

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	NAKNEMYLQMNSLKPEDTAVYVCAAAVLTNKP SWNFWGQG TQVTVSSH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 861) EPL58	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGPIFS DTIRTMGWYRQAAGKQREL VATIASFPRTNYVDSVKGRFTIS RDI AKNTVYLQMDSLKPEDTAVYYCNVDLASIPTKTYWGQGTQ VTVSSH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 862) EPL74	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGSI FGI NAMGWYRQAPGKQRESVAFITIGGNTNYLDSVKGRFTISRDN AKNTVYLQMNGLKPEDTAVYYCNTNPPLILTAGGLYWGQGTQ VTVSSH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 863) EPL78	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCATSANRPN INVMGWYRQAPGQREL VATINIGGSTDYADSVKGRFTISR NAKNTVYLQLSDLKPEDTAVYYCNVLRVSGPTGPNVYWGQG TQVTVSSH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 864) EPL82	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLKLSCTASGTILST MAWYRQAPGKQREL VATISRGGTNYSDSVKGRFAISRDSKTN TVYLQMNLSLKPEDTAVYYCNTPLTDYGMGYNWGQGTQVTVS SH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 865) EPL83	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAVSGSIFSL NTLAWYRQAPGRQDLIARITGGGT VYADSVKGRFTISR DNA KNTVYLQMNLSLKPEDTAVYYCNLMVRHPSGSTYEWGQGTQ VTVSSH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 866) EPL86	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGSGGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFS DWAMSWVRQAPGKGLEWVSGIHYGDHTHYADFVKGRFTIS RDDAKNTLYLQMNLSLKPEDTAVYYCARGSTKGQGTQVTVSSH HHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 867) EPL90	VVGGGGTQT VVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGKTCLTVLGGGSGGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFR AASMAWYRQSPGNERELVASISSGAFTNYADSVKARFTISRDN AKNTVYLQMNSLKPEDTAVYFCGATFLRSDGHHTINGQTQV TVSSHHHHHH*
Exemplary anti-CD3/anti- EpCAM fusion protein(SEQ ID NO: 868) EPL97	VVGGGGTQTQVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTKFLVPGTTPARFSGSLGGAALTLTG VQPEDEAEYYCTLWYSNRWVFGGGLVQAGGSLRLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVLT VSSGGGGSGGGSQVQLQESGGGLVQAGGSLRLSCAASGIIIFRG TTMGWFRQAPGKQRESVASISPLGTTSYSGSVEGRFTVSRDNA KNTLFLQMNSLKSEDVAYYCNAIQVTNVGPRVYWGGTQVT VSSHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 869) EPL10	GGGSQVQLQESGGGLVQAGGSLRLSCAASGSVFRANVMGWY RQAPGKQHELVARIDPGGTTTYADPVKGRFTISRDNAAKTQVYL QMNSLKPDDTAVYYCNAIIILLSGGPKDYWGQGTQVTVSSGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VTVSSGGGGSGGGSGGGGSGTQVVTQEPSLTVSPGGTVTLTC ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTTPARF SGSLGGAALTLTGQVQPEDEAEYYCTLWYSNRWVFGGGLT VLHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 870) EPL109	GGGSQVQLQESGGGLVQPGGSLRLSCASSGFTLDDYTIGWFR QAPGKEREGVSCISRDDSTTYADSVKGRFTISRDNAAKTQVYL MISLRPEDTAVYYCAATPRSYTLRCLGKFDQGGTQVTVSSG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSGTQVVTQEPSLTVSPGGTVTL TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTTPA RFGSLGGAALTLTSGVQPEDEAEYYCTLWYSNRWVFGGGLT LTVLHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 871) EPL117	GGGSQVQLQESGGGLVQAGGSLRLSCAASGNIVRMTNMAW YRQAPGKQREFVATISAGGSTTYVDSVKDRFTISRDNKNTVYL QMNYLKPEDTAVYYCATGSLTNRGAI PGSWGHGTQVTVSSG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSGTQVVTQEPSLTVSPGGTVTL TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTTPA RFGSLGGAALTLTSGVQPEDEAEYYCTLWYSNRWVFGGGLT LTVLHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 872) EPL120	GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASISWYRQ SPGNERELVATINSGGFTNYADSVLGRFTISRDNAAKTGYLQM NSLKPEDTAVYFCAATFLRSDGQPPIWGGTQVTVSSGGGG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVLT VSSGGGGSGGGSGGGGSGTQVVTQEPSLTVSPGGTVTLTCAS STGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTTPARFSG SLGGAALTLTSGVQPEDEAEYYCTLWYSNRWVFGGGLT HHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 873) EPL125	GGGSQVQLQESGGGLVQAGGSLRLSCAASGNVFRAATMAWY RQVPEKQREMVATIASGGTNYADFVKGRFTISRDNAAKTQVYL QMNTLKPEDTAVYYCNALYLSLGPKEYWGQGTQVTVSSGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLT VTVSSGGGGSGGGSGGGGSGTQVVTQEPSLTVSPGGTVTLTC ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTKFLVPGTTPARF SGSLGGAALTLTSGVQPEDEAEYYCTLWYSNRWVFGGGLT VLHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 874) EPL127	GGGSQVQLQESGGGLVQAGGSLRLSCAAPGFANDHAILWFR QAPGKEREGVSEICRDGTTYTDSVKGRFTISSDNAKNTVYLQ MNSVKTDVTAVYYCAVDRRRYYCSGNRAPSSDYWYWGQGTQ VTVSSGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTF

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	NKYAINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFT ISRDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAY WGGGTLVTVS SGGGGSGGGSGGGGSGTQVTVTQEPSTVSPG GTVTLTCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLV PGTAPRFGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVF GGGTKLTVLHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 875) EPL13	GGGSQVQLQESGGGLVQPGGSLRLSCAASGFAFGNHWMYW YRQAPGRGRELVASISSGGSTNYVDSVKGRFTISRDNARNTVYL QMYSLKPEDTAVYYCGTSDNWGGGTQVTVSSGGGGSGGGSE VQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGK GLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN NNLKTEDTAVYYCVRHANFGNSYISYWAYWGGGTLVTVS SGG GGGGGGGSGGGGSGTQVTVTQEPSTVSPGGTTLTCASSTGAV TSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTAPRFGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 876) EPL136	GGGSQVQLQESGGGLVQPGGSLRLSCAASGDTFLRYAMGWF RQAPGKEREFVAAITWNGNTDYAGSLKGRFTISRDNKTNTVY LQMNSLKPEDTAVYYCAADLTFLGLASSHYQDYWGGGTQVTV SGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKY AINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR DDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYW GGGTTLVTVS SGGGGSGGGSGGGGSGTQVTVTQEPSTVSPGGT VTLCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPG TPARFGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGG GTKLTVLHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 877) EPL138	GGGSQVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMDWY RQPPGNERESIATISSGGFTNYADSVKGRFTISRDNKNTVYVQ MNSLKPEDTAVYFCGATFLRSDGHHTINGGTQVTVSSGGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGGGTTLV VSSGGGGSGGGSGGGGSGTQVTVTQEPSTVSPGGTTLTCAS STGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTAPRFG SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 878) EPL145	GGGSQVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMGWF RQSPGNERELVATVSSGGFTNYADSVKGRFTISRDNKNTVYL QMNSLKPEDTAVYFCGATFVRSDGHHTIYGGGTQVTVSSGGG GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGGGTTL VTVSSGGGGSGGGSGGGGSGTQVTVTQEPSTVSPGGTTLTC ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTAPR FSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKL TVLHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 879) EPL152	GGGSQVQLQESGGGLVQAGGSLRLSCVHSGSIFRASTMAYR QAPGKQRELVAQIMSGGGTNYAGSVKGRFTISRDNANNTVYL QMNSLKPEDTAVYYCNAAQITSWGPKVYWGQTQVTVSSGG GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGGG TLVTVS SGGGGSGGGSGGGGSGTQVTVTQEPSTVSPGGTTL TCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTAP RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 880) EPL153	GGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWV RQAPGKGLEWVSSIHYGDHTHYADFVKGRFTISRDDAKNTLY LQMNSLKPEDTAVYYCEKGTTRGGGTQVTVSSGGGGSGGGSE VQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGK GLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMN NNLKTEDTAVYYCVRHANFGNSYISYWAYWGGGTLVTVS SGG GGGGGGGSGGGGSGTQVTVTQEPSTVSPGGTTLTCASSTGAV TSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTAPRFGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 880) EPL153	GGGSQVQLQESGGGLVQPGASLRVSCAASEYILSMYRMAWY RQAPGKVRELVDMSGGTNYADFVKGRFTISRDNDRNTVY

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
NO: 881) EPL156	LQMNRLQPEDTAAYYCNVAGRTGPPSYDAFNNWQGQTQVT VSSGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNK YAINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISR DDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYW GQGTTLTVSSGGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGT VTLTCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPG TPARFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGG GTKLTVLHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 882) EPL164	GGGSQVQLQESGGGLVQAGGSLRLSCAASGFI FRAASMDWY RQSPGTQPELVATISSTGFTNYANSVKGRFTISRDNKNTVYLQ MNSLKPEDTAVYFCGATFLRSDGQHSIYGQGTQVTVSSGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTTLV VSSGGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCAS STGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPARFSG SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 883) EPL189	GGGSQVQLQESGGGLVQPGGSLRLSCAASGRINSINTMGWYR QAPGNQRELVAEITRGTTNYADSVQGRYAIRDNKLNLYLQ MNSLKPEDTDVYYCNAQTFFTFSRPTGLDYWGQGTQVTVSSG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPA RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 884) EPL2	GGGSQVQLQESGGGLVQPGGSLRLSCAASESISFIAVGWYRQ APGKERELVAGINRSGFTYYTDSVKGRFISRDNKNTVLLQMT SLKPEDTAVYYCNAAGLYFSNAYTQGDYWGQGTQVTVSSGG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPA RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 885) EPL20	GGGSQVQLQESGGGLVQAGGSLKLSCAASGNVFRAATMAWY RQAPEKQREMVATIASGGTNYADFVKGRFTISRDNKNTVYL QMNTLKPEDTAVYYCNAAGLYTSLGPKNYWGQGTQVTVSSGG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPA RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHHH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 886) EPL34	GGGSQVQLQESGGGSVQAGGSLRLSCAASGFTFSSYYMSWVR QAPGKGLEWVSGIHYTGDWNTYADSVKGRFTISRDNKNELY LEMNMLKPEDTAVYYCARGSDKQGTQVTVSSGGGGSGGG EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPG KGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQ MNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTTLTVSSG GGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTLTCASSTGAV TSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH*
Exemplary anti-EpCAM/anti- EpCAM fusion protein(SEQ ID NO: 887) EPL49	GGGSQVQLQESGGGLVQAGGSLRLSCAPSGRTSSIFGMGWFR QAPGKEREFVASINWGGSTSYADSVKGRFTISRDNKNEMYL QMNSLKFEDTAVYCAAVALTNKPSWNFWGQGTQVTVSSG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSQTVVTQEPSTLTVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPA RFSGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHHH*

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 888) EPL58	GGGSQVQLQESGGGLVQAGGSLRLSCAASGPIFSDTIRTMGW YRQAAGKQRELVATIASFPSTNYVDSVKGRFTISRDIKNTVYL QMDSLKPEDTAVYYCNVDLASIPTKTYWGQGTQVTVSSGGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGGGGGGSGGGGSQTVVTQEPSTLVSPGGTVTLTCAS STGAVTSGNYPNWVQKPGQAPRGLIGGTFKFLVPGTPARFSG SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 889) EPL74	GGGSQVQLQESGGGLVQAGGSLRLSCAASGSIFGINAMGWYR QAPGKQRESVAFITIGGNTNYLDSVKGRFTISRDNKNTVYLQ MNLKLPEDTAVYYCNTNPPLIITAGGLYWGQGTQVTVSSGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN TAYLQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQGL VTVSSGGGGSGGGSGGGGSQTVVTQEPSTLVSPGGTVTLTC ASSTGAVTSGNYPNWVQKPGQAPRGLIGGTFKFLVPGTPARF SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKL VLHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 890) EPL78	GGGSQVQLQESGGGLVQPGGSLRLSCATSANRFNINVMGWY RQAPGQRELVATINIGGSTDYADSVKGRFTISRDNKNTVYL QLSDLKPEDTAVYYCNVLRVSGPTGPNVYWGQGTQVTVSSG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSQTVVTQEPSTLVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTFKFLVPGTPA RFSGLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 891) EPL82	GGGSQVQLQESGGGLVQAGGSLKLSCTASGTILSTMAWYRQA PGKQRELVATISRGTTNYSVSKGRFAISRSTKNTVYLQMN LKPEDTAVYYCNTPLTDYGMGYNWGGTQVTVSSGGGGSGG GSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAP GKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLTVTVSS GGGGGGGGSGGGGSQTVVTQEPSTLVSPGGTVTLTCASSTG AVTSGNYPNWVQKPGQAPRGLIGGTFKFLVPGTPARFSGSLL GGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHH HHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 892) EPL83	GGGSQVQLQESGGGLVQAGGSLRLSCAVSGSIFSLNTLAWYR QAPGRQRLDIARITGGGTTVYADSVKGRFTISRDNKNTVYLQ MNSLKPEDTAVYYCNLMVRHPSGSTEYEWGQGTQVTVSSGG GGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAIN WVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDS KNTAYLQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQ TLVTVSSGGGGSGGGSGGGGSQTVVTQEPSTLVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTFKFLVPGTPA RFSGLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTK LTVLHHHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 893) EPL86	GGGSQVQLQESGGGLVQPGGSLRLSCAASGFTFSDWAMSWV RQAPGKGLEWVSGIHYGDHHTHYADFVKGRFTISRDDAKNTLY LQMNSLKPEDTAVYYCARGSTKGGTQVTVSSGGGGSGGGSE VQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGK GLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLTVTVSSG GGSGGGSGGGGSQTVVTQEPSTLVSPGGTVTLTCASSTGAV TSGNYPNWVQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHH HH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein(SEQ ID NO: 894) EPL90	GGGSQVQLQESGGGLVQAGGSLRLSCAASGFIFRAASMAWYR QSPGNERELVASISSGAFTNYADSVKARFTISRDNKNTVYLQ MNSLKPEDTAVYFCGATFLRSDGHHTINGQGTQVTVSSGGGG SGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNLLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLVT VSSGGGGSGGGSGGGGSQTVVTQEPSTLVSPGGTVTLTCAS

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	STGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARFSG SLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVL HHHHH*
Exemplary anti-EpCAM/anti-EpCAM fusion protein (SEQ ID NO: 895) EPL97	GGGSQVQLQESGGGLVQAGGSLRLSCAASGIIFRGTTMGWFR QAPGKQRESVASISPLGTTSYSGSVGRFTVSRDNAKNTLFLQ MNSLKSEDTAVYYCNAIQVTNVGPRVYWGQGTQVTVSSGGG GSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINW VRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKN TAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGT LTVSSGGGGSGGGSGGGGSGTQVVTQEPSTLTVSPGGTVTLT ASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARF SGSLLGGKAALTLSGVQPEDEAEYYCTLWYSNRWVFGGGTKL VLHHHHH*
Exemplary humanized anti-EpCAM sequence (SEQ ID No. 896)	EVQLVESGGGLVQPGGSLTSLCAASGFAPGNHWMYWRQAP GRGRELVASISSGGSTNYVDSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCGTSDNWGQGLTVTVSS
Exemplary humanized anti-EpCAM sequence (SEQ ID No. 897)	EVQLLESGGGLVQPGGSLTSLCAASGFIFRAASMDWYRQFP NERESIATISSGGFTNYADSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCGATFLRSDGHHTINGQGLTVTVSS
Exemplary humanized anti-EpCAM sequence (SEQ ID No. 898)	EVQLLESGGGLVQPGGSLTSLCAASGFIFRAASMAWYRQSP NERELVASISSGAFTNYADSVKARFTISRDNKNTLYLQMN SLRAEDTAVYYCGATFLRSDGHHTINGQGLTVTVSS
Exemplary Humanized Anti-CD3/Anti-EpCAM sequence name H13 (SEQ ID No. 899)	VVGGGGTQTQVVTQEPSTLTVSPGGTVTLTCAASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLTV TVSSGGGGSGGGSEVQLVESGGGLVQPGGSLTSLCAASGFAP GNHWMYWRQAPGRGRELVASISSGGSTNYVDSVKGRFTISR DNKNTLYLQMNSLRAEDTAVYYCGTSDNWGQGLTVTVSSH HHHH*
Exemplary Humanized Anti-CD3/Anti-EpCAM sequence name H138 (SEQ ID No. 900)	VVGGGGTQTQVVTQEPSTLTVSPGGTVTLTCAASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLTV TVSSGGGGSGGGSEVQLLESGGGLVQPGGSLTSLCAASGFIFRA ASMDWYRQFPGNRESIATISSGGFTNYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCGATFLRSDGHHTINGQGLTV TVSSHHHHHH*
Exemplary Humanized Anti-CD3/Anti-EpCAM sequence name H90 (SEQ ID No. 901)	VVGGGGTQTQVVTQEPSTLTVSPGGTVTLTCAASSTGAVTSGNYPN WVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLSG VQPEDEAEYYCTLWYSNRWVFGGGTKLTVLGGGGSGGGSG GGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVR QAPGKGLEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTA YLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLTV TVSSGGGGSGGGSEVQLLESGGGLVQPGGSLTSLCAASGFIFRA ASMAWYRQSPGNERELVASISSGAFTNYADSVKARFTISRDN SKNTLYLQMNSLRAEDTAVYYCGATFLRSDGHHTINGQGLTV TVSSHHHHHH*
SEQ ID No. 902	QDGNEEMGGITQ
SEQ ID No. 903 AB loop oligo WT	LVQPGN
SEQ ID No. 904 C'D loop oligo WT	DSVKGR
SEQ ID No. 905 EF loop oligo WT	SLRPED
EpCAM H90 TriTAC C2854 (SEQ ID No. 906)	EVQLVESGGGLVQPGNSLRLSCAASGFTFSKFGMSWVRQ APGKGLEWVSSISGSGRDTLYADSVKGRFTISRDNKNT LYLQMNSLRPEDTAVYYCTIGGSLSVSSQGLTVTVSSGG GGSGGGSTQVVTQEPSTLTVSPGGTVTLTCAASSTGAVTSG

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SEQUENCE TABLE	
DESCRIPTION	SEQUENCE
	NYPNWVQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGG KAALTLSGVQPEDEAEYYCTLWYSNRWVFGGKTLTVLG GGGSGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAAS GFTENKYAINWVRQAPGKGLEWVARIRSKYNNYATYYAD QVKDRFTISRDDSKNTAYLQMNNLKTEDTAVYYCVRHAN FGNSYISYWAYWGQGLTVTVSSGGGGSGGSEVQLLESG GGLVQPGGSLTSCAASGFI FRAASMAWYRQSPGNEREL VASISSGAFTNYADSVKARFTISRDN SKNTLYLQMNLSR AEDTAVYYCGATFLRSDGHHTINGQGLTVTVSSHHHHHH
EpCAM H90 ProTriTAC (L040) C2704 (SEQ ID No. 907)	QTVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQ QKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALTLS GVQPEDEAEYYCTLWYSNRWVFGGKTLTVLGGGGSGG GSGGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTENKY AINWVRQAPGKGLEWVARIRSKYNNYATYYADQVKDRFT ISRDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYIS YWAYWGQGLTVTVSSGGGGSGGSEVQLLESGGGLVQPG GSLTSCAASGFI FRAASMAWYRQSPGNERELVASISSG AFTNYADSVKARFTISRDN SKNTLYLQMNLSRAEDTAVY YCGATFLRSDGHHTINGQGLTVTVSSHHHHHH
EpCAM H90 ProTriTAC (NCLV) C2302 (SEQ ID No. 908)	EVQLVESGGGLVQPGNSLRSLSCAASGFTFSKFGMSWVRQ GGGGGLDGN EEPGGLEWVSSI SGSGRDTLYADSVKGRFT ISRDN AKTTLYLQMNLSRPEDTAVYYCTIGGSLSVSSQG TLTVTVSSGGGGSGGGSGGVVGGGTQTVVTQEPSLTVS PGGTVTLTCASSTGAVTSGNYPNWVQKPGQAPRGLIGG TKFLVPGTPARFSGSLLGGKAALTLSGVQPEDEAEYYCT LWYSNRWVFGGKTLTVLGGGGSGGGSGGGSEVQLVE SGGGLVQPGGSLKLSCAASGFTENKYAINWVRQAPGKGL EWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQ MNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGLTVTV SSGGGSGGGSEVQLLESGGGLVQPGGSLTSCAASGFI FRAASMAWYRQSPGNERELVASISSGAFTNYADSVKARF TISRDN SKNTLYLQMNLSRAEDTAVYYCGATFLRSDGHH TINGQGLTVTVSSHHHHHH

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12415860B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A conditionally active binding protein comprising
 - a) a binding moiety (M) which comprises a non-CDR loop, wherein the binding moiety (M) comprises a serum albumin binding domain and the non-CDR loop comprises a binding site specific for a CD3ε domain, and wherein the binding moiety further comprises complementarity determining regions (CDRs),
 - (b) a cleavable linker (L),
 - (c) a first target antigen binding domain (T1) comprising an immunoglobulin molecule that binds to CD3ε, and
 - (d) a second target antigen binding domain (T2),
 wherein the binding moiety (M) is capable of masking the binding of the first target antigen binding domain (T1) to CD3ε.
2. The conditionally active binding protein of claim 1, wherein the binding moiety is capable of binding to a half-life extending protein.
3. The conditionally active binding protein of claim 1, wherein the binding moiety is a natural peptide, a synthetic peptide, an engineered scaffold, or an engineered serum bulk protein.
4. The conditionally active binding protein of claim 3, wherein the engineered scaffold comprises an sdAb, an scFv, a Fab, a VHH, a fibronectin type III domain, an immunoglobulin-like scaffold, a DARPin, a cystine knot peptide, a lipocalin, a three-helix bundle scaffold, a protein G-related albumin-binding module, or a DNA or RNA aptamer scaffold.
5. The conditionally active binding protein of claim 1, wherein the non-CDR loop is from a variable domain, a constant domain, a C1-set domain, a C2-set domain, an I-domain, or any combinations thereof.
6. The conditionally active binding protein of claim 1, wherein the binding site is capable of binding to a human serum albumin.

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7. The conditionally active binding protein of claim 6, wherein the CDRs provide the binding site specific for the serum albumin.

8. The conditionally active binding protein of claim 1, wherein the non-CDR loop provides the binding site specific for binding of the binding moiety to the first target antigen binding domain.

9. The conditionally active binding protein of claim 1, wherein the the second target antigen binding domain binds to a tumor antigen.

10. The conditionally active binding protein of claim 9, wherein the tumor antigen comprises EpCAM, EGFR, HER-2, HER-3, c-Met, FoIR, PSMA, CD38, BCMA, CEA, 5T4, AFP, B7-H3, CDH-6, CAIX, CD117, CD123, CD138, CD166, CD19, CD20, CD205, CD22, CD30, CD33, CD352, CD37, CD44, CD52, CD56, CD70, CD71, CD74, CD79b, DLL3, EphA2, FAP, FGFR2, FGFR3, GPC3, gpA33, FLT-3, gpNMB, HPV-16 E6, HPV-16 E7, ITGA2, ITGA3, SLC39A6, MAGE, mesothelin, Muc1, Muc16, NaPi2b, Nectin-4, CDH-3, CDH-17, EPHB2, ITGAV, ITGB6, NY-ESO-1, PRLR, PSCA, PTK7, RORI, SLC44A4, SLITRK5, SLITRK6, STEAP1, TIM1, Trop2, or WT1.

11. The conditionally active binding protein of claim 1, wherein the first target antigen binding domain binds to a human CD3e.

12. The conditionally active binding protein of claim 1, wherein the binding moiety (M), the cleavable linker (L), the first target antigen binding domain (T1), and the second target antigen binding domain (T2) are in one of the following configurations: M:L:T1:T2, and T2:T1:L:M.

13. The conditionally active binding protein of claim 1, wherein the cleavable linker comprises a cleavage site.

14. The conditionally active binding protein of claim 13, wherein the cleavage site is recognized by a protease.

15. The conditionally active binding protein of claim 14, wherein the protease cleavage site is recognized by a serine protease, a cysteine protease, an aspartate protease, a threonine protease, a glutamic acid protease, a metalloproteinase, a gelatinase, or an asparagine peptide lyase.

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16. The conditionally active binding protein of claim 14, wherein the protease cleavage site is recognized by a Cathepsin B, a Cathepsin C, a Cathepsin D, a Cathepsin E, a Cathepsin K, a Cathepsin L, a kallikrein, a hK1, a hK10, a hK15, a plasmin, a collagenase, a Type IV collagenase, a stromelysin, a Factor Xa, a chymotrypsin-like protease, a trypsin-like protease, a elastase-like protease, a subtilisin-like protease, an actinidain, a bromelain, a calpain, a caspase, a caspase-3, a Mir1-CP, a papain, a HIV-1 protease, a HSV protease, a CMV protease, a chymosin, a renin, a pepsin, a matriptase, a legumain, a plasmepsin, a nepenthesin, a metalloexopeptidase, a metalloendopeptidase, a matrix metalloprotease (MMP), a MMP1, a MMP2, a MMP3, a MMP8, a MMP9, a MMP10, a MMP11, a MMP12, a MMP13, a MMP14, an ADAM10, an ADAM12, an urokinase plasminogen activator (uPA), an enterokinase, a prostate-specific target (PSA, hK3), an interleukin-1 β converting enzyme, a thrombin, a FAP, a dipeptidyl peptidase, or dipeptidyl peptidase IV (DPP/CD26), a type II transmembrane serine protease (TTSP), a neutrophil elastase, a cathepsin G, a proteinase 3, a neutrophil serine protease 4, a mast cell chymase, a mast cell tryptase, a dipeptidyl peptidase, or a dipeptidyl peptidase IV (DPP/CD26).

17. The conditionally active binding protein of claim 11, wherein the binding moiety comprises a binding site specific the first target antigen binding domain (T1), and wherein the binding site comprises at least one of the following motifs: QDGNE (SEQ ID NO: 921), QDGNEE (SEQ ID NO: 801), DGNE (SEQ ID NO: 922), and DGNEE (SEQ ID NO: 923).

18. The conditionally active binding protein of claim 13, wherein upon cleavage of the cleavable site in a tumor microenvironment, the conditionally active binding protein is activated by separation of the binding moiety.

19. The conditionally active binding protein of claim 1, wherein the binding moiety comprises a sequence selected from the group consisting of SEQ ID Nos. 50 and 259-301.

20. The conditionally active binding protein of claim 1, wherein the binding moiety comprises an sdAb.

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